

STAT3006: Statistical Learning – Lecture Notes, 2022

Hien Nguyen

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Contents

Contents

1	Foundations and background	1
1.1	What is statistical learning?	1
1.1.1	What is statistics?	1
1.1.1.1	Three main statistical tasks	4
1.1.2	What is statistical learning?	7
1.2	Some probability theory	12
1.2.1	Sets and such	13
1.2.1.1	A topological detour	17
1.2.1.2	Product of σ -algebras	20
1.2.2	Measure theory	22
1.2.2.1	Distribution functions	26
1.2.2.2	Measurable functions and random variables	27
1.2.3	Integration and expectation	31
1.2.3.1	Products of measure spaces	39

CONTENTS

1.2.4	Sequences of measurable functions and random variables . . .	42
1.2.4.1	Exchanging limits and integrals	46
1.2.5	Bibliographical notes	48
1.3	Optimization Theory	50
1.3.1	Minima and infima	51
1.3.2	Giving $\mathbb{R}^p$ some structure	54
1.3.3	Functions on $\mathbb{R}^p$	60
1.3.3.1	Smoothness	60
1.3.3.2	Convexity	62
1.3.3.3	Some matrix analysis	66
1.3.4	Minimization	69
1.3.4.1	Existence and uniqueness	69
1.3.5	Computing minimizers with the MM algorithm	73
1.3.6	Sample average approximation	78
1.3.7	Bibliographical notes	84
2	Density estimation and regression analysis	85
2.1	Density estimation	85
2.1.1	Density functions	85
2.1.2	Density estimation	86
2.1.2.1	Distances and divergences	87
2.1.2.2	Maximum likelihood estimation	91
2.1.2.3	Some matrix calculus	93

CONTENTS

2.1.2.4	Misspecification	100
2.1.3	Conditional densities and conditional probability	102
2.2	Regression	109
2.2.1	Maximum likelihood estimation	109
2.2.2	Loss-based risk minimization	115

Bibliography	126
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Chapter 1

Foundations and background

1.1 What is statistical learning?

1.1.1 What is statistics?

To begin to define what statistical learning is, we firstly need to define the notion of statistics. This tends to be a very discipline dependent term, with even many statisticians often disagreeing on the full scope and meaning of the term. The definition can range from the simple summarization and presentation of numerical data and information, to the mathematical exploration of abstract notions of information and knowledge. This course takes a position somewhere closer to the latter, although we will not forget the numerical roots of the discipline.

In this course, we shall take a technical and mathematical perspective to statistics. That is, we shall define the notion of statistics via its mathematical objects of interests, its techniques of investigation, and its technical aims. Along with our tech-

nical investigation, we shall also consider examples to which we can apply statistical techniques that we developed that we investigate and develop in this course.

In this section, we will introduce objects of investigation that we will only later properly define. To get things started, we shall assume that we are interest in a **random variable** $X : \Omega \rightarrow \mathbb{X}$, which we usually write as $\omega \mapsto X(\omega)$, with ω endowed with the **probability space** $(\Omega, \mathcal{F}, P)$, where Ω is a set of **events**, $\mathcal{F}$ is a **σ -algebra** of Ω , and P is a **probability measure** of $\mathcal{F}$, respectively. Since $X : \Omega \rightarrow \mathbb{X}$, we can also endow X with its own probability measure, relative to P (and with respect to some σ -algebra $\mathcal{X}$ of $\mathbb{X}$), which we will call the **(image) probability measure** of X , denoted by P_X . We shall switch between $X(\omega)$ and X depending on the level of precision required.

If we know P_X , we would know everything that we could possibly know about X . However, we often do not need to know everything, but only something, that is we often only care about some property of P_X , which we shall write as $\mathcal{P}(P_X)$, or simply $\mathcal{P}(X)$ or $\mathcal{P}$, if it causes no confusion.

Example 1. Consider $X \in \mathbb{X} = \mathbb{R}$, then we may consider properties such as the mean (the **expectation**), written as

$$\mathcal{P}(X) = E(X) = \int_{\mathbb{X}} x dP_X = \int_{\mathbb{X}} x p_X(x) dx,$$

where we shall define the meaning of the integral expressions in the sequel. Alternatively, we may also consider the **median**

$$\mathcal{P}(X) = \text{med}(X) = \left\{ m : P_X(\{x : x \leq m\}) = \frac{1}{2} \right\}.$$

Remark 1. It is notable that our notion of a property is equivalent to what is usually called a parameter in a traditional statistical course. That is, it is some characteristic of the underlying population probability measure. In this course, we reserve the word parameter for a different purpose, but it must also be noted that sometimes, in our context, a parameter is also a property.

Now, let us consider that we have a sequence of replicates of our random variable X . Namely, for $n \in \mathbb{N}$, we call the sequence

$$(X_i(\omega))_{i=1}^n = (X_1(\omega), X_2(\omega), \dots, X_n(\omega))$$

a data set, where each $X_i : \Omega \rightarrow \mathbb{X}$ ($i \in [n] = \{1, 2, \dots, n\}$) is endowed with the same image probability measure as X : P_X . We say that such a data set is identically distributed. As before, we shall drop the dependence on ω and simply write $(X_i)_{i=1}^n = (X_1, X_2, \dots, X_n)$.

Remark 2. Equivalently to $(X_i)_{i=1}^n$, we can also write $(X_i)_{i \in [n]}$. Furthermore, we need not only have sequences of finite length n , but also countably infinite sequences, such as

$$(X_i)_{i=1}^\infty = (X_i)_{i \in \mathbb{N}} = (X_1, X_2, \dots), \text{ and}$$

$$(X_i)_{i=-\infty}^\infty = (X_i)_{i \in \mathbb{Z}} = (\dots, X_{-2}, X_{-1}, X_0, X_1, X_2, \dots).$$

We shall endow these abstract objects with mathematical properties when we require them.

1.1.1.1 Three main statistical tasks

The first task of statistics is to use the sequence of $(X_i)_{i=1}^n$ to construct an estimator or a statistic for our property $\mathcal{P}(X)$. We shall usually write such an estimate as

$$\hat{\mathcal{P}}_n = \hat{\mathcal{P}}(X_1, \dots, X_n) = \hat{\mathcal{P}}\left((X_i)_{i \in [n]}\right).$$

As is suggested by the terminology, $\hat{\mathcal{P}}_n$ estimates $\mathcal{P}$ in some sense, and is required to be accurate and precise, in some sense.

Example 2. Consider again Example 1, and consider the sample mean

$$\hat{\mathcal{P}}(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$$

as an estimator of the mean $\mathcal{P}(X) = E(X)$. If the sample $(X_i)_{i \in [n]}$ is **independent** as well as identically distributed (we shall write IID), then $\hat{\mathcal{P}}_n$ is an accurate estimator of $\mathcal{P}$ in the sense that

$$E\left[\hat{\mathcal{P}}_n - \mathcal{P}\right] = 0,$$

and if X has a finite variance

$$\sigma_X^2 = E\left[\{X - E(X)\}^2\right],$$

then we can show that it becomes more precise as our sample size gets larger by the fact that its variance

$$\sigma_{\hat{\mathcal{P}}}^2 = E\left[\left\{\hat{\mathcal{P}}_n - \mathcal{P}\right\}^2\right] = \frac{\sigma_X^2}{n}$$

is decreasing as a function of n . We shall see in the sequel that this fact actually leads to another notion about the accuracy of $\hat{\mathcal{P}}_n$.

The second task is to use the data $(X_i)_{i \in [n]}$ to decide whether some logical statement regarding its property $\mathcal{P}(X)$ is true or not. Let any statement regarding $\mathcal{P}(X)$ be denoted by $S_j(\mathcal{P})$, for some subscript $j \in \mathbb{Z}$. We can assess the veracity of a statement $S_0(\mathcal{P})$ by conducting a hypothesis tests to decide between the null hypothesis:

$$H_0 : S_0(\mathcal{P}) \text{ is true,}$$

versus the alternative hypothesis:

$$H_1 : S_1(\mathcal{P}) \text{ is true,}$$

where both the statements $S_0(\mathcal{P})$ and $S_1(\mathcal{P})$ cannot be true, at the same time. In order to make a decision between H_0 or H_1 , we require a test statistic

$$\mathcal{T}_n = \mathcal{T}(X_1, \dots, X_n) \in \mathbb{T},$$

for some set $\mathbb{T}$, which has know probability measure $P_{\mathcal{T}_n}$ when $S_0(\mathcal{P})$ is true. Then, we can construct a test for H_0 versus H_1 by specifying a rejection region $\text{Rej}_{S_0} \subset \mathbb{T}$, whereupon event that $\{\mathcal{T}_n \in \text{Rej}_{S_0}\}$ occurs with low probability, under the $P_{\mathcal{T}_n}$. Thus, if $\mathcal{T}_n \in \text{Rej}_{S_0}$, we know that such an event is unlikely given that $S_0(\mathcal{P})$ is true, and we can conclude with some confidence that $S_0(\mathcal{P})$ is false.

Example 3. Again, we look back at Example 1. Consider, that we believe that the mean of X is zero. Then, to confirm or refute our belief, we would like to test the null hypothesis that

$$H_0 : E(X) = 0$$

versus the alternative that

$$H_1 : E(X) \neq 0.$$

Of course, we can verify that both the null and alternative hypotheses cannot be true at the same time! Suppose that $(X_i)_{i \in [n]}$ is an IID sequence and that $\sigma_X^2 = E \left[\{X - E(X)\}^2 \right]$ exists. Then, we can use Chebyshev's inequality to show that when $\mathcal{T}_n = \bar{X}_n$ the event $\text{Rej}_{\mathcal{P}}(\bar{X}_n) = \{|\bar{X}_n| \geq \epsilon\}$ occurs with probability no greater than

$$\frac{\sigma_X^2}{n\epsilon^2}.$$

Therefore, for any $\epsilon > 0$, we can make the upper bound of the probability smaller than any $\alpha \in (0, 1)$ by increasing n , and further, for any α and n , we can choose an ϵ such that the rejection region occurs with probability no greater α .

The third task of statistics is that of constructing confidence sets. That is, using the data $(X_i)_{i \in [n]}$, we wish to determine, with high probability, the region in which $\mathcal{P}(X)$ resides. That is, we wish to construct confidence sets of the form $\text{Conf}_{\mathcal{P},n} = \text{Conf}_{\mathcal{P}} \left((X_i)_{i \in [n]} \right)$, dependent on $(X_i)_{i \in [n]}$, such that

$$\mathcal{P}(X) \in \text{Conf}_{\mathcal{P},n},$$

with high probability.

Example 4. Another look at Example 1. Without the assumption that $E(X) = 0$, Chebyshev's inequality states that we can make the probability of the event $\{|\bar{X}_n - E(X)| \geq \epsilon\}$ is no greater than

$$\frac{\sigma_X^2}{n\epsilon^2}.$$

Thus, since probabilities of something happening and its logical complement happening must sum to 1, we have the fact that the event $\{|\bar{X}_n - E(X)| < \epsilon\}$ occurs with probability no less than

$$1 - \frac{\sigma_X^2}{n\epsilon^2}.$$

Therefore, for any $\alpha \in (0, 1)$, we can construct $\text{Conf}_{\mathcal{P}, n} = \{\mathcal{P} : |\bar{X}_n - \mathcal{P}| < \epsilon\}$, that occurs with probability at least $1 - \alpha$, by either making $\epsilon > 0$ or n sufficiently large.

1.1.2 What is statistical learning?

Like statistics, if you were to ask for the definition of statistical learning from one hundred machine learning practitioners, statisticians, and data scientists, you will end up with one hundred definitions. It is often seen as a catchall term and can cover as large a set as all possible investigations of data analytic and machine learning procedures via the lens of probability theory and statistics. This definition is preferred by many contemporary texts, such as the popular volumes: “The Elements of Statistical Learning” by Hastie, Tibshirani, and Friedman [2009] and “An Introduction to Statistical Learning: with Applications in R” by James, Witten, Hastie, and Tibshirani [2021]. On the other hand, the definition can be as small as that put forth by one of the originators of the discipline, Vladimir Vapnik, who defines all statistical learning problems as risk minimization. One can read of Vapnik’s approach in his excellent monographs: “Statistical Learning Theory” [1998] and “The Nature of Statistical Learning Theory” [2000]. In this course, we lean towards the definition

of Vapnik but we will stray into the broader context on occasion.

When we talk about the (empirical) risk in the sense of statistical learning or machine learning, what we are referring to is some data-based function of the form

$$r_n(\theta) = r\left(\theta; (X_i)_{i \in [n]}\right) = \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i)$$

where $\theta \in \mathbb{T}$ is a parameter that characterizes the loss function $\ell_\theta : \mathbb{X} \rightarrow \mathbb{R}$. With some mathematical assumptions, we have the fact that

$$r_n(\theta) \rightarrow \bar{r}(\theta)$$

in some sense, for all $\theta \in \mathbb{T}$. We simply call $\bar{r}(\theta)$ the risk, since it is now no longer empirical; i.e., it does not depend on the data $(X_i)_{i \in [n]}$.

As alluded to above, the primary problem of statistical learning is the exploration of risk minimization problems. To serve this end, we define the risk minimizer:

$$\theta_* \in \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta),$$

where we can consider θ_* (or more generally $\arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta)$), as a property of the underlying probability measure P_X . Since it is a property, we wish to estimate it, test hypotheses about it, and provide confidence sets that allow us to locate it with high probability! Of particular interest, we are concerned with the relationship between the risk minimizer θ_* and the empirical risk minimizer:

$$\hat{\theta}_n = \hat{\theta}\left((X_i)_{i \in [n]}\right) \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta).$$

Here, we can then consider $\hat{\theta}_n$ as an estimator of θ_* , and again we can consider $\arg \min_{\theta \in \mathbb{T}} r_n(\theta)$ as a set estimator of the property $\arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta)$.

Remark 3. For some function $r : \mathbb{T} \rightarrow \mathbb{R}$, we write

$$\arg \min_{\theta \in \mathbb{T}} r(\theta) = \left\{ \tau \in \mathbb{T} : r(\tau) = \min_{\theta \in \mathbb{T}} r(\theta) \right\}.$$

Similarly, we can also write

$$\arg \max_{\theta \in \mathbb{T}} r(\theta) = \left\{ \tau \in \mathbb{T} : r(\tau) = \max_{\theta \in \mathbb{T}} r(\theta) \right\}.$$

Further, with the notion of the infimum and supremum, i.e.,

$$\inf_{\theta \in \mathbb{T}} r(\theta) = \{b \in \mathbb{R} : \text{for all } \theta \in \mathbb{T}, b \leq r(\theta) \text{ and } b \geq c, \text{ for all } c \leq r(\theta)\}$$

and

$$\sup_{\theta \in \mathbb{T}} r(\theta) = \{b \in \mathbb{R} : \text{for all } \theta \in \mathbb{T}, b \geq r(\theta) \text{ and } b \leq c, \text{ for all } c \geq r(\theta)\},$$

we can define the respective sets $\arg \inf_{\theta \in \mathbb{T}} r(\theta)$ and $\arg \sup_{\theta \in \mathbb{T}} r(\theta)$.

Example 5. Let $\mathbb{X} = \mathbb{R}$, and let $\mathbb{T} = \mathbb{R}$. We can consider the loss function

$$\ell_{\theta}(x) = (\theta - x)^2 \tag{1.1.1}$$

and its associated risk and empirical risks:

$$\bar{r}(\theta) = \mathbb{E} \left[(\theta - X)^2 \right] \text{ and } r_n(\theta) = \frac{1}{n} \sum_{i=1}^n (\theta - X_i)^2.$$

Using calculus, we can show that

$$\begin{aligned} \hat{\theta}_n &= \arg \min_{\theta \in \mathbb{T}} r_n(\theta) \\ &= \arg \min_{\theta \in \mathbb{T}} \frac{1}{n} \sum_{i=1}^n (\theta - X_i)^2 \\ &= \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n. \end{aligned}$$

A bit more calculus shows that

$$\begin{aligned}\theta_* &= \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta) \\ &= \arg \min_{\theta \in \mathbb{T}} \mathbb{E} \left[(\theta - X)^2 \right] \\ &= \mathbb{E}(X).\end{aligned}$$

Thus, the risk minimization problem that is characterized by the loss function 1.1.1 is simply the problem of estimating the mean!

Example 6. Let $\mathbb{X} = \mathbb{R}$, and let $\mathbb{T} = \mathbb{R}$. We can consider instead the loss function

$$\ell_\theta(x) = |\theta - x|$$

and its associated risk and empirical risks

$$\bar{r}(\theta) = \mathbb{E} |\theta - X| \quad \text{and} \quad r_n(\theta) = \frac{1}{n} \sum_{i=1}^n |\theta - X_i|.$$

It can be shown that

$$\theta_* = \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta) = \arg \min_{\theta \in \mathbb{T}} \mathbb{E} |\theta - X| = \text{med}(X)$$

and

$$\arg \min_{\theta \in \mathbb{T}} r_n(\theta) = \left\{ \theta \in \mathbb{T} : \sum_{i=1}^n [X_i \leq \theta] = \sum_{i=1}^n [X_i \geq \theta] \right\},$$

is the set of sample median values!

Remark 4. For a logical statement S , we use the Iverson bracket notation:

$$[S] = \begin{cases} 1 & \text{if } S \text{ is true,} \\ 0 & \text{if } S \text{ is false.} \end{cases}$$

Example 7. As an extension to Example 5, we may consider data $\mathbb{X} = \mathbb{R}^2$ and $\mathbb{T} = \mathbb{R}^2$, where $X = (W, Y) \in \mathbb{X}$ and $\theta = (\beta_0, \beta_1) \in \mathbb{T}$. We propose the loss function

$$\ell_{\theta}(w, y) = (y - \beta_0 - \beta_1 w)^2$$

with corresponding risk and empirical risk:

$$\bar{r}(\theta) = \mathbb{E} \left[(Y - \beta_0 - \beta_1 W)^2 \right] \text{ and } r_n(\theta) = \frac{1}{n} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 W_i)^2.$$

This scenario corresponds to the simple linear regression problem, where the estimator

$$\hat{\theta}_n = \arg \min_{\theta \in \mathbb{T}} r_n(\theta)$$

is comprised of the elements:

$$\begin{aligned} \hat{\beta}_{1n} &= \frac{\left(\frac{1}{n} \sum_{i=1}^n W_i Y_i \right) - \left(\frac{1}{n} \sum_{i=1}^n W_i \right) \left(\frac{1}{n} \sum_{i=1}^n Y_i \right)}{\left(\frac{1}{n} \sum_{i=1}^n W_i^2 \right) - \left(\frac{1}{n} \sum_{i=1}^n W_i \right)^2}, \\ \hat{\beta}_{0n} &= \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) - \hat{\beta}_{1n} \left(\frac{1}{n} \sum_{i=1}^n W_i \right), \end{aligned}$$

which estimate the property of P_X :

$$\theta_* = \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta),$$

comprising of the elements:

$$\beta_1 = \frac{\mathbb{E}(WY) - \mathbb{E}(W) \mathbb{E}(Y)}{\mathbb{E}(W^2) - [\mathbb{E}(W)]^2}$$

and

$$\beta_0 = \mathbb{E}(Y) - \beta_1 \mathbb{E}(W).$$

Example 8. It is typical for us to hypothesize that the probability measure P_X corresponding to our data $(X_i)_{i \in [n]}$ by a probability density function (PDF) of $p_\theta : \mathbb{X} \rightarrow \mathbb{R}$, for some particular value $\theta_0 \in \mathbb{T}$. One way to estimate the value of θ_0 is to construct the maximum likelihood estimator

$$\hat{\theta}_n \in \arg \max_{\theta \in \mathbb{T}} \frac{1}{n} \sum_{i=1}^n \log p_\theta(X_i),$$

where the function $\log p_\theta(x)$, where $\log p_\theta(X)$ is usually called the log likelihood of X and correspondingly

$$\frac{1}{n} \sum_{i=1}^n \log p_\theta(X_i)$$

is usually called the sample log likelihood function. Of course, since the maximization of a function is simply the minimization of its negative, we can rephrase the maximum likelihood estimator as

$$\arg \min_{\theta \in \mathbb{T}} r_n(\theta) = \arg \min_{\theta \in \mathbb{T}} \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i),$$

where

$$\ell_\theta(x) = -\log p_\theta(x),$$

and in fact θ_0 can be described via log likelihood expression

$$\arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta) = \arg \min_{\theta \in \mathbb{T}} -\mathbb{E}[\log p_\theta(X)].$$

1.2 Some probability theory

We have already alluded to the notions of probability, random variables, probability spaces, events, probability measures, expectation, and independence, which are all

concepts that are rooted in the theory of probability and measures. We now provide a rigorous technical grounding of these concepts which are required to fully appreciate the theory of statistical learning.

1.2.1 Sets and such

We begin with the humble set, which is simply a mathematical method for listing collections of objects, or more technically, elements. We shall usually (although not always) denote a set by blackboard bold symbols: $\mathbb{A}, \mathbb{B}, \mathbb{C}$, etc. Elements of sets will typically be denoted by lower case letters a, b, c , etc. (again, not always). We will use the following set notation:

- The element operation $\in$: $x \in \mathbb{A}$ states that the element x is in set $\mathbb{A}$.
 - We may also write $x \notin \mathbb{A}$ to say that the element x is not in $\mathbb{A}$.
- The intersect (cap): $x \in \mathbb{A} \cap \mathbb{B}$ states that $x \in \mathbb{A}$ and $x \in \mathbb{B}$.
- The union (cup): $x \in \mathbb{A} \cup \mathbb{B}$ states that $x \in \mathbb{A}$ or $x \in \mathbb{B}$.
- Set exclusion: $x \in \mathbb{B} \setminus \mathbb{A}$ states that $x \in \mathbb{B}$ and $x \notin \mathbb{A}$.
- Subset: $\mathbb{A} \subset \mathbb{B}$ states that for all $x \in \mathbb{A}$, it is also true that $x \in \mathbb{B}$.
- Superset: $\mathbb{A} \supset \mathbb{B}$ states that for all $x \in \mathbb{B}$, it is also true that $x \in \mathbb{A}$.
- Set equality: $\mathbb{A} = \mathbb{B}$ states $\mathbb{A} \subset \mathbb{B}$ and $\mathbb{B} \subset \mathbb{A}$.
- The empty set: $\emptyset$ denotes a set with no elements.

- If $\mathbb{A} \neq \emptyset$, then we say that $\mathbb{A}$ is nonempty.

We write a set containing a finite number of elements $x_1, \dots, x_n$, for some $n \in \mathbb{N}$, as $\{x_1, \dots, x_n\}$. If we wish to define a set by the elements $x \in \mathbb{X}$ that satisfies some logical statement $\mathbf{S}(x)$, then we can use set builder notation and write:

$$\{x \in \mathbb{X} : \mathbf{S}(x) \text{ is true}\}.$$

Example 9. Consider the sets

$$\mathbb{A} = \{a, b, c\}, \mathbb{B} = \{d, e, f\}, \mathbb{C} = \{a, b, c, d, e, f\}.$$

We can write the following:

$$a \in \mathbb{A},$$

$$a \notin \mathbb{B},$$

$$\mathbb{A} \cap \mathbb{B} = \emptyset,$$

$$\mathbb{A} \cup \mathbb{B} = \mathbb{C},$$

$$\mathbb{A} \subset \mathbb{C},$$

$$\mathbb{C} \supset \mathbb{B},$$

$$\mathbb{C} \setminus \mathbb{A} = \mathbb{B}.$$

Remark 5. In some texts, the set exclusion operation $\mathbb{B} \setminus \mathbb{A}$ is written as $\mathbb{B} - \mathbb{A}$. Another potential difference between texts is the use of the subset and superset

symbols $\subset$ and $\supset$. Some authors write $\subseteq$ and $\supseteq$ for subset and superset, respectively. Here, they reserve the use of the symbols $\subset$ and $\supset$ for the proper subset and proper superset operations, where they write $\mathbb{A} \subset \mathbb{B}$ to state that for every $x \in \mathbb{A}$ it is true that $x \in \mathbb{B}$ and there exists a $y \in \mathbb{B}$ such that $y \notin \mathbb{A}$.

We will reserve some special symbols for important sets. We use the following notation:

$\mathbb{N}$	The natural number $\{1, 2, 3, \dots\}$.
$\mathbb{Z}$	The integers $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
$\mathbb{Z}_{\geq 0}$	The non-zero integers $\mathbb{N} \cup \{0\}$.
$\mathbb{R}$	The real numbers: $(-\infty, \infty)$.
$\mathbb{R}_{\geq 0}$	The non-negative real numbers: $[0, \infty)$.
$\mathbb{R}_{> 0}$	The positive real numbers: $(0, \infty)$.
$\bar{\mathbb{R}}$	The extended real numbers $[-\infty, \infty] = \mathbb{R} \cup \{-\infty, \infty\}$.
$\bar{\mathbb{R}}_{\geq 0}$	The non-negative extended real numbers $[0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$.
$\bar{\mathbb{R}}_{> 0}$	The positive extended real numbers $(0, \infty] = \mathbb{R}_{> 0} \cup \{\infty\}$.

For any two sets $\mathbb{A}$ and $\mathbb{B}$, we define the Cartesian product as

$$\mathbb{A} \times \mathbb{B} = \{(a, b) : a \in \mathbb{A} \text{ and } b \in \mathbb{B}\}$$

and more generally, for a sequence of sets $(\mathbb{A}_i)_{i \in [n]}$ ($n \in \mathbb{N}$), we write the Cartesian product as

$$\prod_{i=1}^n \mathbb{A}_i = \{(a_1, \dots, a_n) : a_1 \in \mathbb{A}_1, \dots, a_n \in \mathbb{A}_n\}.$$

For a set $\mathbb{A}$, we will write $2^{\mathbb{A}}$ to denote its power set: the set of all subsets of $\mathbb{A}$; i.e.,

$$2^{\mathbb{A}} = \{\mathbb{B} : \mathbb{B} \subset \mathbb{A}\}.$$

Furthermore, for any set $\mathbb{B} \subset 2^{\mathbb{A}}$, we say that the complement of $\mathbb{B}$, in $2^{\mathbb{A}}$, is $\mathbb{B}^c = \mathbb{A} \setminus \mathbb{B}$.

Example 10. Let $\mathbb{A} = \{a, b, c\}$, then

$$2^{\mathbb{A}} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \mathbb{A}\}.$$

Here, we can write $\mathbb{A}^c = \emptyset$ and $\{a\}^c = \{b, c\}$.

Definition 1. We say that $\mathcal{F} \subset 2^{\Omega}$ is an algebra over the nonempty set Ω , if:

- (i) $\mathcal{F}$ is nonempty;
- (ii) If $\mathbb{A} \in \mathcal{F}$, then $\mathbb{A}^c \in \mathcal{F}$;
- (iii) If $\mathbb{A}, \mathbb{B} \in \mathcal{F}$, then $\mathbb{A} \cup \mathbb{B} \in \mathcal{F}$.

Exercise 1. Show that Definition 1 also implies that if $\mathbb{A}, \mathbb{B} \in \mathcal{F}$, then $\mathbb{A} \cap \mathbb{B} \in \mathcal{F}$.

Furthermore, show that the definition also implies that if $(\mathbb{A}_i)_{i \in [n]} \subset \mathcal{F}$, for some $n \in \mathbb{N}$, then

$$\bigcup_{i=1}^n \mathbb{A}_i = \mathbb{A}_1 \cup \mathbb{A}_2 \cup \dots \cup \mathbb{A}_n \in \mathcal{F} \text{ and}$$

$$\bigcap_{i=1}^n \mathbb{A}_i = \mathbb{A}_1 \cap \mathbb{A}_2 \cap \cdots \cap \mathbb{A}_n \in \mathcal{F}.$$

Example 11. We say that $\mathcal{F} = \{\emptyset, \Omega\}$ is the trivial algebra, since it is the smallest algebra. For a more substantial example, consider the set $\Omega = \{a, b, c\}$. An algebra on Ω is $\{\emptyset, \{a\}, \{b, c\}, \Omega\}$.

Definition 2. We say that $\mathcal{F} \subset 2^\Omega$ is an σ -algebra over the nonempty set Ω , if:

- (i) $\mathcal{F}$ is nonempty;
- (ii) If $\mathbb{A} \in \mathcal{F}$, then $\mathbb{A}^c \in \mathcal{F}$;
- (iii) If $(\mathbb{A}_i)_{i \in \mathbb{N}} \subset \mathcal{F}$, then $\bigcup_{i=1}^{\infty} \mathbb{A}_i \in \mathcal{F}$.

Example 12. Every algebra on Ω is also a σ -algebra on Ω . As such, $\mathcal{F} = \{\emptyset, \Omega\}$ is also the trivial σ -algebra. For every $\mathbb{A} \in 2^\Omega$, we must also have $\mathbb{A}^c \in 2^\Omega$, thus $\mathcal{F} = 2^\Omega$ is also a σ -algebra, known as the discrete σ -algebra. Like algebras, the expression $\bigcup_{i=1}^{\infty} \mathbb{A}_i \in \mathcal{F}$ can be replaced by $\bigcap_{i=1}^{\infty} \mathbb{A}_i \in \mathcal{F}$ in Definition 2.

1.2.1.1 A topological detour

Definition 3. We say that a function $\text{dist} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}_{\geq 0}$ is a metric on the set $\mathbb{X}$ if, for every $x, y, z \in \mathbb{X}$:

- (i) $\text{dist}(x, y) = 0$ if and only if $x = y$; (identity)
- (ii) $\text{dist}(x, y) = \text{dist}(y, x)$; (symmetry)

(iii) $\text{dist}(x, z) \leq \text{dist}(x, y) + \text{dist}(y, z)$. (triangle inequality)

For a set $\mathbb{X}$ and a metric dist on $\mathbb{X}$, we say that $(\mathbb{X}, \text{dist})$ is a metric space, with open balls and closed balls, centered at $x \in \mathbb{X}$ with radius $\rho > 0$, defined by

$$\mathbb{B}_\rho(x) = \{y \in \mathbb{X} : \text{dist}(x, y) < \rho\} \text{ and}$$

$$\bar{\mathbb{B}}_\rho(x) = \{y \in \mathbb{X} : \text{dist}(x, y) \leq \rho\},$$

respectively. We further say that a set $\mathbb{A} \subset \mathbb{X}$ is open if and only if, for each $a \in \mathbb{A}$, there exists a $\rho > 0$, such that $\mathbb{B}_\rho(a) \subset \mathbb{A}$. We can then say that $\mathbb{B} \subset \mathbb{X}$ is closed if its complement $\mathbb{B}^c = \mathbb{X} \setminus \mathbb{B}$ is open.

Remark 6. In many (most) situations, the words metric and distance can be interchanged. However, some authors reserve different meanings for the two words, so one should double check within the text of interest for specifications of the terminologies.

Exercise 2. Prove that $\mathbb{B}_\rho(x)$ is open and $\bar{\mathbb{B}}_\rho(x)$ is closed, for any $\rho > 0$ and $x \in \mathbb{X}$.

Example 13. Let $\mathbb{X} = \mathbb{R}$ be equipped with the metric $\text{dist}(x, y) = |x - y|$. Then, $(\mathbb{X}, \text{dist})$ has open and closed balls, centered at x with radius $\rho > 0$, of the forms

$$\mathbb{B}_\rho(x) = \{y \in \mathbb{X} : |x - y| < \rho\} = \{y : x - \rho < y < x + \rho\} \text{ and}$$

$$\bar{\mathbb{B}}_\rho(x) = \{y \in \mathbb{X} : |x - y| \leq \rho\} = \{y : x - \rho \leq y \leq x + \rho\}.$$

Recall that a set $\mathbb{A}$ is said to be finite if it has fewer elements than $\mathbb{N}$, countable if it has the same number of elements as $\mathbb{N}$, and uncountable if it has more elements than $\mathbb{N}$; e.g., the set $\mathbb{R}$ is uncountable.

Proposition 1. *Let $\{\mathbb{A}_i \subset \mathbb{X} : i \in \mathbb{I}\}$ be a family of subsets of $\mathbb{X}$, indexed by a potentially uncountable set $\mathbb{I}$.*

- *If each $\mathbb{A}_i$ is open, then $\bigcup_{i \in \mathbb{I}} \mathbb{A}_i$ is also open.*
- *If each $\mathbb{A}_i$ is closed, then $\bigcap_{i \in \mathbb{I}} \mathbb{A}_i$ is also closed.*

Remark 7. For a potentially uncountable set $\mathbb{I}$, and indexed family of sets $\{\mathbb{A}_i : i \in \mathbb{I}\}$, De Morgan's law states that

$$\left(\bigcap_{i \in \mathbb{I}} \mathbb{A}_i \right)^c = \bigcup_{i \in \mathbb{I}} \mathbb{A}_i^c \text{ and}$$

$$\left(\bigcup_{i \in \mathbb{I}} \mathbb{A}_i \right)^c = \bigcap_{i \in \mathbb{I}} \mathbb{A}_i^c.$$

Exercise 3. For $(\mathbb{A}_i)_{i \in [n]} \subset \mathbb{X}$, show that

- If each $\mathbb{A}_i$ is open, then $\bigcap_{i=1}^n \mathbb{A}_i$ is open.
- If each $\mathbb{A}_i$ is closed, then $\bigcup_{i=1}^n \mathbb{A}_i$ is closed.

For any $\mathcal{C} \subset 2^\Omega$, not necessarily a σ -algebra, we write $\sigma(\mathcal{C})$ to denote the minimum σ -algebra containing $\mathcal{C}$, and define it as

$$\sigma(\mathcal{C}) = \bigcap_{i \in \mathbb{I}} \mathcal{F}_i,$$

where $\{\mathcal{F}_i : i \in \mathbb{I}\}$ denotes the set of all σ -algebras such that $\mathcal{C} \subset \mathcal{F}_i$. For a metric space (Ω, dist) , let $\mathcal{O} \subset 2^\Omega$ be defined as

$$\mathcal{O} = \{\mathbb{O} \subset \Omega : \mathbb{O} \text{ is an open set in } \Omega\}.$$

Then, we say that $\mathcal{B}(\Omega) = \sigma(\mathcal{O})$ is the Borel σ -algebra Ω .

Example 14. Let $\Omega = \mathbb{R}$ and $\text{dist}(x, y) = |x - y|$. Then, the set of open sets $\mathcal{O}$ can be defined as

$$\mathcal{O} = \left\{ \bigcup_{i \in \mathbb{N}} (a_i, b_i) : a_i < b_i, a_i, b_i \in \mathbb{R} \right\},$$

and thus the Borel σ -algebra of Ω is

$$\begin{aligned} \mathcal{B}(\Omega) &= \sigma(\mathcal{O}) \\ &= \sigma(\{(a, b) : a < b, a, b \in \mathbb{R}\}). \end{aligned}$$

In fact, $\mathcal{B}(\Omega)$ must also contain all intervals of the form (a, b) , $[a, b]$, $[a, b)$, and $(a, b]$, for all $a < b$.

Exercise 4. Based on Example 14, show that $\mathcal{B}(\Omega)$ must additionally contain all intervals of the form (a, ∞) , $[a, \infty)$, $(-\infty, a)$, and $(-\infty, a]$, for $a \in \mathbb{R}$.

1.2.1.2 Product of σ -algebras

Let Ω_1 and Ω_2 be two sets, equipped with their respective σ -algebras $\mathcal{F}_1$ and $\mathcal{F}_2$. Then, we intuitively can combine the two sets Ω_1 and Ω_2 via Cartesian product, i.e., we can write:

$$\Omega_1 \times \Omega_2 = \{\omega = (\omega_1, \omega_2) : \omega_1 \in \Omega_1 \text{ and } \omega_2 \in \Omega_2\}.$$

Since $\mathcal{F}_1$ and $\mathcal{F}_2$ are also sets, it is tempting to say that $\mathcal{F}_1 \times \mathcal{F}_2$ is a σ -algebra of $\Omega_1 \times \Omega_2$. However, this is not the case as we can see $\mathcal{F}_1 \times \mathcal{F}_2 \subset 2^{\Omega_1} \times 2^{\Omega_2}$, whereas

what we actually want is to define a system of sets of from $2^{\Omega_1 \times \Omega_2}$. A sufficiently rich σ -algebra for $\Omega_1 \times \Omega_2$ is the set

$$\mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma \{ \mathbb{A}_1 \times \mathbb{A}_2 : \mathbb{A}_1 \in \mathcal{F}_1 \text{ and } \mathbb{A}_2 \in \mathcal{F}_2 \},$$

containing rectangles of subsets, rather than just pairs of subsets, as in $\mathcal{F}_1 \times \mathcal{F}_2$. In the same way, if $(\Omega_i)_{i \in [n]}$ and $(\mathcal{F}_i)_{i \in [n]}$ are sequences of sets with their σ -algebras, then define a σ -algebra for the product $\prod_{i=1}^n \Omega_i$ via the minimum σ -algebra

$$\bigotimes_{i=1}^n \mathcal{F}_i = \sigma \left(\left\{ \prod_{i=1}^n \mathbb{A}_i : \mathbb{A}_i \in \mathcal{F}_i \text{ for each } i \in [n] \right\} \right).$$

Example 15. The set $\mathbb{R}^d = \prod_{i=1}^d \mathbb{R}$ of d -dimensional vectors $x = (x_1, \dots, x_d)$ can be equipped with the distance

$$\text{dist}(x, y) = \|x - y\|_1 = \sum_{i=1}^d |x_i - y_i|,$$

for each $x \in \mathbb{R}^d$ and $y = (y_1, \dots, y_d) \in \mathbb{R}^d$. We have

$$\begin{aligned} \mathcal{B}(\mathbb{R}^d) &= \sigma \left(\left\{ \prod_{i=1}^d (a_i, b_i) : a_i < b_i, \text{ for each } i \in [d] \right\} \right) \\ &= \bigotimes_{i=1}^d \sigma(\{(a_i, b_i) : a_i < b_i\}), \end{aligned}$$

which is the smallest set that contain all open n -dimensional cubes. Of course, we may more directly define $\mathcal{B}(\mathbb{R}^d)$ as

$$\sigma(\{\mathbb{O} \in \mathbb{R}^d : \mathbb{O} \text{ is an open set of } \mathbb{R}^d\}),$$

where

$$\{(a_i, b_i) : a_i < b_i\} \subset \{\mathbb{O} \in \mathbb{R}^d : \mathbb{O} \text{ is an open set of } \mathbb{R}^d\}.$$

1.2.2 Measure theory

Let Ω be a set equipped with its σ -algebra $\mathcal{F}$. Together, the pair $(\Omega, \mathcal{F})$ is often called a measurable space. Here, we say that a set $\mathbb{A} \subset \Omega$ is measurable if $\mathbb{A} \in \mathcal{F}$. We can now provide the measurable sets in $\mathcal{F}$ with a measurement or a valuation by defining a measure $M : \mathcal{F} \rightarrow \bar{\mathbb{R}}_{\geq 0}$.

Definition 4. M is a measure if:

- (i) For every $\mathbb{A} \in \mathcal{F}$, $M(\mathbb{A}) \geq 0$; (non-negativity)
- (ii) For every sequence of pairwise disjoint sets $(\mathbb{A}_i)_{i \in \mathbb{N}}$,

$$M\left(\bigcup_{i=1}^{\infty} \mathbb{A}_i\right) = \sum_{i=1}^{\infty} M(\mathbb{A}_i); \text{ } (\sigma\text{-additivity})$$

- (iii) $M(\emptyset) = 0$. (nullity)

Remark 8. We say that sets $\mathbb{A}$ and $\mathbb{B}$ are disjoint if $\mathbb{A} \cap \mathbb{B} = \emptyset$. Similarly, we say that a sequence of sets $(\mathbb{A}_i)_{i \in \mathbb{N}}$ is pairwise disjoint if $\mathbb{A}_i \cap \mathbb{A}_j = \emptyset$ for all $i, j \in \mathbb{N}$, such that $i \neq j$.

Together, the tuple of the set Ω , its σ -algebra $\mathcal{F}$, and the measure M form a measure space $(\Omega, \mathcal{F}, M)$. When $M(\Omega) < \infty$, the measure is said to be finite, and when $M(\Omega) = 1$, we call M a probability measure, and typically write $M = P$. We then say that the tuple $(\Omega, \mathcal{F}, P)$ is a probability space, and we say that Ω is the set of events.

We further say that the measure M is σ -finite if the underlying set Ω is the countable union of sets from $\mathcal{F}$ of finite measure. That is, we can write

$$\Omega = \bigcup_{i \in \mathbb{N}} A_i,$$

where $A_i \in \mathcal{F}$ and $M(A_i) < \infty$, for each $i \in \mathbb{N}$.

Exercise 5. Let $(\Omega, \mathcal{F}, M)$ be a measure space and for each $A, B \in \mathcal{F}$, such that $A \subset B$, show that $M(A) \leq M(B)$. Then argue that if M is finite, then $M(A) < \infty$, for all $A \in \mathcal{F}$. Finally, for $A, B \in \mathcal{F}$ with $M(A \cap B) < \infty$, show that

$$M(A \cup B) = M(A) + M(B) - M(A \cap B).$$

Exercise 6. An infinite sequence of sets $(A_i)_{i \in \mathbb{N}}$ is said to be *increasing* if

$$A_1 \subset A_2 \subset A_3 \subset \dots$$

and an infinite sequence of sets $(B_i)_{i \in \mathbb{N}}$ is said to be *decreasing* if

$$B_1 \supset B_2 \supset B_3 \supset \dots$$

We say that a set function M is continuous from below if for each increasing sequence $(A_i)_{i \in \mathbb{N}}$

$$M(A_i) \uparrow M\left(\lim_{i \rightarrow \infty} A_i\right),$$

and continuous from above if for each decreasing sequence $(B_i)_{i \in \mathbb{N}}$,

$$M(B_i) \downarrow M\left(\lim_{i \rightarrow \infty} B_i\right),$$

where the up and down arrows are used to remind us that the sequences are increasing or decreasing and $\lim_{i \rightarrow \infty} \mathbb{A}_i$ and $\lim_{i \rightarrow \infty} \mathbb{B}_i$ can be replaced by $\bigcup_{i \in \mathbb{N}} \mathbb{A}_i$ and $\bigcap_{i \in \mathbb{N}} \mathbb{B}_i$, respectively. If a set function M is both continuous from below and continuous from above then we simply say that it is continuous. By Definition 4, show that every measure is continuous.

Example 16. A very important measure when dealing with countable sets Ω is the counting measure. Let $\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}$ and equip Ω with the discrete σ -algebra $\mathcal{F} = 2^\Omega$, then for each $\mathbb{A} \in \Omega$,

$$M(\mathbb{A}) = |\mathbb{A}|$$

defines the counting measure, where we write $|\mathbb{A}|$ to denote the cardinality of $\mathbb{A}$: the number of elements in the set $\mathbb{A}$. As a more concrete example, suppose that M is the counting measure on the set $\Omega = \mathbb{N}$. Then, we may write statements such as

$$M(\{1, 2, 3\}) = 3$$

or

$$M(\{\omega : 0 < \omega < 11\}) = 10.$$

Example 17. Let us consider $\Omega = \mathbb{R}^d$ equipped with its Borel σ -algebra $\mathcal{B}(\mathbb{R}^d)$. Then, we can define Lebesgue measure as the unique measure on $\mathcal{B}(\mathbb{R}^d)$ with the property that

$$M\left(\left\{\prod_{i=1}^d (a_i, b_i) : a_i < b_i, \text{ for each } i \in [d]\right\}\right) = \prod_{i=1}^d (b_i - a_i).$$

Concretely, for each interval $(a, b) \subset \mathbb{R}$, the Lebesgue measure for $\mathbb{R}$ measures the length of the interval:

$$M((a, b)) = b - a.$$

Similarly, for each rectangle $(a_1, b_1) \times (a_2, b_2) \subset \mathbb{R}^2$, the Lebesgue measure for $\mathbb{R}^2$ endows the rectangle with an area:

$$M((a_1, b_1) \times (a_2, b_2)) = (b_1 - a_1)(b_2 - a_2),$$

and for each rectangular prism $(a_1, b_1) \times (a_2, b_2) \times (a_3, b_3) \subset \mathbb{R}^3$, the Lebesgue measure for $\mathbb{R}^3$ provides it with a volume:

$$M((a_1, b_1) \times (a_2, b_2) \times (a_3, b_3)) = (b_1 - a_1)(b_2 - a_2)(b_3 - a_3).$$

Thus, in any $\mathbb{R}^d$ for $d > 3$, the Lebesgue measure extends the notion of lengths, areas, and volumes to that dimension. So important is the Lebesgue measure in this course that we will often refer to it by name by using the notation Leb .

Example 18. Lebesgue measure on Leb on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is σ -finite, since

$$\mathbb{R} = \bigcup_{i \in \mathbb{N}} (-i, i),$$

where each $(-i, i)$ has finite measure $\text{Leb}(\omega \in (-i, i)) = 2i < \infty$. Similarly the counting measure on $(\mathbb{N}, 2^{\mathbb{N}})$ is also σ -finite, since

$$\mathbb{N} = \bigcup_{i \in \mathbb{N}} \{i\},$$

and the counting measure of each $\{i\}$ is $M(\{i\}) = 1$.

Finally suppose that we try to apply the counting measure on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We of course cannot cover $\mathbb{R}$ via a countable union of singletons, since there is an uncountable number of singletons in $\mathbb{R}$. However, if we try to cover $\mathbb{R}$ via a countable number of intervals, then the counting measure of each interval will be infinite, and thus we would again violate the conditions of σ -finiteness. Thus, the counting measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is not σ -finite.

1.2.2.1 Distribution functions

We shall specialize our attention towards probability measures for a moment, and consider a function measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P)$ or $(\mathbb{Z}, 2^{\mathbb{Z}}, P)$. We shall say that a function

$$t \in \mathbb{R} \mapsto F(t) = P(\{\omega : \omega \leq t\}) \quad (1.2.1)$$

is the cumulative distribution function (or CDF) of ω , or of its probability measure P .

Exercise 7. Using the conclusion of Exercise 6, prove that

$$\lim_{t \rightarrow -\infty} F(t) = 0$$

and

$$\lim_{t \rightarrow \infty} F(t) = 1.$$

Remark 9. For sets such as $\{\omega : S(\omega) \text{ is true}\}$, we should properly write its measure

as

$$M(\{\omega : S(\omega) \text{ is true}\}).$$

However, when there is no threat of misunderstanding, we can simplify our notation by writing

$$M(\omega : S(\omega) \text{ is true}), \text{ or } M(S(\omega) \text{ is true}),$$

or even $M(S(\omega))$. For example, our definition for the CDF (1.2.1) can be simplified to

$$F(t) = P(\omega \leq t).$$

For the space $(\Omega, \mathcal{F}, P)$, where Ω is either $\mathbb{R}^d$ or $\mathbb{Z}^d$ and $\mathcal{F}$ is an appropriate σ -algebra, we can also define the so-called multivariate CDF as

$$(t_1, \dots, t_d) \mapsto F(t_1, \dots, t_d) = P(\{\omega : \omega_1 \leq t_1, \dots, \omega_d \leq t_d\}),$$

where $\omega = (\omega_1, \dots, \omega_d)$ is in Ω .

1.2.2.2 Measurable functions and random variables

Recall that for a measure space $(\Omega, \mathcal{F}, M)$, we say that a set $\mathbb{A}$ is measurable if $\mathbb{A} \in \mathcal{F}$. Now, let $X : \Omega \rightarrow \mathbb{X}$ be a function, where $(\mathbb{X}, \mathcal{X})$ is a measurable space. We will write the inverse image of X as $X^{-1} : 2^{\mathbb{X}} \rightarrow 2^{\Omega}$, where

$$X^{-1}(\mathbb{A}) = \{\omega \in \Omega : X(\omega) \in \mathbb{A}\}$$

for each $\mathbb{A} \in \mathcal{X}$.

Definition 5. A function X is said to be measurable (with respect to $\mathcal{F}$ and $\mathcal{X}$) if for all $\mathbb{A} \in \mathcal{X}$,

$$X^{-1}(\mathbb{A}) \in \mathcal{F}.$$

In the case where X is a measurable function, we can define its image measure of X as M_X , where for every set $\mathbb{A} \in \mathcal{X}$, we may define

$$M_X(\mathbb{A}) = M(X^{-1}(\mathbb{A})) = M(\omega \in \Omega : X(\omega) \in \mathbb{A}).$$

Thus, M_X provides elements of $\mathcal{X}$ with a measurement using the measurements endowed on elements of $\mathcal{F}$. Together, $(\mathbb{X}, \mathcal{X}, M_X)$ forms a measure space.

Remark 10. For some function $X : \Omega \rightarrow \mathbb{X}$, suppose that we have some subset of $2^{\mathbb{X}}$ which we want to make measurable. In such a situation, for such a subset $\mathcal{C} \subset 2^{\mathbb{X}}$, we may generate a σ -algebra $\mathcal{X} = \sigma(\mathcal{C})$, and provide Ω with a σ -algebra after the fact by endowing it with

$$\begin{aligned} \mathcal{F} &= \sigma(X^{-1}(\mathcal{C})) \\ &= X^{-1}(\mathcal{X}) \\ &= \{\mathbb{B} \subset \Omega : \mathbb{B} = X^{-1}(\mathbb{A}) \text{ for some } \mathbb{A} \in \mathcal{X}\}. \end{aligned}$$

Remark 11. When $M = P$ is a probability measure we typically call the function X a (measurable) random function. Furthermore, we replace the image measure M_X by the notation P_X and call it the (image) probability measure of X . We additionally note that when the set $\mathbb{X} = \mathbb{R}$ and we take $\mathcal{X} = \mathcal{B}(\mathbb{R})$, we say that X is a (measurable) random variable.

Exercise 8. Let $(\Omega, \mathcal{B}(\Omega), \mathbb{M})$ be a measure space where (Ω, dist_1) is a metric space, and let $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$ be a measurable space, where $(\mathbb{X}, \text{dist}_2)$ is also a metric space. Using the fact that a continuous function from $\omega \mapsto X(\omega)$ maps open sets to open sets, prove that if $X : \Omega \rightarrow \mathbb{X}$ is continuous then it is also measurable.

Exercise 9. Let $(\Omega, \mathcal{F})$, $(\mathbb{X}, \mathcal{X})$, and $(\mathbb{Y}, \mathcal{Y})$ each be measurable spaces and let $X : \Omega \rightarrow \mathbb{X}$ and $Y : \mathbb{X} \rightarrow \mathbb{Y}$ both be measurable. Then, show that $Y \circ X : \Omega \rightarrow \mathbb{Y}$ is measurable.

We now specialize our attention to the case where $\mathbb{X} = \mathbb{R}$ and $\mathcal{X} = \mathcal{B}(\mathbb{R})$. Let us consider the characteristic function that evaluates the veracity of statements of the form $S(x) = \{x \in \mathbb{A}\}$ for each $\mathbb{A} \subset \Omega$:

$$\chi_{\mathbb{A}}(x) = \begin{cases} 1 & \text{if } x \in \mathbb{A}, \\ 0 & \text{if } x \notin \mathbb{A}. \end{cases}$$

Exercise 10. Show that for each $\mathbb{A} \in \mathcal{F}$, the characteristic function $\chi_{\mathbb{A}} : \Omega \rightarrow \mathbb{R}$ is measurable with respect to $\mathcal{F}$ and $\mathcal{B}(\mathbb{R})$.

Proposition 2. *Let $(\Omega, \mathcal{F})$ and $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable spaces. Then, a function $X : \Omega \rightarrow \mathbb{R}$ is measurable with respect to $\mathcal{F}$ and $\mathcal{B}(\mathbb{R})$ if any of the following statements are true: for every $x \in \mathbb{R}$,*

- $X^{-1}((x, \infty)) = \{\omega \in \Omega : X(\omega) > x\}$ is measurable.
- $X^{-1}((-\infty, x)) = \{\omega \in \Omega : X(\omega) < x\}$ is measurable.

- $X^{-1}([x, \infty)) = \{\omega \in \Omega : X(\omega) \geq x\}$ is measurable.
- $X^{-1}((-\infty, x]) = \{\omega \in \Omega : X(\omega) \leq x\}$ is measurable.

Once we have established measurability for very simple functions, we can construct more complex measurable functions from the building blocks via the very generous closure properties of measurable functions. To be concrete, if $X, Y : \Omega \rightarrow \mathbb{R}$ are measurable functions with respect to $\mathcal{F}$ and $\mathcal{B}(\mathbb{R})$, and if $a \in \mathbb{R}$, then the following function operations are measurable:

- $X + Y$ (sums)
- $X \times Y$ (products)
- $a \times X$ (scalar multiplication)
- $|X|$ (absolute values)
- $\min\{X, Y\}$ (pairwise minimum)
- $\max\{X, Y\}$ (pairwise maximum)

Furthermore, if in addition $(X_i)_{i \in \mathbb{N}}$ are a sequence of measurable functions with respect to $\mathcal{F}$ and $\mathcal{B}(\mathbb{R})$, then the following operations are also measurable:

- $\lim_{i \rightarrow \infty} X_i$ (limits)
- $\sup_{i \in \mathbb{N}} X_i$ (suprema)
- $\inf_{i \in \mathbb{N}} X_i$ (infima)

- $\limsup_{i \rightarrow \infty} X_i = \lim_{i \rightarrow \infty} \sup_{j \geq i} X_j$ (limit superior)
- $\liminf_{i \rightarrow \infty} X_i = \lim_{i \rightarrow \infty} \inf_{j \geq i} X_j$ (limit inferior)

1.2.3 Integration and expectation

For a σ -algebra $\mathcal{F}$ on Ω , we say that a set of disjoint subsets $\mathcal{P}$ of Ω , is a partition of Ω , if

$$\mathcal{P} = \{\mathbb{A}_1, \dots, \mathbb{A}_m\}$$

for some $m \in \mathbb{N}$, such that $\mathbb{A}_i \in \mathcal{F}$ for each $i \in [m]$, $\mathbb{A}_i \cap \mathbb{A}_j = \emptyset$ for each $i \neq j$, and

$$\bigcup_{i=1}^m \mathbb{A}_i = \Omega.$$

Let $(\Omega, \mathcal{F}, M)$ be a measure space, and let $X : \Omega \rightarrow \bar{\mathbb{R}}_{\geq 0}$ be a non-negative function. Similar to the Riemann sum for functions on real domains, for each partition $\mathcal{P}$ of Ω , we can define the so called lower Lebesgue sum for measurable functions on arbitrary spaces by

$$L(X, \mathcal{P}) = \sum_{i=1}^m M(\mathbb{A}_i) \left[\inf_{\omega \in \mathbb{A}_i} X(\omega) \right].$$

Using this notion, we can now define the integral of a non-negative function $X : \Omega \rightarrow \bar{\mathbb{R}}_{\geq 0}$, with respect to the set Ω and its σ -algebra $\mathcal{F}$, by:

$$\int X dM = \sup \{L(X, \mathcal{P}) : \mathcal{P} \subset \mathcal{F} \text{ is a partition of } \Omega\}. \quad (1.2.2)$$

Remark 12. We notice that since the integral requires the evaluation of $L(X, \mathcal{P})$ for arbitrary partitions $\mathcal{P} \subset \mathcal{F}$, and it appears to require only the measurability of the

sets of $\mathcal{P}$. However, (1.2.2) is only one of many equivalent definitions of the integral that is only made permissible via the assumption of measurability on X .

Example 19. Consider the set $\Omega = \{1, 2, 3\}$ equipped with the discrete σ -algebra $\mathcal{F} = 2^\Omega$ and the measure $M, A \mapsto |A|$. Then, the only partitions of Ω are

$$\mathcal{P}_1 = \{\{1, 2, 3\}\}, \mathcal{P}_2 = \{\{1, 2\}, \{3\}\}, \mathcal{P}_3 = \{\{1\}, \{2, 3\}\},$$

$$\mathcal{P}_4 = \{\{1, 3\}, \{2\}\}, \mathcal{P}_5 = \{\{1\}, \{2\}, \{3\}\}.$$

Let $X(\omega) = \omega$, for each $\omega \in \Omega$, and compute the lower Lebesgue sums:

$$L(X, \mathcal{P}_1) = 3 \times 1 = 3, L(X, \mathcal{P}_2) = 2 \times 1 + 3 = 5, L(X, \mathcal{P}_3) = 1 + 2 \times 2 = 5,$$

$$L(X, \mathcal{P}_4) = 2 \times 1 + 2 = 4, L(X, \mathcal{P}_5) = 1 + 2 + 3 = 6.$$

Thus, the largest of the sums:

$$\int X dM = L(X, \mathcal{P}_5) = 6.$$

Remark 13. There are many ways you will see the integral expression being written. For example, when one wants to be reminded that the $X(\omega)$ is a function of elements ω , we can write

$$\int X(\omega) dM(\omega) \text{ or } \int X(\omega) M(d\omega).$$

Furthermore, to be reminded that $\omega \in \Omega$, we can also write

$$\int_{\Omega} X(\omega) dM(\omega).$$

In the sequel, we will see that there are in fact several other ways of writing the integral as well, when M is endowed with some additional properties.

Exercise 11. Given the measure space $(\Omega, \mathcal{F}, M)$, prove that the characteristic function $\chi_{\mathbb{A}} : \Omega \rightarrow \bar{\mathbb{R}}_{\geq 0}$ has integral

$$\int \chi_{\mathbb{A}} dM = M(\mathbb{A}),$$

for each $\mathbb{A} \in \mathcal{F}$.

Given a function $X : \Omega \rightarrow \bar{\mathbb{R}}$, we can define the its positive part and the absolute value of its negative part via the functions

$$X^+(\omega) = \begin{cases} X(\omega) & \text{if } X(\omega) \geq 0, \\ 0 & \text{if } X(\omega) < 0, \end{cases}$$

and

$$X^-(\omega) = \begin{cases} 0 & \text{if } X(\omega) \geq 0, \\ -X(\omega) & \text{if } X(\omega) < 0. \end{cases}$$

Notice that both the positive and negative parts are non-negative functions and thus we can use the definition from (1.2.2) to obtain a new definition for the integral of an extended real valued function on $(\Omega, \mathcal{F}, M)$:

$$\int X dM = \int X^+ dM - \int X^- dM.$$

We observe, again, that the integral is defined only when both X^+ and X^- are measurable.

Remark 14. It is notable that when either $\int X^+ dM$ or $\int X^- dM$ goes to infinity, the integral can take on values of ∞ or $-\infty$, respectively. However, when both terms

go to infinity, the integral becomes undefinable. Furthermore, we observe that the integral is finite if and only if both $\int X^+ dM$ and $\int X^- dM$ are finite. In such a case, we say that the function X is *Lebesgue integrable* (or often *summable*).

The integral, as we have defined it, admits the usual compositional identities that one may expect. Namely, for two measurable functions X and Y on the space $(\Omega, \mathcal{F}, M)$, and for constant $a \in \mathbb{R}$, we have the following facts:

- $\int aX dM = a \int X dM$, (homogeneity)
- If $X(\omega) \leq Y(\omega)$, for each $\omega \in \Omega$, then $\int X dM \leq \int Y dM$, (order preservation)
- If $\int |X| dM < \infty$ and $\int |Y| dM < \infty$, then $\int (X + Y) dM = \int X dM + \int Y dM$,
(additive closure)

Exercise 12. For a measurable function X on the space $(\Omega, \mathcal{F}, M)$, prove that

$$\left| \int X dM \right| \leq \int |X| dM.$$

Furthermore, show that the absolute integrability condition $\int |X| dM < \infty$ is equivalent to the definition of Lebesgue integrability.

Remark 15. Like the probabilistic language applied to probability spaces $(\Omega, \mathcal{F}, P)$ and their random variables, X , we also have the fact that the integral of the random variable X , with respect to the measure P is called the expectation of X , denoted by

$$E(X) = \int X dP = \int_{\Omega} X(\omega) dP(\omega).$$

Since we can write $X : \Omega \rightarrow \mathbb{R}$, we can endow X with its image probability measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$, and further write

$$E(X) = \int_{\mathbb{R}} x \, dP_X(x).$$

Exercise 13. Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $X : \Omega \rightarrow \mathbb{R}$ be a (measurable) random variable such that $E|X| < \infty$. Prove the famous Markov's inequality:

$$P(\{\omega \in \Omega : |X(\omega)| \geq \epsilon\}) \leq \frac{E|X|}{\epsilon},$$

for any $\epsilon > 0$. Then, use this fact to derive Chebyshev's inequality:

$$P(\{\omega \in \Omega : |X(\omega) - E(X)| \geq \epsilon\}) \leq \frac{\sigma_X^2}{\epsilon^2},$$

for any $\epsilon > 0$, where

$$\sigma_X^2 = E[(X - E(X))^2] < \infty.$$

Definition 6. Suppose that for each $i \in [n]$, the random variable X_i is measurable on $(\Omega, \mathcal{F}, P)$. Then, we say that the sequence of random variables $(X_i)_{i \in [n]}$ is independent if for each sequence of sets $(\mathbb{A}_i)_{i \in [n]} \subset \mathcal{F}$, we have

$$P(\omega : X_1(\omega) \in \mathbb{A}_1, \dots, X_n(\omega) \in \mathbb{A}_n) = P(\omega : X_1(\omega) \in \mathbb{A}_1) \dots P(\omega : X_n(\omega) \in \mathbb{A}_n).$$

Remark 16. Using the definition of the multivariate CDF for the sequence $(X_i)_{i \in [n]}$:

$$F(x_1, \dots, x_n) = P(\{\omega : X_1(\omega) \leq x_1, \dots, X_n(\omega) \leq x_n\})$$

we can also prove, albeit tediously, that $(X_i)_{i \in [n]}$ is independent if and only if the familiar condition

$$F(x_1, \dots, x_n) = P(\omega : X_1(\omega) \leq x_1) \dots P(\omega : X_n(\omega) \leq x_n)$$

is satisfied. Another equivalent definition is that

$$E(X_1 X_2 \dots X_n) = E(X_1) E(X_2) \dots E(X_n).$$

Definition 7. For two measures M and N on the measurable space $(\Omega, \mathcal{F})$, we say that N is absolutely continuous with respect to M , which we write as $N \ll M$, if

$$N(\mathbb{A}) = 0$$

whenever $\mathbb{A} \in \mathcal{F}$ and $M(\mathbb{A}) = 0$.

Theorem 1 (Radon–Nikodym). *Let M and N be measures on a measurable space $(\Omega, \mathcal{F})$, such that $N \ll M$. Then, there exists a function $p : \Omega \rightarrow \mathbb{R}_{\geq 0}$ such that*

$$M(\mathbb{A}) = \int \chi_{\mathbb{A}}(\omega) p(\omega) dM(\omega). \quad (1.2.3)$$

Remark 17. Notice here that we write can also write the integral

$$\int \chi_{\mathbb{A}}(\omega) p(\omega) dM(\omega)$$

as

$$\int [\omega \in \mathbb{A}] p(\omega) dM(\omega) = \int_{\mathbb{A}} p dM.$$

Remark 18. In the language of measure theory, $\mathbf{p}$ is often called the *Radon–Nikodym derivative*, and can be written as

$$p = \frac{dN}{dM},$$

which conveniently allows us to rigorously apply the heuristic

$$\int \chi_{\mathbb{A}} p dM = \int \chi_{\mathbb{A}} \frac{dN}{dM} dM = \int \chi_{\mathbb{A}} dN.$$

When P is a probability measure, such that $P \ll \text{Leb}$, we say that P is a continuous probability measure and we call $p = dP/d\text{Leb}$ the *probability density function (PDF)* of P . Alternatively, when P is a probability measure, such that $P \ll M$, where M is the counting measure, we say that P is a discrete probability measure and call $p = dP/dM$ the *probability mass function (PMF)* of P .

Example 20. On the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we say that the measure $P \ll \text{Leb}$ is the *standard normal probability measure* if

$$P(\mathbb{A}) = \int_{\mathbb{A}} \phi(\omega) d\text{Leb}(\omega),$$

for each $\mathbb{A} \in \mathcal{B}(\mathbb{R})$, where

$$\phi(\omega) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

is its PDF. Alternatively, on the measurable space $(\mathbb{Z}_{\geq 0}, 2^{\mathbb{Z}_{\geq 0}})$, we say that the measure P , absolutely continuous with respect to the counting measure M , is the *Bernoulli probability measure* with parameter $\theta \in [0, 1]$ if

$$\begin{aligned} P(\mathbb{A}) &= \int_{\mathbb{A}} p(\omega) dM(\omega) \\ &= \sum_{\omega \in \mathbb{A}} p(\omega), \end{aligned}$$

where

$$p(\omega) = \theta^{[\omega=1]} (1 - \theta)^{[\omega=0]}$$

is its PMF.

Remark 19. For historic and practical reasons, the integral measure symbol $d\text{Leb}(\omega)$ is often written simply as $d\omega$, and the integral expression $\int X(\omega) d\omega$ behaves exactly as the traditional Riemann integral does, when both exist. For example, let $X(\omega) = \sqrt{1 - \omega^2}$ be a measurable real function on the space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})$. Then, we have the fact that

$$\int \chi_{(-1,1)} X(\omega) d\text{Leb}(\omega) = \int_{-1}^1 \sqrt{1 - \omega^2} d\omega = \frac{\pi}{2},$$

as expected.

Exercise 14. Let $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P)$ be a probability space, where $P \ll \text{Leb}$. Then, verify that the CDF of P can be written as

$$F(t) = \int \chi_{(-\infty, t]} P(\omega) d\text{Leb}(\omega).$$

Now consider the probability space $(\mathbb{Z}, 2^{\mathbb{Z}}, P)$, where P is absolutely continuous with respect to the counting measure (on $\mathbb{Z}$). Verify that we have the similar expression for the CDF of P :

$$F(t) = \sum_{i=-\infty}^t P(\omega = i).$$

Definition 8. For a measure space $(\Omega, \mathcal{F}, M)$, we say that a set $A \in \mathcal{F}$ contains (M) -almost every element of Ω if $M(\Omega \setminus A) = 0$. We often replace the phrase almost every by the abbreviation a.e., or even by the statement almost all. When $M = P$ is a probability measure, we say that A is an event that occurs with probability one, and the definition $P(\Omega \setminus A) = 0$ can equivalently be written as $P(A) = 1$.

Remark 20. Via the definition above, the Radon–Nikodym Theorem provides the additional guarantee that the Radon–Nikodym derivative is unique for almost every

ω . That is, if there exists two functions p_1 and p_2 that both satisfy (1.2.2) then there also exists a set $\mathbb{A} \subset \mathcal{F}$, such that

$$p_1(\omega) = p_2(\omega)$$

for all $\omega \in \mathbb{A}$, where $M(\Omega \setminus \mathbb{A}) = 0$.

1.2.3.1 Products of measure spaces

Recall that if $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ are measurable spaces, then we can construct a product measurable space by considering the set $\Omega_1 \times \Omega_2$ and equipping it with the minimal σ -algebra $\mathcal{F}_1 \otimes \mathcal{F}_2$ to generate the measurable space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$. We now consider the manner in which we can usefully combine the measure spaces $(\Omega_1, \mathcal{F}_1, M_1)$ and $(\Omega_2, \mathcal{F}_2, M_2)$. This requires a notion of a product measure $M_1 \times M_2$ that is a function of elements in $\mathbb{A} \in \mathcal{F}_1 \otimes \mathcal{F}_2$. One such definition is via the integral expression:

$$(M_1 \times M_2)(\mathbb{A}) = \int_{\Omega_1} \int_{\Omega_2} \chi_{\mathbb{A}}(\omega_1, \omega_2) dM_2(\omega_2) dM_1(\omega_1),$$

where $\chi_{\mathbb{A}}(\omega_1, \omega_2)$ is simply the bivariate characteristic function:

$$\chi_{\mathbb{A}}(\omega_1, \omega_2) = \begin{cases} 1 & \text{if } (\omega_1, \omega_2) \in \mathbb{A}, \\ 0 & \text{if } (\omega_1, \omega_2) \notin \mathbb{A}. \end{cases}$$

We can verify, albeit with some effort, that $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, M_1 \times M_2)$.

Exercise 15. Let $(\Omega_1, \mathcal{F}_1, M_1)$ and $(\Omega_2, \mathcal{F}_2, M_2)$ be measure spaces, and let $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$. Clearly, $A \times B \in \mathcal{F}_1 \otimes \mathcal{F}_2$. Prove that

$$(M_1 \times M_2)(A \times B) = M_1(A) M_2(B).$$

Theorem 2 (Fubini's Theorem). *Let $(\Omega_1, \mathcal{F}_1, M_1)$ and $(\Omega_2, \mathcal{F}_2, M_2)$ be σ -finite measure spaces, and suppose that $X : \Omega_1 \times \Omega_2 \rightarrow \bar{\mathbb{R}}$ is measurable with respect to $\mathcal{F}_1 \otimes \mathcal{F}_2$ and $\mathcal{B}(\bar{\mathbb{R}})$, such that*

$$\int_{\Omega_1 \times \Omega_2} |X| d(M_1 \times M_2) < \infty. \quad (1.2.4)$$

Then:

(i) *For almost every $\omega_1 \in \Omega_1$,*

$$\int_{\Omega_2} |X(\omega_1, \omega_2)| dM_2(\omega_2) < \infty.$$

(ii) *For almost every $\omega_2 \in \Omega_2$,*

$$\int_{\Omega_1} |X(\omega_1, \omega_2)| dM_1(\omega_1) < \infty.$$

(iii) *The function*

$$X_1(\omega_1) = \int_{\Omega_2} X(\omega_1, \omega_2) dM_2(\omega_2)$$

is measurable with respect to $\mathcal{F}_1$ and $\mathcal{B}(\bar{\mathbb{R}})$.

(iv) *The function*

$$X_2(\omega_2) = \int_{\Omega_1} X(\omega_1, \omega_2) dM_1(\omega_1)$$

is measurable with respect to $\mathcal{F}_2$ and $\mathcal{B}(\bar{\mathbb{R}})$.

(v) *The following equalities hold:*

$$\begin{aligned} \int_{\Omega_1 \times \Omega_2} X d(\mathbf{M}_1 \times \mathbf{M}_2) &= \int_{\Omega_1} \int_{\Omega_2} X(\omega_1, \omega_2) d\mathbf{M}_2 d\mathbf{M}_1 \\ &= \int_{\Omega_2} \int_{\Omega_1} X(\omega_1, \omega_2) d\mathbf{M}_1 d\mathbf{M}_2. \end{aligned}$$

Remark 21. In fact, conclusions (iii)–(v) of Fubini's Theorem remains true, even without the finiteness condition (1.2.4), but only when $X : \Omega_1 \times \Omega_2 \rightarrow \bar{\mathbb{R}}_{\geq 0}$ is non-negative and measurable with respect to $\mathcal{F}_1 \otimes \mathcal{F}_2$ and $\mathcal{B}(\bar{\mathbb{R}})$. This result is known as Tonelli's Theorem.

Remark 22. A classic counter example to Fubini's Theorem is the integral:

$$\int_{\mathbb{R}^2} \chi_{[0,1]^2} \frac{\omega_1^2 - \omega_2^2}{(\omega_1^2 + \omega_2^2)^2} d(\text{Leb} \times \text{Leb}) = \int_{[0,1]^2} \frac{\omega_1^2 - \omega_2^2}{(\omega_1^2 + \omega_2^2)^2} d(\text{Leb} \times \text{Leb}).$$

In this scenario, we find that the two sequential integrals are not equal; i.e.,

$$\begin{aligned} \int_0^1 \int_0^1 \frac{\omega_1^2 - \omega_2^2}{(\omega_1^2 + \omega_2^2)^2} d\omega_2 d\omega_1 &= \frac{\pi}{4}, \text{ but} \\ \int_0^1 \int_0^1 \frac{\omega_1^2 - \omega_2^2}{(\omega_1^2 + \omega_2^2)^2} d\omega_1 d\omega_2 &= -\frac{\pi}{4}. \end{aligned}$$

This is because we can check that

$$\int_{[0,1]^2} \left| \frac{\omega_1^2 - \omega_2^2}{(\omega_1^2 + \omega_2^2)^2} \right| d(\text{Leb} \times \text{Leb}) = \infty,$$

which violates condition (1.2.4).

Exercise 16. Let $\{a_{ij} : i, j \in \mathbb{N}\}$ be set of real numbers. Establish conditions under which the following statement is true:

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

1.2.4 Sequences of measurable functions and random variables

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions from $(\Omega, \mathcal{F}, M)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and let $X : \Omega \rightarrow \mathbb{R}$ be a measurable function on the same spaces. We wish to characterize various notions about the convergence of X_n to X , as $n \rightarrow \infty$.

Definition 9. We say that X_n converges to X , everywhere on Ω , if for each $\omega \in \Omega$:

$$X_n(\omega) \rightarrow X(\omega), \text{ as } n \rightarrow \infty.$$

Definition 10. We say that X_n converges to X , almost everywhere on Ω , if for each $\omega \in \mathbb{A}$:

$$X_n(\omega) \rightarrow X(\omega), \text{ as } n \rightarrow \infty,$$

where $M(\Omega \setminus \mathbb{A}) = 0$.

Remark 23. Notice that Definitions 9 and 10 are nearly identical, except that almost everywhere convergence allows for $X_n(\omega) \nrightarrow X(\omega)$ for some $\omega \in \Omega \setminus \mathbb{A}$, which occur with measure zero. This allowance comes at the cost of the requirement of an underlying measure space, i.e., everywhere convergence requires neither the definition of a σ -algebra nor a measure in order to be defined, and is thus suitable for more generic mathematical settings. Both everywhere and almost everywhere convergence are generally referred to as point-wise convergence, since both notions are concerned with the convergence of X_n to X on a point-by-point basis (i.e., only requiring that X_n at ω goes to X at ω , for each ω , individually).

Remark 24. In probability theory, i.e., when $M = P$ is a probability measure, we often say that the random variables X_n converges to X , almost surely (or with probability one), instead of almost everywhere. Furthermore, almost sure convergence is often characterized by the probabilistic expression:

$$P \left(\left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\} \right) = 1,$$

or more verbosely: the probability of the set containing all values of ω for which $X_n(\omega)$ converges to $X(\omega)$ is equal to 1.

Definition 11. We say that X_n converges to X , uniformly on Ω , if:

$$\sup_{\omega \in \Omega} |X_n(\omega) - X(\omega)| \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Unlike the point-wise modes of convergence, uniform convergence requires that largest difference between functions X_n and X converge to zero, thus making the convergence of $X_n(\omega)$ to $X(\omega)$ occur at the same rate, regardless of ω . This is more obvious when we write Definition 11 in “ δ - ϵ ” form:

For every $\epsilon > 0$, there exists a $\delta \in \mathbb{N}$, such that, for all $n \geq \delta$,

$$|X_n(\omega) - X(\omega)| < \epsilon, \text{ for all } \omega \in \Omega.$$

Theorem 3 (Egorov’s Theorem). *Let $(\Omega, \mathcal{F}, M)$ be a measure space such that $M(\Omega) < \infty$. If $(X_n)_{n \in \mathbb{N}}$ converges to X , almost everywhere on Ω , then for every $\epsilon > 0$, there exists a set $\mathbb{A}_\epsilon \in \mathcal{F}$, such that $M(\Omega \setminus \mathbb{A}_\epsilon) < \epsilon$, on which $\sup_{\omega \in \mathbb{A}_\epsilon} |X_n(\omega) - X(\omega)| \rightarrow 0$.*

Remark 25. It is obvious that when X_n converges to X , uniformly on Ω , the convergence between the functions, almost everywhere, is implied. Egorov's Theorem provides a kind of converse, whereupon we can claim that when X_n converges to X , almost everywhere on Ω , the convergence between the function is uniform on a set $\mathbb{A}_\epsilon$ that excludes a total measure of no more than ϵ , for any $\epsilon > 0$. This type of uniform convergence, except for a set of small measure, is usually referred to as almost uniform convergence.

Definition 12. We say that X_n converges to X , in measure with respect to M , if for every $\epsilon > 0$:

$$M(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Definition 13. For each $q \in \mathbb{R}_{\geq 0}$, we say that X_n converges to X , in L_q -norm, with respect to M , if:

$$\int_{\Omega} |X_n(\omega) - X(\omega)|^q dM(\omega) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Definitions 12 and 13 are similar in that they both require evaluations of quantities related to the measure M . In the case of convergence in measure, we require that the measure of the set on which the difference in the functions X_n and X becomes small for any amount of difference, and the convergence in L_q norm requires that the integrated difference between the functions X_n and X become small as n increases, even when the difference is raised to some power q .

Remark 26. The name L^q norm refers to the fact that when X and Y are measurable functions from $(\Omega, \mathcal{F}, M)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$,

$$\|X - Y\|_q = \left[\int_{\Omega} |X(\omega) - Y(\omega)|^q dM(\omega) \right]^{1/q}$$

is a norm on the space of such functions with the properties $\|X\|_q < \infty$ and $\|Y\|_q < \infty$, up to an almost everywhere equivalence. In fact, when X is a measurable functions from $(\Omega, \mathcal{F}, M)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we say that X is in the space $L_q(\Omega, \mathcal{F}, M)$, if $\|X\|_q < \infty$. When the particular specifications of the measure space is implied or does not cause confusion, we also have the short hands: $L_q(\Omega, M)$, $L_q(M)$, or even L_q .

Remark 27. In probability theory, when $M = P$ is a probability measure, convergence in measure is often referred to as convergence in probability, and convergence in L^q norm is often referred to as convergence in q th mean. In particular, when $q = 1$ or $q = 2$, we often say that X_n converges to X in mean value or in mean squared, respectively. The q th mean convergence is also often expressed in terms of expectation, whereupon we write:

$$E |X_n - X|^q \rightarrow 0, \text{ as } n \rightarrow \infty,$$

or even

$$E(X_n) \rightarrow E(X), \text{ as } n \rightarrow \infty,$$

when $q = 1$.

Exercise 17. Use Markov's inequality to show that if X_n converges in q th mean, for any $q \in \mathbb{R}_{\geq 0}$, then X_n also converges in probability.

1.2.4.1 Exchanging limits and integrals

A constant question when manipulating sequences of measurable functions is whether one can safely exchange the integral and limit symbols. That is, if $(X_n)_{n \in \mathbb{N}}$ is a sequence of measurable functions from $(\Omega, \mathcal{F}, M)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then what additional information do we need in order to say that

$$\lim_{n \rightarrow \infty} \int_{\Omega} X_n dM = \int_{\Omega} \lim_{n \rightarrow \infty} X_n dM?$$

The following theorems provide some answers.

Theorem 4 (Monotone Convergence Theorem). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions that converge to X , almost everywhere. If for each $n \in \mathbb{N}$,*

$$0 \leq X_n(\omega) \leq X_{n+1}(\omega),$$

for almost all $\omega \in \Omega$, then

$$\lim_{n \rightarrow \infty} \int_{\Omega} X_n dM = \int_{\Omega} X dM.$$

Theorem 5 (Dominated Convergence Theorem). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of measurable functions that converge to X , either in measure or almost everywhere. If there exists a $Y(\omega)$ such that,*

$$(i) \quad |X_n(\omega)| \leq Y(\omega), \text{ for almost all } \omega \in \Omega, \text{ and for each } n \in \mathbb{N},$$

$$(ii) \quad Y(\omega) \text{ is integrable in the sense that } \int_{\Omega} Y(\omega) dM(\omega) < \infty,$$

then X_n and X are integrable,

$$\lim_{n \rightarrow \infty} \int_{\Omega} |X_n - X| dM = 0, \text{ and}$$

$$\lim_{n \rightarrow \infty} \int_{\Omega} X_n dM = \int_{\Omega} X dM.$$

Exercise 18. Another simple, albeit not too widely applicable result is available if we can assume that $(X_n)_{n \in \mathbb{N}}$ converges uniformly. Namely, show that if $M(\Omega) < \infty$ and $(X_n)_{n \in \mathbb{N}}$ converges uniformly to X , then $\int_{\Omega} X dM < \infty$ and $\lim_{n \rightarrow \infty} \int_{\Omega} X_n dM = \int_{\Omega} X dM$.

Exercise 19. Via Theorem 5, prove that if $M(\Omega) < \infty$, then almost everywhere convergence implies convergence in measure.

Remark 28. Specializing to the case of $M = P$, where P is a probability measure, a powerful means for establishing convergence in probability is via the notion of *continuous mapping*. That is, if $Y : \mathbb{R}^d \rightarrow \mathbb{R}$ is an almost surely continuous function from $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, in the sense that

$$P(\{\omega \in \Omega : Y \text{ is continuous at } X(\omega)\}) = 1,$$

then $Y \circ X_n$ converges in probability to $Y \circ X^*$ under the condition that $X_n = (X_{1n}, \dots, X_{dn})$ converges in probability to $X^* = (X_1^*, \dots, X_d^*)$, in the sense that $X_{in} : \Omega \rightarrow \mathbb{R}$ converges in probability to X_i^* , for each $i \in [d]$.

From this fact, we also have the more specific statements that if $X_{1n}, X_{2n} : \Omega \rightarrow \mathbb{R}$ converge in probability to X_1 and X_2 , respectively, then

- $X_{1n} + X_{2n}$ converges to $X_1 + X_2$, in probability.
- $X_{1n}X_{2n}$ converges to X_1X_2 , in probability.
- X_{1n}/X_{2n} converges to X_1/X_2 , in probability, if $X_2 \neq 0$ almost surely.

Exercise 20. Prove that Remark 28 holds true if convergence in probability is replaced by almost sure convergence.

1.2.5 Bibliographical notes

The majority of of Section 1.2 was written using combined elements from the texts of Cinlar [2011], Makarov and Podkorytov [2013], Roussas [2014], Shorack [2017], and Axler [2020]. I of course would not recommend that anyone reads all of these text (especially not cover to cover!); this is merely my own bookkeeping regarding the relevant resources from which this section was sourced.

Of use, I shall list out some of the helpful texts that may provide additional insight at each of the subsection level. Firstly, for Section 1.2.1, the theory of sets and σ -algebra can be found in Cinlar [2011, Sec. 1], Makarov and Podkorytov [2013, Sec. 1.1], and Roussas [2014, Secs. 1.1 and 1.2]. For Section 1.2.2, the basic construction of measures and measurable functions can be found in Cinlar [2011, Secs. 1.2 and 1.3], Makarov and Podkorytov [2013, Secs. 1.3 and 3.1], Roussas [2014, Secs. 1.3 and 2.1], Shorack [2017, Ch. 1 and Secs. 2.1 and 2.2], and Axler [2020, Secs. 2B and 2C]. For a probabilistic treatment of measures and measurable functions, we can refer to Cinlar [2011, Sec. 2.1], Shorack [2017, Sec. 2.4], and Axler [2020, Ch. 12].

For Section 1.2.3, integration theory is covered in Cinlar [2011, Sec. 1.4], Makarov and Podkorytov [2013, Ch. 4], Roussas [2014, Secs. 4.1 and 4.2], Shorack [2017, Ch. 4], and Axler [2020, Sec. 3A]. For expectations of random variables, specifically, we can refer to Cinlar [2011, Sec. 2.2], and Roussas [2014, Sec. 4.3]. For Section 1.2.4, resources regarding sequences of measurable functions, in general, can be found in Makarov and Podkorytov [2013, Secs. 3.3 and 4.8, and Ch. 7], Roussas [2014, Secs.5.2], Shorack [2017, Ch. 2.3], and Axler [2020, Sec. 3B]. Specializing to the theory of of random sequences, we can read Cinlar [2011, Secs. 1.2–1.4], Roussas [2014, Secs.6.2], and Shorack [2017, Ch. 2.4].

Of course, there are a great number of results that I have not presented in this course, for the sake of brevity. If I later find that we require some of the results from the aforementioned sections of texts, then I will provide them in the Sections where they are required, or if I find it necessary, appendices may also be generated to contain such content.

1.3 Optimization Theory

We recall from Section 1.1.2 that we are particularly interested in *statistics* of the form

$$\begin{aligned}\hat{\theta}_n(\omega) &= \hat{\theta}\left((X_i)_{i \in [n]}\right) \\ &= \arg \min_{\theta \in \mathbb{T}} r_n(\theta),\end{aligned}$$

where the *risk function*

$$r_n(\theta) = r\left(\theta; (X_i)_{i \in [n]}\right) = \frac{1}{n} \sum_{i=1}^n \ell_{\theta}(X_i),$$

is defined by the *loss function* $\ell_{\theta} : \mathbb{X} \rightarrow \mathbb{R}$. Here, we assume, as usual, that $(X_i)_{i \in [n]}$ is a sequence of IID random functions from $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$, where we may further assume that $\mathbb{X}$ is endowed with some distance dist to form the metric space $(\mathbb{X}, \text{dist})$. In fact, we shall make the further simplification that $\mathbb{X} \subset \mathbb{R}^d$ and that $\mathbb{T} \subset \mathbb{R}^p$, for natural numbers $d, p \in \mathbb{N}$, and that dist can be taken to be the **Euclidean norm**.

However, before we proceed too far with the probabilistic problem, above, we may wish to simply consider the nature of problems requiring the computation of

$$\arg \min_{\theta \in \mathbb{T}} r(\theta) \quad \text{or} \quad \arg \max_{\theta \in \mathbb{T}} r(\theta) \tag{1.3.1}$$

for generic functions $r : \mathbb{T} \rightarrow \bar{\mathbb{R}}$. Such problems are generally referred to as *optimization problems*, in mathematical parlance. *Optimization theory* is branch of mathematics that is most concerned with optimization problems, where the existence,

uniqueness, and computability of sets of the form (1.3.1) are considered under different assumptions on the function r and the space $\mathbb{T}$.

1.3.1 Minima and infima

For a set $\mathbb{A} \subset \mathbb{R}$, we may define the *minimum* and *infimum* set functions as:

- $\min(\mathbb{A}) = m$ if and only if there exists an $m \in \mathbb{A}$, such that $m \leq a$, for all $a \in \mathbb{A}$; and
- $\inf(\mathbb{A}) = m$ if and only if $m \leq a$, for all $a \in \mathbb{A}$, and $m \geq b$ for all $b \in \bar{\mathbb{R}}$, where $b \leq a$ for all $a \in \mathbb{A}$.

Similarly, we may define the maximum and the supremum set functions as:

- $\max(\mathbb{A}) = m$ if and only if there exists an $m \in \mathbb{A}$, such that $m \geq a$, for all $a \in \mathbb{A}$; and
- $\sup(\mathbb{A}) = m$ if and only if $m \geq a$, for all $a \in \mathbb{A}$, and $m \leq b$ for all $b \in \bar{\mathbb{R}}$, where $b \geq a$ for all $a \in \mathbb{A}$.

Remark 29. Using the definitions above, we can show that the infimum and supremum must both always exist. Furthermore, when the minimum exists $\inf(\mathbb{A}) = \min(\mathbb{A})$, and when $\max(\mathbb{A})$ exists, $\sup(\mathbb{A}) = \max(\mathbb{A})$.

Remark 30. We can consider the set

$$\mathbb{B} = \{b \in \mathbb{R} : b \leq a, \text{ for all } a \in \mathbb{A}\}$$

as being the set of lower bounds of the set $\mathbb{A}$. Then, the infimum of $\mathbb{A}$ is the maximum element of $\mathbb{B}$; i.e., $\inf(\mathbb{A}) = \max(\mathbb{B})$. For this reason, the infimum is also often referred to as the *greatest lower bound*. Similarly, the supremum of is also often referred to as the *least upper bound*.

The set function definitions of minima, infima, maxima, and suprema when applied to a function $r : \mathbb{T} \rightarrow \mathbb{R}$ then produces the following definitions:

$$\begin{aligned} \min_{\theta \in \mathbb{T}} / \max_{\theta \in \mathbb{T}} / \inf / \sup r(\theta) &= \min / \max / \inf / \sup (r(\mathbb{T})) \\ &= \min / \max / \inf / \sup (\{a \in \mathbb{R} : r(\theta) = a, \theta \in \mathbb{T}\}). \end{aligned}$$

The function forms of the minimum and maximum pair are related via the equation:

$$\min_{\theta \in \mathbb{T}} r(\theta) = - \max_{\theta \in \mathbb{T}} [-r(\theta)],$$

and similarly, the infimum and supremum pair are related via the equation:

$$\inf_{\theta \in \mathbb{T}} r(\theta) = - \sup_{\theta \in \mathbb{T}} [-r(\theta)], \quad (1.3.2)$$

which implies that we need only study the behavior of the minimum and infimum, since the behavior of the maximum and supremum are identical, up to some sign differences.

Remark 31. Observe that the definitions above are the same as those from Remark (3), albeit more precise.

Let $a \in \mathbb{R}$ and define functions $r, s : \mathbb{T} \rightarrow \mathbb{R}$, then the function infimum has the following properties:

1. $\inf_{\theta \in \mathbb{T}} [a \times r(\theta)] = a \times \inf_{\theta \in \mathbb{T}} r(\theta)$, for $a \geq 0$,
2. $\inf_{\theta \in \mathbb{T}} [a + r(\theta)] = a + \inf_{\theta \in \mathbb{T}} r(\theta)$,
3. If $r(\theta) \leq s(\theta)$, for each $\theta \in \mathbb{T}$, then $\inf_{\theta \in \mathbb{T}} r(\theta) \leq \inf_{\theta \in \mathbb{T}} s(\theta)$,
4. $\inf_{\theta \in \mathbb{T}} |r(\theta)| \leq |\inf_{\theta \in \mathbb{T}} r(\theta)|$,
5. $\inf_{\theta \in \mathbb{T}} r(\theta) + \inf_{\theta \in \mathbb{T}} s(\theta) \leq \inf_{\theta \in \mathbb{T}} [r(\theta) + s(\theta)]$,
6. $\inf_{\theta \in \mathbb{T}} [r(\theta) - s(\theta)] \leq \inf_{\theta \in \mathbb{T}} r(\theta) - \inf_{\theta \in \mathbb{T}} s(\theta)$.

Exercise 21. Use the the relationship between the function infimum and supremum (1.3.2) in order to derive equivalent facts for the supremum, to those stated above.

Remark 32. Suppose that $\mathbb{T} = \mathbb{T}_1 \times \mathbb{T}_2$ is a product of spaces $\mathbb{T}_1$ and $\mathbb{T}_2$, and write $\theta = (\theta_1, \theta_2) \in \mathbb{T}_1 \times \mathbb{T}_2$. Then, minimization with respect to the space $\mathbb{T}$ is equivalent to iteratively minimizing with respect to the spaces $\mathbb{T}_1$ and $\mathbb{T}_2$, in the sense that

$$\inf_{\theta \in \mathbb{T}} r(\theta) = \inf_{(\theta_1, \theta_2) \in \mathbb{T}_1 \times \mathbb{T}_2} r(\theta_1, \theta_2) = \inf_{\theta_1 \in \mathbb{T}_1} \inf_{\theta_2 \in \mathbb{T}_2} r(\theta_1, \theta_2) = \inf_{\theta_2 \in \mathbb{T}_2} \inf_{\theta_1 \in \mathbb{T}_1} r(\theta_1, \theta_2).$$

For completeness, we recall the argument functions from Remark (3):

$$\arg \min_{\theta \in \mathbb{T}} / \arg \max_{\theta \in \mathbb{T}} / \arg \inf_{\theta \in \mathbb{T}} / \arg \sup_{\theta \in \mathbb{T}} r(\theta) = \left\{ \tau \in \mathbb{T} : r(\tau) = \min / \max / \inf / \sup_{\theta \in \mathbb{T}} r(\theta) \right\}.$$

Similarly to the relationship between the function minimum and maximum, and infimum and supremum pairs, we also have the relationships

$$\arg \min_{\theta \in \mathbb{T}} r(\theta) = \arg \max_{\theta \in \mathbb{T}} [-r(\theta)], \text{ and}$$

$$\arg \inf_{\theta \in \mathbb{T}} r(\theta) = \arg \sup_{\theta \in \mathbb{T}} [-r(\theta)].$$

1.3.2 Giving $\mathbb{R}^p$ some structure

As noted earlier, we will restrict our attention to the case of $\mathbb{X} \subset \mathbb{R}^d$ and $\mathbb{T} \subset \mathbb{R}^p$ throughout this section. We will clarify what this restriction actually affords us with respect to useful structures from which we can derive interesting optimization theoretic results. Firstly, we consider a fact that is inherent to $\mathbb{R}^p$, that it is a *vector space*, in the sense that for any $\theta, \tau \in \mathbb{R}^p$ and $a, b \in \mathbb{R}$:

1. $1 \times \theta = \theta$.
2. $0 \times \theta = 0 \in \mathbb{R}^p$.
3. $(a + b) \theta = a\theta + b\theta \in \mathbb{R}^p$.
4. $a(b\theta) = (ab) \theta \in \mathbb{R}^p$.
5. $a(\theta + \tau) = a\theta + a\tau \in \mathbb{R}^p$.

To be more specific, the space $\mathbb{R}^p$ is a *finite dimensional vector space* (p dimensional, to be precise), in the sense that there exists a sequence of vectors $(\beta_i)_{i \in [p]}$, where $\beta_i \in \mathbb{R}^p$ for each $i \in [p]$, such that for any $\theta \in \mathbb{R}^p$, there exists a sequence of real numbers $(a_i)_{i \in [p]}$, such that we can write

$$\theta = \sum_{i=1}^p a_i \beta_i.$$

Any sequence $(\beta_i)_{i \in [p]}$ that can be used, above, is called a *basis* for $\mathbb{R}^p$.

On $\mathbb{R}^p$, we can define a *norm*, denoted as $\theta \mapsto \|\theta\|$, for $\theta \in \mathbb{R}^p$, as any operation that satisfies the properties, for any $\theta, \tau \in \mathbb{R}^q$ and $a \in \mathbb{R}$:

1. $\|\theta + \tau\| \leq \|\theta\| + \|\tau\|$, (*subadditivity*)
2. $\|a\theta\| = |a| \|\theta\|$,
3. $\|\theta\| = 0$ if and only if $\theta = 0$.

Example 21. The most popular choices for norms on $\mathbb{R}^p$ are the so called l_q norms, which have the form:

$$\|\theta\|_q = \left(\sum_{i=1}^p |\theta_i|^q \right)^{1/q},$$

for each $q \in [1, \infty)$, where $\theta = (\theta_1, \dots, \theta_p) \in \mathbb{R}^p$. From this family, we can obtain many of our typical norm choices, such as

- the *taxicab* norm ($q = 1$): $\|\theta\|_1 = \sum_{i=1}^p |\theta_i|$,
- the *Euclidean* norm ($q = 2$): $\|\theta\|_2 = \sqrt{\sum_{i=1}^p \theta_i^2}$.

Exercise 22. A related norm to the l_q family is the *maximum* norm:

$$\|\theta\|_\infty = \max(|\theta_1|, \dots, |\theta_n|),$$

which can be seen as the limit of the l_q norms, in the sense that

$$\|\theta\|_\infty = \lim_{q \rightarrow \infty} \|\theta\|_q.$$

Show that $\theta \mapsto \|\theta\|_\infty$ satisfies the definition of a norm on $\mathbb{R}^p$.

Together $\mathbb{R}^p$ and $\|\cdot\|$ forms the *normed vector space* $(\mathbb{R}^p, \|\cdot\|)$, which in turn is also a metric space $(\mathbb{R}^p, \text{dist})$, where $\text{dist}(\theta, \tau) = \|\theta - \tau\|$, for any $\theta, \tau \in \mathbb{R}^p$. Thus, $\|\cdot\|$

can be used to define a *topology* on $\mathbb{R}^p$ that is characterized by the open and closed balls (as defined in Section 1.2.1.1):

$$\mathbb{B}_\rho(\theta) = \{\tau \in \mathbb{R}^p : \|\theta - \tau\| < \rho\} \text{ and}$$

$$\bar{\mathbb{B}}_\rho(\theta) = \{\tau \in \mathbb{R}^p : \|\theta - \tau\| \leq \rho\}.$$

Exercise 23. Prove that $\text{dist}(\theta, \tau) = \|\theta - \tau\|$ is indeed a metric on $\mathbb{R}^p$.

It would appear that there are an (uncountably) infinite choices of norms on $\mathbb{R}^p$, by considering the l_q norms alone! Thus, this would imply that there are an infinite number of topologies that can be endowed onto $\mathbb{R}^p$, with a norm structure. However, as luck would have it, all norms on $\mathbb{R}^p$ are *equivalent* and generate the same topology. That is, for any two norms $\theta \mapsto \|\theta\|_{\text{one}}$ and $\theta \mapsto \|\theta\|_{\text{two}}$, there exists constants $a, b \in \mathbb{R}_{>0}$ such that

$$a \|\theta\|_{\text{two}} \leq \|\theta\|_{\text{one}} \leq b \|\theta\|_{\text{two}},$$

for every $x \in \mathbb{R}^p$. Thus, it does not matter which norm we choose to endow our $\mathbb{R}^p$ with, as all choices of norms yield the same metric space topology.

Exercise 24. Show that the taxicab, Euclidean, and maximum norms, $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$, satisfy the inequalities

$$\|\theta\|_2 \leq \|\theta\|_1 \leq \sqrt{p} \|\theta\|_2,$$

$$\|\theta\|_\infty \leq \|\theta\|_2 \leq \sqrt{p} \|\theta\|_\infty,$$

$$\|\theta\|_\infty \leq \|\theta\|_1 \leq p \|\theta\|_\infty.$$

Since all choice of norms are effectively the same, we should choose to endow our $\mathbb{R}^p$ with the norm that provides us with the most structure to work with, namely, the Euclidean norm $\|\cdot\|_2$, because it corresponds to the *Hilbert space* defined by the *inner product* $\langle \theta, \tau \rangle = \sum_{i=1}^p \theta_i \tau_i$. We shall now define what an inner product is, and subsequently what a Hilbert space is.

We say that $(\theta, \tau) \mapsto \langle \theta, \tau \rangle$ is an inner product on $\mathbb{R}^p$, if the following properties are satisfied, for each $\theta = (\theta_1, \dots, \theta_p) \in \mathbb{R}^p$, $\tau = (\tau_1, \dots, \tau_p) \in \mathbb{R}^p$ and $t = (t_1, \dots, t_p) \in \mathbb{R}^p$ and $a, b \in \mathbb{R}$:

1. $\langle \theta, \tau \rangle = \langle \tau, \theta \rangle$, (*symmetry*)
2. $\langle a\theta + b\tau, t \rangle = a \langle \theta, t \rangle + b \langle \tau, t \rangle$, (*linearity*)
3. $\langle \theta, \theta \rangle \geq 0$,
4. $\langle \theta, \theta \rangle = 0$ if and only if $\theta = 0$.

Together, $\mathbb{R}^p$ and $\langle \cdot, \cdot \rangle$ form an *inner product space* $(\mathbb{R}^p, \langle \cdot, \cdot \rangle)$, which in turn defines a *normed vector space* $(\mathbb{R}^p, \|\cdot\|)$, where $\|\theta\| = \sqrt{\langle \theta, \theta \rangle}$ (and hence endows a topology on $\mathbb{R}^p$).

A generic inner product space $(\mathbb{T}, \langle \cdot, \cdot \rangle)$ is also a Hilbert space if it is *complete*. Here, completeness is usually intuit as the idea that the underlying metric space has “no missing points”. Let $(\mathbb{T}, \text{dist})$ be a metric space induced by the *inner product* $\langle \cdot, \cdot \rangle$ and let any subset $\mathbb{A} \subset \mathbb{T}$ have *diameter* Δ , if

$$\Delta = \text{diam}(\mathbb{A}) = \sup_{\theta, \tau \in \mathbb{A}} \text{dist}(\theta, \tau).$$

Suppose that $(\mathbb{A}_i)_{i \in \mathbb{N}}$ is a decreasing sequence of subsets of $\mathbb{X}$, which implies that

$$\mathbb{A}_1 \supset \mathbb{A}_2 \supset \mathbb{A}_3 \dots$$

from Exercise 6. Further suppose that $\mathbb{A}_i$ is closed, for each $i \in \mathbb{N}$, and that $\Delta_i = \text{diam}(\mathbb{A}_i) \rightarrow 0$ as $i \rightarrow \infty$. We say that the metric space $(\mathbb{T}, \text{dist})$ is complete if

$$\bigcap_{i \in \mathbb{N}} \mathbb{A}_i \neq \emptyset$$

for any sequence $(\mathbb{A}_i)_{i \in \mathbb{N}}$ satisfying the aforementioned properties.

Exercise 25. Show that $\mathbb{T} = (0, 1)$ equipped with the metric $\text{dist}(\theta, \tau) = |\theta - \tau|$, for each $\theta, \tau \in \mathbb{T}$ is not complete.

Remark 33. In fact, on $\mathbb{R}^p$, any metric space defined via a normed vector space is complete. In fact, if an arbitrary normed vector space $(\mathbb{T}, \|\cdot\|)$ is complete, then we call it a *Banach space*.

Let $\mathbb{A} \subset \mathbb{R}^p$, where $\mathbb{R}^p$ is equipped with the Euclidean norm (which we shall now take to be the default norm structure on $\mathbb{R}^p$, unless stated otherwise), where we write $\|\cdot\| = \|\cdot\|_2$, for brevity. Then, we say that $\mathbb{A}$ is *bounded* if there exists a $\rho > 0$, such that

$$\mathbb{A} \subset \mathbb{B}_\rho(0).$$

We say that a sequence $(\theta_i)_{i \in \mathbb{N}}$, with $\theta_i \in \mathbb{R}^p$ for each $i \in [n]$, on the normed vector space $(\mathbb{R}^p, \|\cdot\|)$, *converges* to $\theta \in \mathbb{R}^p$, if

$$\lim_{i \rightarrow \infty} \|\theta_i - \theta\| = 0.$$

Suppose that $(\theta_i)_{i \in \mathbb{N}}$ is a *bounded sequence* in the sense that $\|\theta_i\| < b$ for each $i \in \mathbb{N}$, such that $b < \infty$. If $(\theta_i)_{i \in \mathbb{N}}$ is a bounded sequence, then the *Bolzano–Weierstrass Theorem* states that there exists a subsequence $(\theta_{i_k})_{k \in \mathbb{N}}$ of $(\theta_i)_{i \in \mathbb{N}}$, such that $(\theta_{i_k})_{k \in \mathbb{N}}$ converges to some limit $\theta \in \mathbb{R}^p$. We call such a subsequence a *convergent subsequence*, and we often refer to the limits of these convergent subsequences as *limit points*, *cluster points*, or *accumulation points*.

Remark 34. We note the alternative definition of convergence, in terms of subsequences, which states that $(\theta_i)_{i \in \mathbb{N}}$ converges to θ if and only if every subsequence $(\theta_{i_k})_{k \in \mathbb{N}}$ converges to θ .

Exercise 26. Prove that the sequence $(\theta_i)_{i \in \mathbb{N}}$, defined by

$$\theta_i = \sin\left(\frac{\pi}{20}i\right)$$

has a convergent subsequence, and provide an example of one such convergent subsequence.

An implication of the Bolzano–Weierstrass Theorem is that a subset $\mathbb{A}$ of $\mathbb{R}^p$ (equipped with $\|\cdot\|$) is compact if and only if $\mathbb{A}$ is closed and bounded, where we say that $\mathbb{A}$ is compact if every sequence $(\theta_i)_{i \in \mathbb{N}}$ in $\mathbb{A}$ has a convergent subsequence with limit $\theta \in \mathbb{A}$.

1.3.3 Functions on $\mathbb{R}^p$

1.3.3.1 Smoothness

Let $r : \mathbb{T} \rightarrow \mathbb{R}$ be a function on $\mathbb{T} \subset \mathbb{R}^p$. We shall say that r is *continuous* if

$$r^{-1}(\mathbb{A}) = \{\theta \in \mathbb{T} : r(\theta) \in \mathbb{A}\}$$

is an open set, whenever $\mathbb{A} \subset \mathbb{R}$ is open. That is, r is continuous if r maps open sets $(r^{-1}(\mathbb{A}))$ to open sets $(\mathbb{A})$. Since we have endowed $\mathbb{R}^p$ with the norm $\|\cdot\|$, we can also obtain the more familiar “ δ - ϵ ” definition of continuity. That is, we say that r is *continuous at* $\theta \in \mathbb{T}$ if for every $\epsilon > 0$, there exists an $\delta > 0$ such that $|r(\theta) - r(\tau)| < \epsilon$ whenever $\tau \in \mathbb{B}_\delta(\theta)$. We then say that r is continuous if it is continuous at θ , for all $\theta \in \mathbb{T}$.

Exercise 27. Use the definition above to show that the function $r : \mathbb{R} \rightarrow \mathbb{R}$ is not continuous, where r is defined by

$$r(\theta) = \begin{cases} -1 & \text{if } \theta < 0, \\ 1 & \text{if } \theta \geq 0. \end{cases}$$

An important consequence of the continuity of r is that “limits can pass through r ”. That is, if $r : \mathbb{T} \rightarrow \mathbb{R}$ is continuous and $(\theta_i)_{i \in [n]}$ converges to $\theta \in \mathbb{T}$, then

$$\lim_{i \rightarrow \infty} r(\theta_i) = r\left(\lim_{i \rightarrow \infty} \theta_i\right) = r(\theta).$$

If we view continuity as a kind of “smoothness” and “local predictability” of the function, then we can strengthen such characteristics by demanding that our

functions are also *differentiable*. For an open set $\mathbb{T} \subset \mathbb{R}^p$, we say that $r : \mathbb{T} \rightarrow \mathbb{R}$ is differentiable at $\tau \in \mathbb{T}$ if there exists a $\mathfrak{D}_\tau \in \mathbb{R}^p$ such that

$$\lim_{\|v\| \rightarrow 0} \frac{|r(\tau + v) - r(\tau) - \langle \mathfrak{D}_\tau, v \rangle|}{\|v\|}.$$

We call $\mathfrak{D}_\tau$ the *derivative* (or *full derivative*, or *total derivative*) of r at $\tau \in \mathbb{T}$. We can will also use the notation $r' : \mathbb{T} \rightarrow \mathbb{R}^p$, where

$$r'(\tau) = \mathfrak{D}_\tau.$$

Exercise 28. Show that if the derivative $\mathfrak{D}_\tau$ exists at $\tau \in \mathbb{T}$ then it is unique.

If $\mathfrak{D}_\tau$ exists for all $\tau \in \mathbb{T}$, then we say that r is differentiable on $\mathbb{T}$ (in the *Fréchet* sense). If r' is a continuous function from $\mathbb{T}$ to $\mathbb{R}^p$, then we say that r is *continuously differentiable*.

Exercise 29. Show that if r is differentiable at $\tau \in \mathbb{T}$ then it is also continuous.

Remark 35. There are many notational conventions for the derivative. We will often write:

$$r'(\tau) = \nabla r(\tau) = \left. \frac{\partial r}{\partial \theta} \right|_{\theta=\tau} = \left. \frac{\partial}{\partial \theta} r \right|_{\theta=\tau} = \frac{\partial r}{\partial \theta}(\tau).$$

We define the directional derivative of r in the direction $v \in \mathbb{R}^p$ at $\tau \in \mathbb{T}$ as

$$\mathfrak{d}(\tau; v) = \lim_{h \downarrow 0} \frac{r(\tau + hv) - r(\tau)}{h},$$

and if $\mathfrak{d}(\tau; v)$ exists for every $v \in \mathbb{R}^p$, then we say that r is *directionally differentiable* at $\tau \in \mathbb{T}$. Further, if there exists a $\mathfrak{d}_\tau \in \mathbb{R}^p$, such that

$$\mathfrak{d}(\tau; v) = \langle \mathfrak{d}_\tau, v \rangle,$$

then we say that r is *Gateaux differentiable* at $\tau \in \mathbb{T}$. If r is Gateaux differentiable at every $\tau \in \mathbb{T}$, then we say that it is Gateaux differentiable on $\mathbb{T}$.

Exercise 30. Show that if r is Fréchet differentiable, then it is also Gateaux differentiable, in the sense that

$$\langle \mathfrak{D}_\tau, v \rangle = \langle \mathfrak{d}_\tau, v \rangle,$$

for each $\tau \in \mathbb{T}$ and $v \in \mathbb{R}^p$.

The notions of Gateaux differentiability and directional differentiability give rise to our traditional notion of partial derivatives. That is, for each $j \in [p]$, let

$$\iota_j = \left(0, \dots, 0, \underbrace{1}_{j\text{th position}}, 0, \dots, 0 \right).$$

Then, we say that

$$\frac{\partial r}{\partial \theta_j}(\tau) = \nabla_j r(\tau) = \langle \mathfrak{d}_\tau, \iota_j \rangle$$

is the *partial derivative along the j th coordinate* (or *in the j th direction*) at $\tau \in \mathbb{T}$.

1.3.3.2 Convexity

We say that the set $\mathbb{T} \subset \mathbb{R}^p$ is *convex* if for every $\theta, \tau \in \mathbb{T}$ and $\lambda \in [0, 1]$,

$$\lambda\theta + (1 - \lambda)\tau = t \in \mathbb{T}.$$

An important property of convexity is that the potentially uncountable intersection of convex sets is convex. That is, for a family of convex sets $\{\mathbb{T}_i \subset \mathbb{R}^p : i \in \mathbb{I}\}$, indexed by a potentially uncountable set $\mathbb{I}$, the intersect

$$\bigcap_{i \in \mathbb{I}} \mathbb{T}_i$$

is convex.

Example 22. The following subsets of $\mathbb{R}^p$ are convex:

- The line passing through $\tau \in \mathbb{R}^p$ in the direction of $v \in \mathbb{R}^p$:

$$\{\theta \in \mathbb{R}^p : \theta = \tau + hv, h \in \mathbb{R}\}.$$

- The hyperplane defined by coefficients $a \in \mathbb{R}^p$ and $b \in \mathbb{R}$:

$$\{\theta \in \mathbb{R}^p : \langle a, \theta \rangle = b\}.$$

- The half-plane generated by coefficients $a \in \mathbb{R}^p$ and $b \in \mathbb{R}$:

$$\{\theta \in \mathbb{R}^p : \langle a, \theta \rangle \leq b\}.$$

Exercise 31. Show that the ball

$$\bar{\mathbb{B}}_\rho(0) = \{\theta \in \mathbb{R}^p : \|\theta\| \leq \rho\}$$

is convex, for each $\rho > 0$. Assess whether the sphere

$$\bar{\mathbb{B}}_\rho(0) \setminus \mathbb{B}_\rho(0) = \{\theta \in \mathbb{R}^p : \|\theta\| = \rho\}$$

is convex or not.

For a sequence $(\theta_i)_{i \in [n]}$, with $\theta_i \in \mathbb{T}$ for each $i \in [n]$, we say that

$$t = \sum_{i=1}^n \lambda_i \theta_i \tag{1.3.3}$$

is a *convex combination* if $\lambda_i \geq 0$ for each $i \in [n]$, and $\sum_{i=1}^n \lambda_i = 1$.

Exercise 32. Show that if $\mathbb{T}$ is a convex set then the convex combination t , defined by 1.3.3, is in $\mathbb{T}$.

For a convex set $\mathbb{T} \subset \mathbb{R}^p$, we say that a function $r : \mathbb{T} \rightarrow \mathbb{R}$ is *convex* if for each $\theta, \tau \in \mathbb{T}$ and $\lambda \in [0, 1]$:

$$r(\lambda\theta + (1 - \lambda)\tau) \leq \lambda r(\theta) + (1 - \lambda)r(\tau).$$

We further say that r is *strictly convex* if

$$r(\lambda\theta + (1 - \lambda)\tau) < \lambda r(\theta) + (1 - \lambda)r(\tau),$$

whenever $\theta \neq \tau$, and $\lambda \in (0, 1)$.

Remark 36. A complementary property to convexity is concavity, where, for a convex set $\mathbb{T} \subset \mathbb{R}^p$, we say that a function $r : \mathbb{T} \rightarrow \mathbb{R}$ is *concave* if for each $\theta, \tau \in \mathbb{T}$ and $\lambda \in [0, 1]$:

$$r(\lambda\theta + (1 - \lambda)\tau) \geq \lambda r(\theta) + (1 - \lambda)r(\tau).$$

Similarly, r is *strictly concave* if

$$r(\lambda\theta + (1 - \lambda)\tau) > \lambda r(\theta) + (1 - \lambda)r(\tau),$$

whenever $\theta \neq \tau$, and $\lambda \in (0, 1)$.

Exercise 33. Show that if r is convex, then $-r$ is concave.

When $\mathbb{T} \subset \mathbb{R}^p$ is both open and convex, and r is differentiable on $\mathbb{T}$, then we have the fact that $r : \mathbb{T} \rightarrow \mathbb{R}$ is convex, if and only if

$$r(\theta) \geq r(\tau) + \langle r'(\tau), \theta - \tau \rangle,$$

for each $\theta, \tau \in \mathbb{T}$, and furthermore, r is strictly convex, if and only if

$$r(\theta) > r(\tau) + \langle r'(\tau), \theta - \tau \rangle,$$

whenever $\theta \neq \tau$.

Example 23. The following functions are convex:

- $r(\theta) = e^{a\theta}$, for any $a \in \mathbb{R}$, on $\mathbb{R}$.
- $r(\theta) = |\theta|^q$, for any $q \geq 1$, on $\mathbb{R}$.
- $r(\theta) = -\log \theta$, on $\mathbb{R}_{>0}$.
- $r(\theta) = \|\theta\|$, on $\mathbb{R}^p$, for any norm $\|\cdot\|$.
- $r(\theta) = \max(\{\theta_1, \dots, \theta_p\})$, on $\mathbb{R}^p$.
- $r(\theta) = \log(\sum_{i=1}^p \exp(\theta_i))$, on $\mathbb{R}^p$.

Remark 37. It is not immediately clear what the connection between convex sets and functions is, besides the use of the convex combination device: “ $\lambda\theta + (1 - \lambda)\tau$ ”. However, a deep connection is revealed when we look at the *graph* of a convex function:

$$\{(\theta, r(\theta)) : \theta \in \mathbb{T}\}.$$

In particular if we consider the *epigraph*:

$$\text{epi } r = \{(\theta, a) : \theta \in \mathbb{T}, r(\theta) \leq a\},$$

then we can observe that it is convex if and only if r is convex.

When establishing convexity, we need not start from scratch, and can rely on algebraic properties of convex functions for verification. For instance, we have the following rules:

- If $(r_i)_{i \in [n]}$ is a sequence of convex functions on $\mathbb{T}$ and $(a_i)_{i \in [n]}$ is a sequence of non-negative constants (i.e., $a_i \in \mathbb{R}_{\geq 0}$, for each $i \in [n]$), then $r(\theta) = \sum_{i=1}^n a_i r_i(\theta)$ is convex.
- If $(r_i)_{i \in [n]}$ is a sequence of convex functions on $\mathbb{T}$, then $r(\theta) = \max(r_1(\theta), \dots, r_n(\theta))$ is convex.
- If $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex and non-decreasing, then $r(\theta) = (g \circ f)(\theta)$ is convex.
- If $f : \mathbb{R} \rightarrow \mathbb{R}$ is concave and $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex and non-increasing, then $r(\theta) = (g \circ f)(\theta)$ is convex.

1.3.3.3 Some matrix analysis

We now consider some computation aspects of working with elements in $\mathbb{R}^p$ as matrix analytic vectors rather than as *p-tuples*. That is, instead of conceptualizing elements $\theta = (\theta_1, \dots, \theta_p) \in \mathbb{R}^p$ as a list of numbers, we shall instead conceptualize them as column vectors. That is, we identify θ with the column vector (which we denote in

boldface):

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix} \in \mathbb{R}^{p \times 1}.$$

In this way, we can consider other p -tuple objects as column vectors, as well, such as the derivative ∇r , which we can write as

$$\nabla r = \begin{bmatrix} \partial r / \partial \theta_1 \\ \partial r / \partial \theta_2 \\ \vdots \\ \partial r / \partial \theta_p \end{bmatrix} \in \mathbb{R}^{p \times 1},$$

and thus, when r is differentiable, we can write the directional derivative at $\boldsymbol{\tau} \in \mathbb{R}^{p \times 1}$ in the direction $\boldsymbol{v} \in \mathbb{R}^{p \times 1}$, as

$$\mathfrak{d}(\tau; v) = \langle \nabla r(\tau), v \rangle = [\nabla r(\tau)]^\top \boldsymbol{v},$$

where $\top$ is the *transposition* symbol.

Now, for an open set $\mathbb{T} \subset \mathbb{R}^p$, we define the *Hessian* of a function $r : \mathbb{T} \rightarrow \mathbb{R}$, evaluated at $\tau \in \mathbb{T}$ as being the matrix

$$\mathbf{H}_r(\tau) = \begin{bmatrix} \frac{\partial^2 r}{\partial \theta_1 \partial \theta_1}(\tau) & \frac{\partial^2 r}{\partial \theta_1 \partial \theta_2}(\tau) & \cdots & \frac{\partial^2 r}{\partial \theta_1 \partial \theta_p}(\tau) \\ \frac{\partial^2 r}{\partial \theta_2 \partial \theta_1}(\tau) & \frac{\partial^2 r}{\partial \theta_2 \partial \theta_2}(\tau) & \cdots & \frac{\partial^2 r}{\partial \theta_2 \partial \theta_p}(\tau) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 r}{\partial \theta_p \partial \theta_1}(\tau) & \frac{\partial^2 r}{\partial \theta_p \partial \theta_2}(\tau) & \cdots & \frac{\partial^2 r}{\partial \theta_p \partial \theta_p}(\tau) \end{bmatrix},$$

where

$$\frac{\partial^2 r}{\partial \theta_i \partial \theta_j}(\tau) = \frac{\partial}{\partial \theta_j} \left(\frac{\partial}{\partial \theta_i} r \right)(\tau),$$

for each pair $i, j \in [p]$.

Remark 38. For $i, j \in [p]$, such that $i \neq j$, if both

$$\frac{\partial^2 r}{\partial \theta_i \partial \theta_j} : \mathbb{T} \rightarrow \mathbb{R} \text{ and } \frac{\partial^2 r}{\partial \theta_j \partial \theta_i} : \mathbb{T} \rightarrow \mathbb{R}$$

are continuous, then

$$\frac{\partial^2 r}{\partial \theta_i \partial \theta_j} = \frac{\partial^2 r}{\partial \theta_j \partial \theta_i}.$$

Definition 14. A matrix $\mathbf{A} \in \mathbb{R}^{p \times p}$ is *positive semi-definite* if

$$\boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta} \geq 0$$

for each $\boldsymbol{\theta} \in \mathbb{R}^{p \times 1}$, and further, it is *positive definite* if

$$\boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta} > 0$$

for each $\boldsymbol{\theta} \in \mathbb{R}^{p \times 1} \setminus \{\mathbf{0}\}$.

Using the Hessian and Definition 14, we obtain an additional characterization of convexity. Namely, let $\mathbb{T} \subset \mathbb{R}^p$ be open and $r : \mathbb{T} \rightarrow \mathbb{R}$ be a function on which the Hessian $\mathbf{H}_r(\tau)$ exists for each $\tau \in \mathbb{T}$. Then, r is convex on $\mathbb{T}$ if $\mathbf{H}_r(\tau)$ is positive semi-definite, for each $\tau \in \mathbb{T}$. Further, r is strictly convex if $\mathbf{H}_r(\tau)$ is positive definite, for each $\tau \in \mathbb{T}$.

Remark 39. We also obtain another useful composition tool for verifying the convexity of a function. Namely, if $s : \mathbb{R}^p \rightarrow \mathbb{R}$ is convex, $\mathbf{A} \in \mathbb{R}^{p \times p}$, and $\mathbf{b} \in \mathbb{R}^{p \times 1}$, then

$$r(\theta) = s(\mathbf{A}\theta + \mathbf{b})$$

is convex.

1.3.4 Minimization

We consider now the original task of identifying the existence, uniqueness and computability of the set of *minimizers* of r :

$$\arg \min_{\theta \in \mathbb{T}} r(\theta).$$

1.3.4.1 Existence and uniqueness

We begin with the classic *Extreme Value Theorem* of Weierstrass.

Theorem 6 (Extreme Value Theorem). *If $\mathbb{T} \subset \mathbb{R}^p$ is compact and $r : \mathbb{T} \rightarrow \mathbb{R}$ is continuous, then*

1. Both $\min_{\theta \in \mathbb{T}} r(\theta)$ and $\max_{\theta \in \mathbb{T}} r(\theta)$ exist.
2. Both $\arg \inf_{\theta \in \mathbb{T}} r(\theta)$ and $\arg \sup_{\theta \in \mathbb{T}} r(\theta)$ are subsets of $\mathbb{T}$.

For $r : \mathbb{T} \rightarrow \mathbb{R}$ that is differentiable on an open $\mathbb{T} \subset \mathbb{R}^p$, we can define the set of *stationary points* a

$$\text{sp}_r(\mathbb{T}) = \{\tau \in \mathbb{T} : \nabla r(\tau) = 0\}.$$

The *Fermat's rule* then states that the minimizers and maximizers of r are in $\text{sp}_r(\mathbb{T})$.

That is,

$$\arg \min_{\theta \in \mathbb{T}} r(\theta) \subset \text{sp}_r(\mathbb{T}) \text{ and } \arg \max_{\theta \in \mathbb{T}} r(\theta) \subset \text{sp}_r(\mathbb{T}).$$

Unfortunately, the set $\text{sp}_r(\mathbb{T})$ contains more than the minimizers and maximizers of r , but also the local minimizers and local maximizers of r . Here, we say that $\theta^* \in \mathbb{T}$ is a *local minimizer* if there exists a $\rho > 0$, such that

$$r(\theta^*) \leq r(\theta),$$

for all $\theta \in \mathbb{B}_\rho(\theta^*)$. Similarly, we say that θ^* is a local maximizer if there exists a $\rho > 0$, such that

$$r(\theta^*) \geq r(\theta),$$

for all $\theta \in \mathbb{B}_\rho(\theta^*)$. Of course, every minimizer is a local minimizer and every maximizer is a local maximizer.

Remark 40. To distinguish minimizers and maximizers from their local counterparts, we shall often refer to them as *global minimizers* and *global maximizers*, instead.

Now, when $r : \mathbb{T} \rightarrow \mathbb{R}$ is continuous but not differentiable, we replace Fermat's rule by *Oresme's rule*, which states that the set of local minimizers of r are contained in the *critical point* set

$$\text{cp}_r(\mathbb{T}) = \{\tau \in \mathbb{T} : \bar{\mathbf{d}}_r(\tau; v) \geq 0, \text{ for all } v \in \mathbb{R}^p\},$$

where $\mathfrak{d}_r(\tau; v)$ and $\bar{\mathfrak{d}}_r(\tau; v)$ differ by the fact that $\bar{\mathfrak{d}}_r(\tau; v) = \infty$ if it does not exist in direction v . Thus, it is then implied that

$$\arg \min_{\theta \in \mathbb{T}} r(\theta) \subset \text{sp}_r(\mathbb{T}).$$

Example 24. Consider the absolute value function $r(\theta) = |\theta|$ on $\mathbb{T} = \mathbb{R}$. Here, we can check that

$$\bar{\mathfrak{d}}_r(\tau; v) = \begin{cases} v & \text{if } \tau > 0, \\ |v| & \text{if } \tau = 0, \\ -v & \text{if } \tau < 0. \end{cases}$$

Thus, at $\theta^* = 0$, $\bar{\mathfrak{d}}_r(\theta^*; v) = |v| \geq 0$, implying that θ^* is a local minimizer, which also turns out to be the unique global minimizer.

Remark 41. Notice that Fermat's rule is only applicable without additional complements when $\mathbb{T}$ is open, whereas Oresme's rule is applicable in all situations. In fact, $\text{cp}_r(\mathbb{T}) \subset \text{sp}_r(\mathbb{T})$, whenever $\mathbb{T}$ is open and r is differentiable. Further note that Oresme's rule, along with the heuristic that $\mathfrak{d}_r(\tau; v) = \infty$, whenever it does not exist, allows us to check the boundaries of a compact set, when identifying minimizers in conjunction with Theorem 6.

Exercise 34. Derive the directional derivative $\bar{\mathfrak{d}}_r(\tau; v)$ of

$$r(\theta) = \begin{cases} -\theta & \text{if } \theta \leq 0, \\ \theta^2 & \text{if } \theta > 0, \end{cases}$$

for each $\theta, v \in \mathbb{R}$, and determine whether r has a local minimizer on $\mathbb{R}$.

When using Oresme's or Fermat's rule to obtain

$$\arg \min_{\theta \in \mathbb{T}} r(\theta),$$

we must sieve through the stationary points $\text{sp}_r(\mathbb{T})$ or critical points $\text{cp}_r(\mathbb{T})$, in order to locate the relevant points of interest. If we can assume that r is convex or strictly convex then the task is made easier.

Proposition 3. *If $\mathbb{T} \subset \mathbb{R}^p$ is convex and $r : \mathbb{T} \rightarrow \mathbb{R}$ is continuous and convex, then if $\theta^* \in \mathbb{T}$ is a local minimizer, then it is also a global minimizer. Moreover, θ^* is a local minimizer if and only if $\theta^* \in \text{cp}_r(\mathbb{T})$, and*

$$\text{cp}_r(\mathbb{T}) = \arg \min_{\theta \in \mathbb{T}} r(\theta).$$

Furthermore, if r is also strictly convex, then the set $\arg \min_{\theta \in \mathbb{T}} r(\theta)$ is either empty or it contains a unique element. That is, if there is a global minimizer then it is unique.

Example 25. Determine that $\theta^* = 0$ is the unique global minimizer of $r(\theta) = |\theta| + \theta^2$ on $\mathbb{T} = \mathbb{R}$, by demonstrating that it is strictly convex. Then, determine whether or not we conclude that the function

$$s(\theta) = \max(0, \theta)$$

has a unique global minimizer on $\mathbb{R}$.

1.3.5 Computing minimizers with the MM algorithm

In many practical situations, we cannot easily compute $\theta^* \in \mathbb{T}$ that solves the *Fermat condition*

$$\nabla r(\theta^*) = 0$$

or that satisfies *Oresme's condition*

$$\bar{\mathfrak{d}}_r(\theta^*; v) \geq 0, \text{ for all } v \in \mathbb{R}^p.$$

Example 26. Consider the function

$$r(\theta) = \log(1 + \exp(2\theta)) + \log(1 + \exp(4\theta)) - \theta$$

on $\mathbb{R}$. The Fermat condition for r is

$$\begin{aligned} \nabla r(\tau) &= \frac{2 \exp(2\tau)}{1 + \exp(2\tau)} + \frac{4 \exp(4\tau)}{1 + \exp(4\tau)} - 1 \\ &= \frac{5e^{6\tau} + 3e^{4\tau} + e^{2\tau} - 1}{(1 + e^{2\tau})(1 + e^{4\tau})} = 0, \end{aligned}$$

which requires the solution of a 6th order polynomial, which is not solvable in closed form (in general).

In order to solve problems such as in Example 26, we require an iterative tool. More specifically, we require an *algorithm* $\mathfrak{A} : \mathbb{T} \rightarrow \mathbb{T}$, together with some *initialization* $\theta^{(0)}$, that generates a sequence of *iterates* $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$

$$\theta^{(\nu)} = \mathfrak{A}(\theta^{(\nu-1)}),$$

for each $\nu \in \mathbb{N}$. Here, we hope that the sequence of iterates converge in some sense to an element θ^* in $\text{sp}_r(\mathbb{T})$, $\text{cp}_r(\mathbb{T})$, or ideally $\arg \min_{\theta \in \mathbb{T}} r(\theta)$.

Towards this goal, we define a function $(\theta, \tau) \in \mathbb{T}^2 \mapsto \mathbf{m}(\theta; \tau)$ to be a *majorizer* of $r(\theta)$ if the following two conditions are satisfied:

1. For each $\theta \in \mathbb{T}$, $r(\theta) = \mathbf{m}(\theta; \theta)$, (*touching*)
2. For each $\theta, \tau \in \mathbb{T}$, $r(\theta) \leq \mathbf{m}(\theta; \tau)$, (*majorizing*)

Using the majorizer, we propose the *majorization-minimization (MM) algorithm*, via the iteration rule:

$$\theta^{(\nu)} = \mathfrak{A}\left(\theta^{(\nu-1)}\right) \in \arg \min_{\theta \in \mathbb{T}} \mathbf{m}\left(\theta; \theta^{(\nu-1)}\right).$$

Remark 42. Here, the goal is to construct, *majorizer* functions that are computationally simpler than the function that it *majorizes*. For example, if r is not convex or differentiable, then we can pick $\mathbf{m}(\cdot; \tau)$ that is convex or differentiable at each $\tau \in \mathbb{T}$. More importantly, if r does not admit a closed form solution to the Fermat or Oresme conditions, then we may pick $\mathbf{m}(\cdot; \tau)$ to be solvable in closed, with a unique solution.

The first and most important property of the MM algorithm is that the sequence $(r(\theta^{(\nu)}))_{\nu \in \mathbb{N}}$ is decreasing, which is desirable for an algorithm that seeks to find the

minimizers of r . This property is easy to demonstrate by observing that

$$\begin{aligned} r\left(\theta^{(\nu)}\right) &= \mathbf{m}\left(\theta^{(\nu)}; \theta^{(\nu)}\right) \text{ (touching)} \\ &\geq \mathbf{m}\left(\theta^{(\nu+1)}; \theta^{(\nu)}\right) \text{ (definition of } \mathfrak{A}) \\ &\geq r\left(\theta^{(\nu+1)}\right) \text{ (majorizing).} \end{aligned}$$

In fact, we can do much better than this result. Namely, we can show that if r and $\mathbf{m}$ are both differentiable, then every limit point of $\left(\theta^{(\nu)}\right)_{\nu \in \mathbb{N}}$ is in $\text{sp}_r(\mathbb{T})$.

Proposition 4. *If $r : \mathbb{T} \rightarrow \mathbb{R}$ is differentiable on $\mathbb{T}$ and $\mathbf{m} : \mathbb{T}^2 \rightarrow \mathbb{R}$ is a majorizer that is differentiable on $\mathbb{T}$ in the sense that $\mathbf{m}(\cdot; \tau)$ is differentiable on $\mathbb{T}$ and $\mathbf{m}(\tau; \cdot)$ is continuous on $\mathbb{T}$, for each $\tau \in \mathbb{T}$, then every cluster point of an MM algorithm sequence $\left(\theta^{(\nu)}\right)_{\nu \in \mathbb{N}}$ is in $\text{sp}_r(\mathbb{T})$.*

Alternatively, if r has directional derivative in all directions $v \in \mathbb{R}^p$ and $\mathbf{m}(\cdot; \tau)$ has directional derivatives in all directions $v \in \mathbb{R}^p$, such that

$$\mathfrak{d}_{\mathbf{m}(\cdot; \tau)}(\tau; v) = \mathfrak{d}_r(\tau; v),$$

for each $\tau \in \mathbb{T}$, then every cluster point of an MM algorithm sequence $\left(\theta^{(\nu)}\right)_{\nu \in \mathbb{N}}$ is in $\text{cp}_r(\mathbb{T})$.

Exercise 35. Prove the first statement of Proposition 4. Hint: by Properties 1 and 2 of a majorizer, if $\theta^* \in \mathbb{T}$ minimizes $\mathbf{m}(\cdot; \theta^*)$, then it also minimizes $\mathbf{m}(\cdot; \theta^*) - r(\cdot)$.

Remark 43. Notice that Proposition 4 says nothing about where the MM algorithm sequence has a cluster point or not. To guarantee that a cluster point exists, we

must provide conditions under which the sequence $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ is bounded. One such addition set of conditions is that the function $r : \mathbb{R}^p \rightarrow \bar{\mathbb{R}}$ is *coercive*, in the sense that

$$\lim_{\|\theta\| \rightarrow \infty} r(\theta) = \infty,$$

and is *proper*, in the sense that $-\infty < r(\theta)$, for all $\theta \in \mathbb{R}^p$, and that $r \neq \infty$. The intuition here is that the level set

$$\{\theta \in \mathbb{R}^p : r(\theta) \leq a\}$$

is bounded, for each $a \in \mathbb{R}$, when r is coercive, and thus the sequence $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ is bounded, since $r(\theta^{(\nu)}) \leq r(\theta^{(0)})$, for each $\nu \in \mathbb{N}$. Another set of conditions is that r and $\mathbf{m}$ are both differentiable (in the sense of proposition) on $\mathbb{S} \supset \mathbb{T}$, where $\mathbb{T}$ is compact and $\mathbb{S}$ is open.

Example 27. Let $\mathbb{T}$ be an open set and $r : \mathbb{T} \rightarrow \mathbb{R}$ be *twice differentiable* on $\mathbb{T} \subset \mathbb{R}^p$ (i.e., the Hessian $\mathbf{H}_r(\tau)$ exists for every $\tau \in \mathbb{T}$). If there exists a matrix $\mathbf{B} \in \mathbb{R}^{p \times p}$, such that $\mathbf{B} - \mathbf{H}_r(\tau)$ is positive definite, for every $\tau \in \mathbb{T}$, then

$$\mathbf{m}(\theta; \tau) = r(\tau) + \langle \nabla r(\tau), \theta - \tau \rangle + \frac{1}{2}(\theta - \tau)^\top \mathbf{B}(\theta - \tau)$$

is a majorizer of r , on $\mathbb{T}$. We call this majorization construction, the *quadratic upper bound*.

Example 28. Proceeding from Example 26, we observe that

$$\begin{aligned} \mathbf{H}_r(\tau) &= 4 \frac{\exp(2\tau)}{1 + \exp(2\tau)} \left(1 - \frac{\exp(2\tau)}{1 + \exp(2\tau)} \right) + 16 \frac{\exp(4\tau)}{1 + \exp(4\tau)} \left(1 - \frac{\exp(4\tau)}{1 + \exp(4\tau)} \right) \\ &\leq 4 \times \frac{1}{4} + 16 \times \frac{1}{4} = 5 = \mathbf{B}, \end{aligned}$$

since $\varpi(1 - \varpi) \leq 1/4$, for every

$$\varpi = \frac{\exp(a)}{1 + \exp(a)} \in (0, 1),$$

where $a \in \mathbb{R}$. Thus, we can use the quadratic upper bound to majorize

$$r(\theta) = \log(1 + \exp(2\theta)) + \log(1 + \exp(4\theta)) - \theta$$

at $\tau \in \mathbb{T} = \mathbb{R}$, by

$$\mathbf{m}(\theta; \tau) = r(\tau) + \left[\frac{5e^{6\tau} + 3e^{4\tau} + e^{2\tau} - 1}{(1 + e^{2\tau})(1 + e^{4\tau})} \right] (\theta - \tau) + \frac{5}{2} (\theta - \tau)^2,$$

which yields the algorithm

$$\begin{aligned} \theta^{(\nu)} &= \mathfrak{A}(\theta^{(\nu-1)}) \\ &= \arg \min_{\theta \in \mathbb{R}} \mathbf{m}(\theta; \theta^{(\nu-1)}) \\ &= \theta^{(\nu-1)} - \frac{1}{5} \left[\frac{5e^{6\theta^{(\nu-1)}} + 3e^{4\theta^{(\nu-1)}} + e^{2\theta^{(\nu-1)}} - 1}{(1 + e^{2\theta^{(\nu-1)}})(1 + e^{4\theta^{(\nu-1)}})} \right]. \end{aligned}$$

Notice that since $\varpi \in (0, 1)$, for each τ , $\mathbf{H}_r(\tau) > 0$, and thus r is strictly convex on $\mathbb{R}$. Thus, since Proposition 4 guarantees that every limit point of the sequence $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ is in $\text{sp}_r(\mathbb{R})$, it must be true that every subsequence converges to the unique global minimum of r . It is not difficult to see that r is also coercive, and thus, the MM algorithm sequence is guaranteed to converge to the global minimum of r , in this scenario!

1.3.6 Sample average approximation

Let us now reintroduce the IID sequence of random functions $(X_i)_{i \in \mathbb{N}}$, each mapping from the probability space $(\Omega, \mathcal{F}, P)$ to the measurable space $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$, where $\mathbb{X} \subset \mathbb{R}^d$, and each X_i has the same image measure P_X , on the space $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$.

We define $\mathbb{T} \subset \mathbb{R}^p$, and let the *loss function* $\ell_\theta(x)$ be a *Caratheodory function* in the sense that $\ell_\theta(\cdot) : \mathbb{X} \rightarrow \mathbb{R}$ is a measurable function on $(\mathbb{X}, \mathcal{B}(\mathbb{X}), P_X)$, for each $\theta \in \mathbb{T}$, and $\ell_{(\cdot)}(x) : \mathbb{T} \rightarrow \mathbb{R}$ is continuous for each $x \in \mathbb{X}$. We shall henceforth always assume the loss function is Caratheodory, unless stated otherwise. Similarly, we say that the *risk function*

$$\begin{aligned} r_n(\theta) &= r_n(\theta; \omega) \\ &= \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i(\omega)) \end{aligned}$$

is Caratheodory on $(\Omega, \mathcal{F}, P)$ and $\mathbb{T}$, if it is continuous in θ and measurable in ω .

We shall say that $r_n(\theta)$ converges to some non-random function $\bar{r} : \mathbb{T} \rightarrow \mathbb{R}$, almost surely, as $n \rightarrow \infty$, uniformly on $\mathbb{T}$, if

$$P \left(\omega \in \Omega : \lim_{n \rightarrow \infty} \sup_{\theta \in \mathbb{T}} |r_n(\theta, \omega) - \bar{r}(\theta)| = 0 \right) = 1.$$

Or alternatively, for all $\omega \in \mathbb{A}$, such that $P(\mathbb{A}) = 1$, we have

$$\lim_{n \rightarrow \infty} \sup_{\theta \in \mathbb{T}} |r_n(\theta, \omega) - \bar{r}(\theta)| = 0.$$

Proposition 5. *If $r_n(\theta; \omega)$ is Caratheodory and converges to a continuous function $\bar{r}$, almost surely, as $n \rightarrow \infty$, uniformly on a compact $\mathbb{T}$, then*

$$r_n(\hat{\theta}_n; \omega) \rightarrow \bar{r}(\theta_*),$$

almost surely, where $(\hat{\theta}_n(\omega))_{n \in \mathbb{T}}$ is a sequence with

$$\hat{\theta}_n(\omega) \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta; \omega), \text{ and } \theta_* \in \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta).$$

Remark 44. For sequences, $(\theta_n)_{n \in \mathbb{N}}$ in $\mathbb{T}$, and $(r_n)_{n \in \mathbb{N}}$ of functions from $\mathbb{T}$ to $\mathbb{R}$, we say that $(r_n)_{n \in \mathbb{N}}$ continuously converges to $\bar{r}$ if

$$r_n(\theta_n) \rightarrow \bar{r}(\theta),$$

whenever $\theta_n \rightarrow \theta$, for each $\theta \in \mathbb{T}$, as $n \rightarrow \infty$. If $\mathbb{T}$ is compact then

$$\lim_{n \rightarrow \infty} \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)| = 0,$$

implies that r_n continuously converges to $\bar{r}$.

Exercise 36. Prove Proposition 5 is true.

Proposition 6. Under the same conditions as Proposition 5, it is also true that any sequence $(\hat{\theta}_n)_{n \in \mathbb{N}}$ converges to the set $\arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta)$, in the sense that

$$\inf_{\theta_* \in \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta)} \left\| \hat{\theta}_n(\omega) - \theta_* \right\| \rightarrow 0,$$

almost surely, as $n \rightarrow \infty$.

Remark 45. The conclusion Proposition 6 is usually referred to as *consistency*. That is, we say that $\hat{\theta}_n$ is a *consistent* estimator for θ_* . Some authors will reserve the terminology for the case where $\theta_* = \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta)$ is unique, but also allow for convergence in probability. In such a case, we say that $\hat{\theta}_n$ is a consistent estimator of θ_* if

$$\hat{\theta}_n(\omega) \rightarrow \theta_*,$$

almost surely (or in probability), as $n \rightarrow \infty$.

Remark 46. In the optimization literature, $r_n(\cdot; \omega)$ is known as the *sample average approximation (SAA)* of $\bar{r}$, and thus statistics $\hat{\theta}_n(\omega) \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta; \omega)$ are called SAA estimators of $\theta_* \in \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta)$. In the econometrics literature, since estimators are often called *extremum estimators*, since they make the function r_n take its most extreme value, and in the statistics literature, such estimators are usually called *M-estimators*, where “M” stands for minimum (or maximum, should it be the case). You will also often see the terminology of *minimum contrast estimators* and *empirical risk minimizers* in the machine learning literature. Here, the words contrast can be taken to mean any general data-dependent function of interest.

Remark 47. Recall that we have the fact that for an IID random variable sequence $(X_i)_{i \in [n]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, where each X_i has the same image measure as X , *Markov’s strong law of large numbers (SLLN)* states that if $\mathbb{E}|X| < \infty$ then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mathbb{E}(X), \text{ a.s..}$$

It may be tempting to apply the SLLN for each $\theta \in \mathbb{T}$, to get

$$r_n(\theta; \omega) = \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i(\omega)) \rightarrow \mathbb{E}[\ell_\theta(X)] = \bar{r}(\theta),$$

almost surely, as $n \rightarrow \infty$, and conclude that that this is sufficient for guaranteeing the uniform convergence required by Propositions 5 and 6. Unfortunately, this is not the case because, in general, the union of an uncountable number of events of probability zero need not have probability zero.

Example 29. On the measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})$, the set of the form $\{c\}$ has measure zero, for each $c \in \mathbb{R}$, since $\text{Leb}(\{c\}) = c - c = 0$. But for $a < b$, we can write

$$(a, b) = \bigcup_{c \in (a, b)} \{c\},$$

which is an uncountable union of sets of measure zero. But, $\text{Leb}((a, b)) = b - a > 0$, since $a < b$.

Theorem 7 (Uniform strong law of large numbers). *Let $\mathbb{T} \subset \mathbb{R}^p$ be compact and nonempty and let*

$$r_n(\theta, \omega) = \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i),$$

where $\ell_\theta(x)$ is Caratheodory function constructed and $(X_i)_{i \in [n]}$ is an IID sequence of random functions mapping from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{X}, \mathcal{B}(\mathbb{X}), P_X)$. Further suppose that for each $\theta \in \mathbb{T}$,

$$|\ell_\theta(x)| \leq \bar{\ell}_\theta(x), \tag{1.3.4}$$

for all $n \in \mathbb{N}$, and almost all $x \in \mathbb{X} \subset \mathbb{R}^d$, where $\int_{\mathbb{X}} \bar{\ell}_\theta(x) dP_X(x) < \infty$. Then, we can conclude that

$$P\left(\omega \in \Omega : \lim_{n \rightarrow \infty} \sup_{\theta \in \mathbb{T}} |r_n(\theta, \omega) - \bar{r}(\theta)| = 0\right) = 1,$$

where $\bar{r}(\theta) = E[\ell_\theta(X)]$ is a continuous function, in $\theta \in \mathbb{T}$.

Example 30. Let $(X_i)_{i \in [n]}$ be an IID sequence of random variables on $(\Omega, \mathcal{F}, P)$, with image measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$, where it is known that

$$\theta_* \in \arg \min_{\theta \in \mathbb{T}} \underbrace{E[|X - \theta|]}_{=\bar{r}(\theta)}$$

is in some compact set $\mathbb{T} = [a, b]$, where $a < b$. Then, using loss function of the form $\ell_\theta(x) = |x - \theta|$, we construct the empirical risk function

$$\begin{aligned} r_n(\theta) &= \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i) \\ &= \frac{1}{n} \sum_{i=1}^n |X_i - \theta|, \end{aligned}$$

from which we can define the sequence $(\hat{\theta}_n)_{n \in \mathbb{N}}$ of estimators

$$\hat{\theta}_n \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta).$$

To determine that this sequence converges to θ_* , we require an application of the uniform SLLN. This requires that we bound $\ell_\theta(x)$ by some $\bar{\ell}_\theta(x)$, for almost all $x \in \mathbb{R}$. To do this, write

$$\ell_\theta(x) = |x - \theta| \leq |x| + |\theta| = \bar{\ell}_\theta(x),$$

where $\int_{\mathbb{X}} \bar{\ell}_\theta(x) dP_X < \infty$ if $\int_{\mathbb{X}} |x| dP_X = E|X| < \infty$. So under the assumption that $E|X| < \infty$, the uniform SLLN applies, and we have the fact that $r_n(\theta)$ almost surely converges to $\bar{r}(\theta) = E[|X - \theta|]$, as $n \rightarrow \infty$, uniformly. Thus, by Propositions 5 and 6, we can conclude that

$$r_n(\hat{\theta}_n) \rightarrow E[|X - \theta^*|],$$

and

$$\inf_{\theta_* \in \arg \min_{\theta \in \mathbb{T}} E[|X - \theta|]} \left\| \hat{\theta}_n - \theta_* \right\| \rightarrow 0,$$

almost surely, as $n \rightarrow \infty$.

Finally, we wish to obtain an expression for $\arg \min_{\theta \in \mathbb{T}} r_n(\theta)$. To do so, we consider Oresme's rule, which states that if $\hat{\theta}_n$ is a local minimizer of $r_n(\theta)$, then for all $v \in \mathbb{R}$

$$\bar{\delta}_{r_n}(\hat{\theta}_n; v) \geq 0.$$

We can firstly observe that for $\ell_\theta(x) = |x - \theta|$

$$\bar{\delta}_{\ell_{(\cdot)}(x)}(\tau; v) = \begin{cases} v & \text{if } \tau > x, \\ |v| & \text{if } \tau = x, \\ -v & \text{if } \tau < x. \end{cases}$$

So for any $v \in \mathbb{R}$,

$$\begin{aligned} \bar{\delta}_{r_n}(\hat{\theta}_n; v) &= \frac{1}{n} \sum_{i=1}^n \bar{\delta}_{\ell_{(\cdot)}(X_i)}(\hat{\theta}_n; v) \\ &= \frac{1}{n} \sum_{i=1}^n v [\hat{\theta}_n > X_i] + \frac{1}{n} \sum_{i=1}^n |v| [\hat{\theta}_n = X_i] - \frac{1}{n} \sum_{i=1}^n v [\hat{\theta}_n < X_i]. \end{aligned} \tag{1.3.5}$$

Suppose that each element of $(X_i)_{i \in [n]}$ is unique, and let $X_{(1)} < X_{(2)} < \dots < X_{(n)}$ be an *increasing ordering* of $(X_i)_{i \in [n]}$. Then, if $n = 2m + 1$ is odd ($m \in \mathbb{N}$), then we can take $\hat{\theta}_n = X_{(m+1)}$. Then, the first and third terms of (1.3.5) are equal to $(m/n)v$ and $-(m/n)v$, respectively, and cancel out. The second term is then equal to $|v|/n \geq 0$, and hence Oresme's rule implies $\hat{\theta}_n = X_{(m+1)}$ is a in $\text{cp}_{r_n}(\mathbb{T})$.

Alternatively, if we suppose that $n = 2m$ is even, then we can take any $X_{(m)} < \hat{\theta}_n < X_{(m+1)}$. Again, we see that the first and third terms of (1.3.5) cancel, but now the second term is also equal to zero, and thus $\bar{\delta}_{r_n}(\hat{\theta}_n; v) = 0$, which also satisfies

Oresme's rule. Finally, to strengthen the conclusion of Oresme's rule, we simply observe that $r_n(\theta)$ is convex and thus every element of $\text{cp}_{r_n}(\mathbb{T})$ is a global minimizer by Proposition 3.

Exercise 37. Use Oresme's rule to deduce the form of $\hat{\theta}_n$, in the problem above, when n is odd but the elements of $(X_i)_{i \in [n]}$ are not unique. You can use the notation $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ for a *non-decreasing* ordering of $(X_i)_{i \in [n]}$.

1.3.7 Bibliographical notes

To come!

Chapter 2

Density estimation and regression analysis

2.1 Density estimation

2.1.1 Density functions

Let $(X_i)_{i \in [n]}$ be an IID sequence of random functions from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$, where $\mathbb{X} \subset \mathbb{R}^d$. Here, we suppose that each X_i has the same image measure P_X of X , for $i \in [n]$. We shall interchangeably use the random sequence terminology, with the much more common terminology of referring to $(X_i)_{i \in [n]}$ as a *sample* or a *data set*.

We shall make the assumption that P_X is absolutely continuous with respect to either the Lebesgue measure or an appropriate counting measure, and say, as is typical, that X is either a *continuous* or *discrete* random variable, respectively.

Of course, there are many ways of characterizing the probability measure of P_X . For example, we have already handled multiple situations where we characterize X

via its CDF:

$$P_X(X \leq x) = F(x),$$

when $\mathbb{X} \subset \mathbb{R}$, for example. In this section, we will primarily concern ourselves with the characterization of X via its Radon–Nikodym derivative $p_X : \mathbb{X} \rightarrow \mathbb{R}$, either (when it exists) with respect to the Lebesgue measure (in which case it is referred to as the *PDF* of X) or with respect to a counting measure (in which case it is referred to as the *PMF*). In either case, we shall refer to p_X as the density of X .

Due to the definition of a probability measure, the density function being required to satisfy

$$P_X(\mathbb{A}) = \int_{\mathbb{A}} p_X(x) dM(x),$$

(where we write M generically for both Lebesgue and counting measure) must inherit certain properties, most important of which are that

1. $p_X(x) \geq 0$ for almost all $x \in \mathbb{X}$, and
2. $\int_{\mathbb{X}} p_X(x) dM(x) = 1$.

Exercise 38. Prove that if P_X is absolutely continuous with respect to a counting measure, then we also have the fact that $p_X(x) \leq 1$, for all $x \in \mathbb{X}$.

2.1.2 Density estimation

In many statistical settings, the IID data set $(X_i)_{i \in [n]}$ is known to have a probability density function p_X from some *parametric* family

$$\mathbb{P} = \{p : \mathbb{X} \rightarrow \mathbb{R} : p(x) = p_\theta(x), \theta \in \mathbb{T}\}$$

where $\mathbb{T} \subset \mathbb{R}^p$ is usually referred to as the *parameter space* or sometimes the *index set* of $\mathbb{P}$. We shall also often write $p_\theta(x) = p(x; \theta)$, when it is more convenient to do so.

Typically, although we know that $p_X \in \mathbb{P}$, that is $p_X = p_{\theta_0}$, for some $\theta_0 \in \mathbb{T}$, we do not know the value of θ_0 . Here, we shall often refer to θ_0 as the *true parameter* or *underlying parameter* of p_X , or the *data-generative parameter* of $(X_i)_{i \in [n]}$, the terminology of which arises from the reference to P_X as the *data generating process* (*DGP*). This section is devoted to the estimation of θ_0 , a property of P_X , via some statistic $\hat{\theta}_n$.

2.1.2.1 Distances and divergences

Let

$$\mathcal{P} = \{P : P \text{ is a probability measure on } \mathbb{X}, P \ll M\}$$

denote the set of all probability measures on $\mathbb{X}$ that are absolutely continuous with respect to M . Then for any $P, Q \in \mathcal{P}$, there exists densities p and q (we shall also write $p \in \mathcal{P}$, when $P \in \mathcal{P}$). Like any other set of things, we can attempt to endow $\mathcal{P}$ with a topology via some notion of a metric. Towards this end, there are a number of choices.

Example 31. The following distance functions $\text{dist}(P, Q) = \text{dist}(p, q)$ can be used as metrics on $\mathcal{P}$:

- The Kolmogorov distance: $\sup_{x \in \mathbb{X}} |F_P(x) - F_Q(x)|$, where F_P and F_Q are the corresponding CDFs of P and Q .
- The (essential) supremum distance:

$$\inf \{c \in \mathbb{R}_{\geq 0} : |p(x) - q(x)| \leq c, \text{ for almost every } x \in \mathbb{X}\}.$$

- The *Hellinger* distance:

$$H(P, Q) = (1/2) \int_{\mathbb{X}} \left(\sqrt{p(x)} - \sqrt{q(x)} \right)^2 dM(x) = 1 - \int_{\mathbb{X}} \sqrt{p(x)q(x)} dM(x).$$

- The *total variation* distance:

$$TV(P, Q) = \sup_{\mathbb{A} \in \mathcal{B}(\mathbb{X})} |P(\mathbb{A}) - Q(\mathbb{A})| = \frac{1}{2} \int_{\mathbb{X}} |p(x) - q(x)| dM(x).$$

- The L_q -norm distance: $L_q(p, q) = \left(\int_{\mathbb{X}} |p(x) - q(x)|^q dM(x) \right)^{1/q}$.

Exercise 39. It is not difficult to see that the total variation distance and the L_q -norm distance are related. Show that the Hellinger and total variation distances are also related via the inequalities:

$$[H(P, Q)]^2 \leq TV(P, Q) \leq \sqrt{2}H(P, Q).$$

Since it is known that $p_X = p_{\theta_0}$, for some $\theta_0 \in \mathbb{T}$, it is tempting to attempt to estimate θ_0 by solving optimization problem:

$$\arg \min_{\theta \in \mathbb{T}} \text{dist}(p_{\theta_0}, p_{\theta}),$$

which, unfortunately, generally depends on the unknown term θ_0 . However, since we are only concerned with estimation with respect to θ , we can hope that any term involving θ_0 is decoupled from terms involving θ , or is of the form

$$\mathbb{E} [\text{term}_\theta (X)] = \int_{\mathbb{X}} \text{term}_\theta (x) p_{\theta_0} (x) dM (x)$$

which we can estimate, using our sample $(X_i)_{i \in [n]}$, via the sample average

$$\frac{1}{n} \sum_{i=1}^n \text{term}_\theta (X_i).$$

Example 32. Let us consider the square of the L_2 -norm distance between p and q , of the form:

$$L_2^2 (p, q) = \int_{\mathbb{X}} |p (x) - q (x)|^2 dM (x),$$

which we can expand to

$$L_2^2 (p, q) = \int_{\mathbb{X}} |p (x)|^2 dM (x) - 2 \int_{\mathbb{X}} p (x) q (x) dM (x) + \int_{\mathbb{X}} |q (x)|^2 dM (x),$$

Thus, we observe that $\arg \min_{\theta \in \mathbb{T}} L_2^2 (p_{\theta_0}, p_\theta)$ is equivalent to

$$\arg \min_{\theta \in \mathbb{T}} \left[-2 \int_{\mathbb{X}} p_\theta (x) p_{\theta_0} (x) dM (x) + \int_{\mathbb{X}} |p_\theta (x)|^2 dM (x) \right],$$

which can be estimate via the empirical statistic

$$\arg \min_{\theta \in \mathbb{T}} \left[-\frac{2}{n} \sum_{i=1}^n p_\theta (X_i) + \int_{\mathbb{X}} |p_\theta (x)|^2 dM (x) \right]. \quad (2.1.1)$$

Computation of (2.1.1) is usually referred to as *least-squares density estimation*. We observe that even though least-squares density estimation does not require knowledge of θ_0 , it is complicated by the integral expression $\int_{\mathbb{X}} |p_\theta (x)|^2 dM (x)$, which must typically be numerically estimated.

Upon inspection, we observe that we do not need many of the properties that a metric provides. Namely, we do not need symmetry, nor the triangle inequality, and only require that we have some function $D(P, Q) = D(p, q)$ such that $D(P, Q) = 0$ if and only if $P = Q$, almost everywhere, with respect to M . We thus turn our attention to *divergences*, which are a class of functions $D : \mathcal{P} \times \mathcal{P} \rightarrow \mathbb{R}_{\geq 0}$, that are metric like, and have the properties:

1. $D(P, Q) \geq 0$, and
2. $D(P, Q) = 0$ if and only if $P = Q$, almost everywhere.

Definition 15. Let $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ be a strictly convex and differentiable function. Then, the *Bregman divergence* define by f can be is

$$B_f(p, q) = \int_{\mathbb{X}} f(p(x)) - f(q(x)) - f'(q(x))(p(x) - q(x)) dM(x).$$

Example 33. For $f(u) = u \log u - u + 1$, we have

$$f'(u) = \log u - 1,$$

which yields

$$\begin{aligned} B_f(p, q) &= \int_{\mathbb{X}} p(x) \log(p(x)) - p(x) \log(q(x)) dM(x) \\ &= \int_{\mathbb{X}} p(x) \log\left(\frac{p(x)}{q(x)}\right) dM(x). \end{aligned}$$

We common name for the divergence above is the *Kullback–Leibler (KL) divergence*, which we shall write as $KL(p, q)$.

Exercise 40. Show that $B_f(p, q) = L_2^2(p, q)$, when $f(u) = u^2 - 2u + 1$.

2.1.2.2 Maximum likelihood estimation

Let us now consider the problem of computing

$$\arg \min_{\theta \in \mathbb{T}} \text{KL} (p_{\theta_0}, p_{\theta}) ,$$

which we can observe is equivalent to the risk minimizer

$$\theta_* = \theta_0 \in \arg \min_{\theta \in \mathbb{T}} \underbrace{-\mathbb{E} [\log (p_{\theta} (X))]}_{\bar{r}(\theta)},$$

since

$$\begin{aligned} \text{KL} (p_{\theta_0}, p_{\theta}) &= \int_{\mathbb{X}} p_{\theta_0} (x) \log (p_{\theta_0} (x)) - p_{\theta_0} (x) \log (p_{\theta} (x)) \, dM \\ &= \text{term}_{\theta_0} - \int_{\mathbb{X}} p_{\theta_0} (x) \log (p_{\theta} (x)) \, dM \\ &= \text{term}_{\theta_0} - \mathbb{E} [\log (p_{\theta} (X))] . \end{aligned}$$

Thus, we can estimate θ_* by the empirical risk minimizer

$$\hat{\theta}_n = \arg \min_{\theta \in \mathbb{T}} r_n (\theta) ,$$

based on the loss function $\ell_{\theta} (x) = -\log (p_{\theta} (x))$, where we can write

$$\begin{aligned} r_n (\theta) &= \frac{1}{n} \sum_{i=1}^n \ell_{\theta} (X_i) \\ &= -\frac{1}{n} \sum_{i=1}^n \log (p_{\theta} (X_i)) . \end{aligned}$$

We that in the statistical literature, $\hat{\theta}_n$ is usually characterized as

$$\hat{\theta}_n = \arg \max_{\theta \in \mathbb{T}} \sum_{i=1}^n \log (p_{\theta} (X_i)) ,$$

which gives it its name of *maximum likelihood estimator (MLE)*, since

$$\arg \max_{\theta \in \mathbb{T}} \sum_{i=1}^n \log (p_{\theta}(X_i)) = \arg \max_{\theta \in \mathbb{T}} \prod_{i=1}^n p_{\theta}(X_i), \quad (2.1.2)$$

and $\prod_{i=1}^n p_{\theta}(x_i)$ is the *likelihood function* of $(X_i)_{i \in [n]}$, in the sense that it is the density function corresponding to the product measure $\prod_{i=1}^n P_X$. Although we will often say MLE, we can also conceptualize $\hat{\theta}_n$ as the minimum KL divergence estimator, which has its philosophical benefits.

Exercise 41. Show that (2.1.2) is true, using the fact that $\log : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is a strictly increasing function.

Example 34. Let us consider an IID sample $(X_i)_{i \in [n]}$ from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, with *normal (Gaussian)* measure P_X , characterized by density function p_{θ_0} in

$$\mathbb{P} = \left\{ p_{\theta} : p_{\theta}(x) = \phi(x; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp \left[-\frac{(\mathbf{x} - \boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2} \right], \theta \in \mathbb{T} \right\},$$

where $\theta = (\boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \mathbb{T}$,

$$\mathbb{T} = \{(\boldsymbol{\mu}, \boldsymbol{\Sigma}) : \boldsymbol{\mu} \in \mathbb{R}^{d \times 1}, \boldsymbol{\Sigma} \in \text{SPD}_d\}.$$

Here, SPD_d denotes the set of *symmetric positive matrices (SPD)* in $\mathbb{R}^{d \times d}$ and we say that $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the *mean* and *covariance* matrices. We can estimate θ_0 via the MLE, defined as

$$\hat{\theta}_n = \arg \min_{\theta \in \mathbb{T}} r_n(\theta),$$

where

$$\begin{aligned} r_n(\theta) &= -\frac{1}{n} \sum_{i=1}^n \log \phi(X_i; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \\ &= \frac{d}{2} \log(2\pi) + \frac{1}{2} \log |\boldsymbol{\Sigma}| + \frac{1}{2n} \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu}). \end{aligned}$$

2.1.2.3 Some matrix calculus

In order to progress with Example 34, we require solving the Fermat condition

$$\nabla r_n(\theta^*) = 0.$$

This requires us to take partial derivatives of r_n with respect to the mean vector $\boldsymbol{\mu} \in \mathbb{R}^{d \times 1}$ and the covariance matrix $\boldsymbol{\Sigma}$. Such a task requires the following matrix differentiation rules:

- For matrices $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{d \times 1}$ and $\mathbf{A} \in \mathbb{R}^{d \times d}$,

$$\frac{\partial}{\partial \mathbf{y}} \left\{ (\mathbf{x} - \mathbf{y})^\top \mathbf{A} (\mathbf{x} - \mathbf{y}) \right\} = -2\mathbf{A} (\mathbf{x} - \mathbf{y}). \quad (2.1.3)$$

- For matrices $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{d \times 1}$ and non-singular $\mathbf{A} \in \mathbb{R}^{d \times d}$,

$$\frac{\partial}{\partial \mathbf{A}} \left\{ \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{y} \right\} = -\mathbf{A}^{-\top} \mathbf{x} \mathbf{y}^\top \mathbf{A}^{-\top}.$$

- For non-singular matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$,

$$\frac{\partial}{\partial \mathbf{A}} \log |\mathbf{A}| = \mathbf{A}^{-1}.$$

Remark 48. Notice that as above, and in Example 34, that we use $|\cdot|$ to denote the *determinant* of a matrix. When this has potential of causing confusion, we can also use the notation $\det$.

Example 35. Continuing from Example 34, we require the derivative of

$$r_n(\theta) = \frac{d}{2} \log(2\pi) + \frac{1}{2} \log |\Sigma| + \frac{1}{2n} \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}).$$

Since differentiation is linear, we can isolate the required partial derivatives as

$$\frac{\partial}{\partial \Sigma} \log |\Sigma| = \Sigma^{-1},$$

$$\frac{\partial}{\partial \Sigma} \left\{ (\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) \right\} = \Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) (\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1},$$

and

$$\frac{\partial}{\partial \boldsymbol{\mu}} \left\{ (\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) \right\} = -2\Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}).$$

We can now set up and solve the Fermat condition

$$\nabla r_n(\theta) = \begin{bmatrix} \frac{\partial}{\partial \boldsymbol{\mu}} r_n(\theta) \\ \frac{\partial}{\partial \Sigma} r_n(\theta) \end{bmatrix} = \mathbf{0},$$

where

$$\begin{aligned} \frac{\partial}{\partial \boldsymbol{\mu}} r_n(\theta) &= \frac{1}{2n} \sum_{i=1}^n \frac{\partial}{\partial \boldsymbol{\mu}} \left\{ (\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) \right\} \\ &= \frac{1}{2n} \sum_{i=1}^n \left\{ -2\Sigma^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) \right\} \\ &= -\frac{1}{n} \Sigma^{-1} \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu}), \end{aligned}$$

and thus

$$\begin{aligned}
-\frac{1}{n}\Sigma^{-1}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu}) = \mathbf{0} &\implies -\frac{1}{n}\Sigma^{-1}\sum_{i=1}^n\mathbf{X}_i + \frac{1}{n}\Sigma^{-1}\sum_{i=1}^n\boldsymbol{\mu} = \mathbf{0} \\
&\implies \Sigma^{-1}\boldsymbol{\mu} = \Sigma^{-1}\frac{1}{n}\sum_{i=1}^n\mathbf{X}_i \\
&\implies \boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^n\mathbf{X}_i.
\end{aligned}$$

Next,

$$\begin{aligned}
\frac{\partial}{\partial \Sigma}r_n(\theta) &= \frac{1}{2}\frac{\partial}{\partial \Sigma}\log|\Sigma| + \frac{1}{2n}\sum_{i=1}^n\frac{\partial}{\partial \Sigma}\left\{(\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{X}_i - \boldsymbol{\mu})\right\} \\
&= \frac{1}{2}\Sigma^{-1} - \frac{1}{2n}\sum_{i=1}^n\Sigma^{-1}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1},
\end{aligned}$$

and thus

$$\begin{aligned}
\frac{1}{2}\Sigma^{-1} - \frac{1}{2n}\sum_{i=1}^n\Sigma^{-1}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top \Sigma^{-1} &= 0 \\
\implies \frac{1}{2}\Sigma^{-1} &= \frac{1}{2}\Sigma^{-1}\left\{\frac{1}{n}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top\right\}\Sigma^{-1} \\
\implies \Sigma &= \frac{1}{n}\sum_{i=1}^n(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top.
\end{aligned}$$

And therefore, we can take $\hat{\theta}_n = (\hat{\boldsymbol{\mu}}_n, \hat{\Sigma}_n)$ as the solution of $\nabla r_n(\hat{\theta}_n) = 0$, where

$$\hat{\boldsymbol{\mu}}_n = \frac{1}{n}\sum_{i=1}^n\mathbf{X}_i, \quad \hat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^n(\mathbf{X}_i - \hat{\boldsymbol{\mu}}_n)(\mathbf{X}_i - \hat{\boldsymbol{\mu}}_n)^\top.$$

To apply the SAA algorithm theory, specifically Theorem 7, we require that $\mathbb{T}$ is a compact set, which it is clearly not. We can instead consider that our optimization

occurs over some set $\bar{\mathbb{T}} \subset \mathbb{T}$, where $\bar{\mathbb{T}}$ is compact. Here, we can specify $\bar{\mathbb{T}}$ by proposing a sufficiently large closed ball $\bar{\mathbb{B}}_\rho(0)$ for which $\boldsymbol{\mu} \in \bar{\mathbb{B}}_\rho(0)$, and constants $e_\downarrow, e_\uparrow > 0$, such that $\boldsymbol{\Sigma} - e_\downarrow \mathbf{I}$ is positive semi-definite and $e_\uparrow \mathbf{I} - \boldsymbol{\Sigma}$ is positively semi-definite (we can write this more concisely as $e_\downarrow \mathbf{I} \preceq \boldsymbol{\Sigma} \preceq e_\uparrow \mathbf{I}$). That is,

$$\bar{\mathbb{T}} = \{(\boldsymbol{\mu}, \boldsymbol{\Sigma}) : \boldsymbol{\mu} \in \bar{\mathbb{B}}_\rho(0), \boldsymbol{\Sigma} \in \text{SPD}_d, e_\downarrow \mathbf{I} \preceq \boldsymbol{\Sigma} \preceq e_\uparrow \mathbf{I}, \text{ and } \rho, e_\downarrow, e_\uparrow > 0\}$$

can always be made to be an arbitrarily large compact set that contains θ_0 . We finally note that showing that this stationary point $\hat{\theta}_n$ is indeed the global minimizer of r_n is rather complicated, due to the SPD space of $\boldsymbol{\Sigma}$. Such a result can be obtained via the strict concavity of log-likelihood functions in the exponential family distributions.

Exercise 42. Show that the dominance condition (1.3.4) of Theorem 7 is satisfied for the example above.

Example 36. We can now consider a slightly more interesting scenario, where the IID sample $(X_i)_{i \in [n]}$ from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ as a Student t_c probability measure P_X . That is, we consider the case of $p_X = p_{\theta_0}$ in the set

$$\begin{aligned} \mathbb{P} &= \left\{ p_\theta : p_\theta(x) = \frac{\Gamma\left(\frac{c+d}{2}\right)}{\Gamma\left(\frac{c}{2}\right) (c\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2} \left[1 + \frac{1}{c} (\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu})\right]^{\frac{(c+d)}{2}}}, \theta \in \mathbb{T} \right\} \\ &= \{p_\theta : p_\theta(x) = t_c(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}), \theta \in \mathbb{T}\}, \end{aligned}$$

where $c > 0$ is a fixed constant (often called the *degree of freedom*), and

$$\mathbb{T} = \{(\boldsymbol{\mu}, \boldsymbol{\Sigma}) : \boldsymbol{\mu} \in \mathbb{R}^{d \times 1}, \boldsymbol{\Sigma} \in \text{SPD}_d\}.$$

We wish again to obtain the MLE from our data, which requires a solution to

$$\nabla r_n(\theta) = 0,$$

where

$$\begin{aligned} & r_n(\theta) \\ = & -\frac{1}{n} \sum_{i=1}^n \log t_c(\mathbf{X}_i; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \\ = & -\log \Gamma\left(\frac{c+d}{2}\right) + \log \Gamma\left(\frac{c}{2}\right) + \frac{d}{2} \log(c\pi) \\ & + \frac{1}{2} \log |\boldsymbol{\Sigma}| + \frac{1}{n} \left(\frac{c+d}{2}\right) \sum_{i=1}^n \log \left[1 + \frac{1}{c} (\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu})\right]. \end{aligned}$$

Given our matrix calculus expressions, above, we cannot compute simple expressions for the partial derivatives in this scenario.

Remark 49. For every concave function $r(\theta)$, the definition of concavity establishes the fact that we can majorize r by its tangent

$$\mathbf{m}(\theta; \tau) = r(\tau) + \langle \nabla r(\tau), \theta - \tau \rangle.$$

Example 37. We continue on from Example 36, and apply Remark 49 to $r(a) = \log(a)$, which yields the majorizer

$$\mathbf{m}(a, b) = \log(b) + \frac{a-b}{b},$$

when $a, b > 0$.

Applying this majorizer to $r_n(\theta)$, yields $\delta(\mathbf{X}_i; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\begin{aligned}
& \mathbf{m}(\theta, \theta^{(\nu)}) \\
&= -\log \Gamma\left(\frac{c+d}{2}\right) + \log \Gamma\left(\frac{c}{2}\right) + \frac{d}{2} \log(c\pi) + \frac{1}{2} \log |\boldsymbol{\Sigma}| \\
&\quad + \frac{1}{n} \left(\frac{c+d}{2}\right) \sum_{i=1}^n \log \left[1 + \frac{1}{c} \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})\right] \\
&\quad + \frac{1}{n} \left(\frac{c+d}{2}\right) \sum_{i=1}^n \frac{\delta(\mathbf{X}_i; \boldsymbol{\mu}, \boldsymbol{\Sigma}) - \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \\
&= \frac{1}{2} \log |\boldsymbol{\Sigma}| + \frac{1}{n} \left(\frac{c+d}{2}\right) \sum_{i=1}^n \frac{(\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu})}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \\
&\quad + \text{term}_{\theta^{(\nu)}},
\end{aligned}$$

where

$$\delta(\mathbf{X}_i; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = (\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu}).$$

Our MM algorithm is then defined by

$$\mathfrak{A}(\theta^{(\nu)}) \in \arg \min_{\theta \in \mathbb{T}} \mathbf{m}(\theta, \theta^{(\nu)}),$$

which requires the solution of

$$\nabla \mathbf{m}(\theta, \theta^{(\nu)}) = \begin{bmatrix} \frac{\partial}{\partial \boldsymbol{\mu}} \mathbf{m}(\theta, \theta^{(\nu)}) \\ \frac{\partial}{\partial \boldsymbol{\Sigma}} \mathbf{m}(\theta, \theta^{(\nu)}) \end{bmatrix} = \mathbf{0},$$

where

$$\begin{aligned}
\frac{\partial}{\partial \boldsymbol{\mu}} \mathbf{m}(\theta, \theta^{(\nu)}) &= \frac{1}{n} \left(\frac{c+d}{2}\right) \sum_{i=1}^n \frac{-2\boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu})}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \\
&= -\frac{1}{n} (c+d) \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n \frac{\mathbf{X}_i - \boldsymbol{\mu}}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})},
\end{aligned}$$

which implies the solution

$$\begin{aligned}
& \frac{1}{n} (c + d) \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n \frac{\mathbf{X}_i - \boldsymbol{\mu}}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} = \mathbf{0} \\
\Rightarrow & \frac{1}{n} \sum_{i=1}^n \frac{(c + d) \mathbf{X}_i}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} = \boldsymbol{\mu} \frac{1}{n} \sum_{i=1}^n \frac{c + d}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \\
\Rightarrow & \boldsymbol{\mu} = \left(\sum_{i=1}^n \frac{c + d}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \right)^{-1} \sum_{i=1}^n \frac{(c + d) \mathbf{X}_i}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})}.
\end{aligned}$$

We then have

$$\begin{aligned}
& \frac{\partial}{\partial \boldsymbol{\Sigma}} \mathbf{m}(\theta, \theta^{(\nu)}) \\
&= \frac{1}{2} \boldsymbol{\Sigma}^{-1} - \frac{1}{n} \left(\frac{c + d}{2} \right) \sum_{i=1}^n \frac{\boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) (\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})},
\end{aligned}$$

which implies the solution

$$\begin{aligned}
& \frac{1}{2} \boldsymbol{\Sigma}^{-1} - \frac{1}{n} \left(\frac{c + d}{2} \right) \sum_{i=1}^n \frac{\boldsymbol{\Sigma}^{-1} (\mathbf{X}_i - \boldsymbol{\mu}) (\mathbf{X}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} = \mathbf{0} \\
\Rightarrow & \boldsymbol{\Sigma}^{-1} = \frac{1}{n} \boldsymbol{\Sigma}^{-1} \left\{ \sum_{i=1}^n \frac{(c + d) (\mathbf{X}_i - \boldsymbol{\mu}) (\mathbf{X}_i - \boldsymbol{\mu})^\top}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \right\} \boldsymbol{\Sigma}^{-1} = \mathbf{0} \\
\Rightarrow & \boldsymbol{\Sigma} = \frac{1}{n} \sum_{i=1}^n \frac{(c + d) (\mathbf{X}_i - \boldsymbol{\mu}) (\mathbf{X}_i - \boldsymbol{\mu})^\top}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})}.
\end{aligned}$$

We therefore have the MM algorithm for computing the MLE of the form

$$\mathfrak{A}(\theta^{(\nu)}) = (\boldsymbol{\mu}^{(\nu+1)}, \boldsymbol{\Sigma}^{(\nu+1)}),$$

where

$$\boldsymbol{\mu}^{(\nu+1)} = \left(\sum_{i=1}^n \frac{c + d}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})} \right)^{-1} \sum_{i=1}^n \frac{(c + d) \mathbf{X}_i}{c + \delta(\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \boldsymbol{\Sigma}^{(\nu)})},$$

and

$$\Sigma^{(\nu+1)} = \frac{1}{n} \sum_{i=1}^n \frac{(c+d) (\mathbf{X}_i - \boldsymbol{\mu}^{(\nu+1)}) (\mathbf{X}_i - \boldsymbol{\mu}^{(\nu+1)})^\top}{c + \delta (\mathbf{X}_i; \boldsymbol{\mu}^{(\nu)}, \Sigma^{(\nu)})}.$$

2.1.2.4 Misspecification

Consider our setup that IID sample $(X_i)_{i \in [n]}$ from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ has density measure P_X , with density $p_X \in \mathbb{Q}$, but we hypothesize that $p_X \in \mathbb{P}$, where $\mathbb{P}$ contains densities of the form p_θ , for some $\theta \in \mathbb{T}$, and $\mathbb{Q} \cap \mathbb{P} = \emptyset$. We say that our hypothesis $p_X \in \mathbb{P}$ is a *misspecification* of the probability measure P_X . As long as the KL divergence $\text{KL}(p_X, p_\theta)$ exists, our property

$$\theta_* \in \arg \min_{\theta \in \mathbb{T}} \text{KL}(p_X, p_\theta)$$

is still meaningful and estimable via

$$\hat{\theta}_n \in \arg \min_{\theta \in \mathbb{T}} -\frac{1}{n} \sum_{i=1}^n \log(p_\theta(X_i)).$$

In such a situation, our parameter θ_* is no longer the data-generative parameter θ_0 , since there is no such thing as a data-generative parameter $\theta_0 \in \mathbb{T}$, since $p_X \notin \mathbb{P}$. However, we can consider θ_* instead as characteristic of the best estimate of p_X from $\mathbb{P}$; i.e. p_{θ_*} is the best estimate of p_X from $\mathbb{P}$, under the KL divergence.

Example 38. Let X be a *Bernoulli* random variable and have density

$$p_X(x) = \theta_0^x (1 - \theta_0)^{1-x}$$

for $x \in \mathbb{X} = \{0, 1\}$, where $\theta_0 \in (1/2, 1)$. We suppose that we wish to estimate p_X via a density in

$$\mathbb{P} = \left\{ p_\theta : p_\theta(x) = \theta^x (1 - \theta)^{1-x}, \theta \in \mathbb{T} = [0, 1/2] \right\}.$$

Then, our KL divergence between p_X and p_θ is

$$\begin{aligned} \text{KL}(p_X, p_\theta) &= \int_{\mathbb{X}} p_X(x) \log(p_X(x)) - p_X(x) \log(p_\theta(x)) \, dM(x) \\ &= \sum_{x \in \{0,1\}} p_X(x) \log(p_X(x)) - \sum_{x \in \{0,1\}} p_X(x) \log(p_\theta(x)) \\ &= (1 - \theta_0) \log(1 - \theta_0) + \theta_0 \log \theta_0 \\ &\quad - (1 - \theta_0) \log(1 - \theta) - \theta_0 \log \theta. \end{aligned}$$

If we differentiate with respect to θ , we get

$$\nabla \text{KL}(p_X, p_\theta) = \frac{1 - \theta_0}{1 - \theta} - \frac{\theta_0}{\theta},$$

which then yields a second derivative

$$\frac{1 - \theta_0}{(1 - \theta)^2} + \frac{\theta_0}{\theta^2} > 0$$

for all $\theta \in \mathbb{T}$, so $\text{KL}(p_X, p_\theta)$ is strictly convex in θ . Solving

$$\nabla \text{KL}(p_X, p_\theta) = 0,$$

yields $\theta = \theta_0$, indicating that $\text{KL}(p_X, p_\theta)$ is decreasing when $\theta < \theta_0$ and increasing when $\theta > \theta_0$. Thus, the best estimate of p_X from $\mathbb{P}$, under the KL divergence, is p_{θ_*} , where $\theta_* = 1/2$.

2.1.3 Conditional densities and conditional probability

Let us consider the product space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, P)$, where $P \ll M_1 \times M_2$ so that there exists a density $p : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}_{\geq 0}$ of P , such that

$$P((\omega_1, \omega_2) \in \mathbb{A}) = \int_{\Omega_1} \int_{\Omega_2} \chi_{\mathbb{A}}(\omega_1, \omega_2) p(\omega_1, \omega_2) dM_2(\omega_2) dM_1(\omega_1).$$

Then, we say that

$$p_1 = \int_{\Omega_2} p(\omega_1, \omega_2) dM_2(\omega_2)$$

is the *marginal* density of ω_1 (and say that P_1 is the *marginal probability measure* of ω_1), and

$$p(\omega_2 | \omega_1^*) = \begin{cases} p(\omega_1^*, \omega_2) / p_1(\omega_1^*) & \text{if } p_1(\omega_1^*) > 0, \\ \underbrace{\int_{\Omega_1} p(\omega_1, \omega_2) dM_1(\omega_1)}_{=p_2} & \text{if } p_1(\omega_1^*) = 0, \end{cases}$$

is referred to as the *condition* density of ω_2 , given $\omega_1 = \omega_1^*$. We may then interpret the integral

$$P(\omega_2 \in \mathbb{A} | \omega_1 = \omega_1^*) = \int_{\mathbb{A}} p(\omega_2 | \omega_1^*) dM_2(\omega_2),$$

as the *conditional probability* of the event $\omega_2 \in \mathbb{A}$, given the event $\omega_1 = \omega_1^*$.

Using the conditional density and the conditional density functions, we can write

$$P((\omega_1, \omega_2) \in \mathbb{A}) = \int_{\Omega_1} \int_{\Omega_2} \chi_{\mathbb{A}}(\omega_1, \omega_2) p(\omega_2 | \omega_1) dM_2(\omega_2) p_1(\omega_1) dM_1(\omega_1), \quad (2.1.4)$$

and thus, we have the decomposition

$$p(\omega_2 | \omega_1) p_1(\omega_1) = p(\omega_1, \omega_2),$$

for almost every $(\omega_1, \omega_2) \in \Omega_1 \times \Omega_2$, with respect to $M_1 \times M_2$, via uniqueness of the Radon–Nikodym derivative.

Remark 50. As observable from (2.1.4), we can also write, for all $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$,

$$P(\omega_1 \in A, \omega_2 \in B) = \int_A P(\omega_2 \in B | \omega_1 = \omega_1^*) dP_1(\omega_1^*). \quad (2.1.5)$$

Less observable is the fact that we can take *conditional expectations* of functions of both ω_1 and ω_2 , by defining

$$E(X(\omega_1, \omega_2) | \omega_1 = \omega_1^*) = \int_{\Omega_2} X(\omega_1^*, \omega_2) p(\omega_2 | \omega_1^*) dM_2(\omega_2),$$

where we can write

$$E(X(\omega_1, \omega_2)) = \int_{\Omega_1} E(X(\omega_1, \omega_2) | \omega_1 = \omega_1^*) dP_1(\omega_1^*). \quad (2.1.6)$$

We further note that both (2.1.5) and (2.1.6) are true without the existence of the conditional density $p(\omega_2 | \omega_1)$, as long as $P(\omega_2 \in B | \omega_1 = \omega_1^*)$ and $E(X(\omega_1, \omega_2) | \omega_1 = \omega_1^*)$ exist. However, we shall not deal with situations where this observation is required.

Example 39. Let $X(\omega) = (X_1(\omega), X_2(\omega)) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ map from $(\Omega, \mathcal{F}, P)$ to

$$(\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}, \mathcal{B}(\mathbb{R}^{d_1}) \otimes \mathcal{B}(\mathbb{R}^{d_2}), P_X),$$

where $P_X \ll \text{Leb}$, be *jointly normal* in the sense that P_X is characterized by a normal density function $p_X(x) = p_X(x_1, x_2) = \phi(x; \boldsymbol{\mu}, \boldsymbol{\Sigma})$, with mean and covariance matrices

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}.$$

Here, $\boldsymbol{\mu}_1 \in \mathbb{R}^{d_1}$, $\boldsymbol{\mu}_2 \in \mathbb{R}^{d_2}$, $\boldsymbol{\Sigma}_{11} \in \mathbb{R}^{d_1 \times d_1}$, $\boldsymbol{\Sigma}_{22} \in \mathbb{R}^{d_2 \times d_2}$, $\boldsymbol{\Sigma}_{21}^\top = \boldsymbol{\Sigma}_{12} \in \mathbb{R}^{d_1 \times d_2}$, and $\boldsymbol{\Sigma} \in \text{SPD}_{d_1+d_2}$. Under this specification, we can write the *marginal density* of X_1 as

$$p_1(x_1) = \phi(x_1; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}),$$

which yields the *conditional density* of X_2 , given $X_1 = x_1$:

$$p(x_2|x_1) = \phi(x_2; \boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}(\mathbf{x}_1 - \boldsymbol{\mu}_1), \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}).$$

We thus observe that the jointly normal random variable $X = (X_1, X_2)$ has normal marginal densities, as well as normal conditional densities. Since the representation

$$p_X(x_1, x_2) = p(x_2|x_1) p_1(x_1)$$

is almost everywhere unique, by the Radon–Nikodym theorem, it also means that if X_1 is a normal random variable, and if X_2 , condition on $X_1 = x_1$, has normal measure, then $X = (X_1, X_2)$ is jointly normal!

Exercise 43. Confirm that $p(x_2|x_1) = p_X(x_1, x_2) / p_1(x_1)$, for any fixed $x_1 \in \mathbb{R}^{d_1}$.

Example 40. Let $X = (W, Y) \in \mathbb{R}^d \times \mathbb{R}$ be a mapping from $(\Omega, \mathcal{F}, P)$ to

$$(\mathbb{R}^d \times \mathbb{R}, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(\mathbb{R}), P_X)$$

where P_X is characterized by

$$p_X(w, y) = p(y|w) p_W(w),$$

with

$$p_W(w) = \phi(w; \boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W)$$

and

$$p(y|w) = \phi(y; \alpha + \boldsymbol{\beta}^\top \mathbf{w}, \sigma^2).$$

Here, $\boldsymbol{\mu}_W \in \mathbb{R}^{d \times 1}$, $\boldsymbol{\Sigma}_W \in \text{SPD}_d$, $\alpha \in \mathbb{R}$, $\boldsymbol{\beta} \in \mathbb{R}^{d \times 1}$, and $\sigma^2 \in \mathbb{R}_{>0}$. Then, X is jointly normal, and

$$p_X(w, y) = \phi(x; \boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_W \\ \alpha + \boldsymbol{\beta}^\top \boldsymbol{\mu}_W \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_W & \boldsymbol{\Sigma}_W \boldsymbol{\beta} \\ \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_W & \sigma^2 \end{bmatrix}.$$

Exercise 44. Verify the correctness of the expressions for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ in Example 40.

Remark 51. The following variant of the matrix derivative (2.1.3) will be useful in the sequel. Let $\mathbf{x} \in \mathbb{R}^{d_1}$, $\mathbf{y} \in \mathbb{R}^{d_2}$, $\mathbf{B} \in \mathbb{R}^{d_2 \times d_1}$ and $\mathbf{A} \in \mathbb{R}^{d_2 \times d_2}$, then

$$\frac{\partial}{\partial \mathbf{B}} \left\{ (\mathbf{y} - \mathbf{B}\mathbf{x})^\top \mathbf{A} (\mathbf{y} - \mathbf{B}\mathbf{x}) \right\} = -2\mathbf{A} (\mathbf{y} - \mathbf{B}\mathbf{x}) \mathbf{x}^\top,$$

which has the implication that when $y \in \mathbb{R}$, $\boldsymbol{\beta} \in \mathbb{R}^{d_1 \times 1}$, $\mathbf{x} \in \mathbb{R}^{d_1 \times 1}$,

$$\frac{\partial}{\partial \boldsymbol{\beta}} \left\{ (y - \boldsymbol{\beta}^\top \mathbf{x})^2 \right\} = -2\mathbf{x} (y - \boldsymbol{\beta}^\top \mathbf{x}).$$

Example 41. Let X have the regression characterization from Example 40, characterized by parameter value $\theta_0 = (\alpha_0, \boldsymbol{\beta}_0, \sigma_0^2, \boldsymbol{\mu}_{W0}, \boldsymbol{\Sigma}_{W0})$ and let $(X_i)_{i \in [n]}$ be an IID sample from the same data generating process as X . We wish to use $(X_i)_{i \in [n]}$ to

estimate θ_0 , via an MLE, which requires minimization of the empirical risk

$$\begin{aligned} r_n(\theta) &= -\frac{1}{n} \sum_{i=1}^n \log \left\{ \underbrace{\phi(y; \alpha + \beta^\top \mathbf{w}, \sigma^2)}_{=p(y|w)} \underbrace{\phi(x; \boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W)}_{=p_W(w)} \right\} \\ &= r_{2,n}(\theta) + r_{1,n}(\theta), \end{aligned}$$

where

$$\begin{aligned} r_{1,n}(\theta) &= \frac{1}{2} \log(2\pi) + \frac{1}{2} \log(\sigma^2) + \frac{1}{2n\sigma^2} \sum_{i=1}^n (Y_i - \alpha - \beta^\top \mathbf{W}_i)^2 \\ &= r_{1,n}(\alpha, \beta, \sigma^2) \end{aligned}$$

and

$$\begin{aligned} r_{2,n}(\theta) &= \frac{d}{2} \log(2\pi) + \frac{1}{2} \log |\boldsymbol{\Sigma}_W| \\ &\quad + \frac{1}{2n} \sum_{i=1}^n (\mathbf{W}_i - \boldsymbol{\mu}_W)^\top \boldsymbol{\Sigma}_W^{-1} (\mathbf{W}_i - \boldsymbol{\mu}_W) \\ &= r_{2,n}(\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W). \end{aligned}$$

We observe that $r_n(\theta)$ is separable into two functions that are themselves functions dependent on partitions of the parameter $\theta = (\alpha, \beta, \sigma^2, \boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W)$. As such,

$$\arg \min_{\theta \in \mathbb{T}} r_n(\theta) = \left\{ \arg \min_{\theta \in \mathbb{T}} r_{1,n}(\theta) \right\} \cap \left\{ \arg \min_{\theta \in \mathbb{T}} r_{2,n}(\theta) \right\}.$$

Of course,

$$\arg \min_{\theta \in \mathbb{T}} r_{1,n}(\theta) = \left(\arg \min_{\alpha, \beta, \sigma^2} r_{1,n}(\alpha, \beta, \sigma^2), \boldsymbol{\mu}'_W, \boldsymbol{\Sigma}'_W \right),$$

for any $\boldsymbol{\mu}'_W \in \mathbb{R}^d$ and $\boldsymbol{\Sigma}'_W \in \text{SPD}_d$, and

$$\arg \min_{\theta \in \mathbb{T}} r_{2,n}(\theta) = \left(\alpha', \boldsymbol{\beta}', (\sigma^2)', \arg \min_{\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W} r_{2,n}(\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W) \right),$$

for any $\alpha' \in \mathbb{R}$, $\boldsymbol{\beta}' \in \mathbb{R}^{d \times 1}$, and $(\sigma^2)' \in \mathbb{R}_{>0}$. Thus, to compute $\arg \min_{\theta \in \mathbb{T}} r_n(\theta)$, we can solve

$$\arg \min_{\alpha, \boldsymbol{\beta}, \sigma^2} r_{1,n}(\alpha, \boldsymbol{\beta}, \sigma^2) \text{ and } \arg \min_{\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W} r_{2,n}(\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W),$$

separately.

Using the results from Example 35, we have the solution

$$\left(\hat{\boldsymbol{\mu}}_{W,n}, \hat{\boldsymbol{\Sigma}}_{W,n} \right) = \arg \min_{\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W} r_{2,n}(\boldsymbol{\mu}_W, \boldsymbol{\Sigma}_W),$$

where

$$\hat{\boldsymbol{\mu}}_{W,n} = \frac{1}{n} \sum_{i=1}^n \mathbf{W}_i, \quad \hat{\boldsymbol{\Sigma}}_{W,n} = \frac{1}{n} \sum_{i=1}^n (\mathbf{W}_i - \hat{\boldsymbol{\mu}}_{W,n})(\mathbf{W}_i - \hat{\boldsymbol{\mu}}_{W,n})^\top.$$

To solve $\arg \min_{\alpha, \boldsymbol{\beta}, \sigma^2} r_{1,n}(\alpha, \boldsymbol{\beta}, \sigma^2)$, we will write

$$\tilde{\boldsymbol{\beta}} = \begin{bmatrix} \alpha \\ \boldsymbol{\beta} \end{bmatrix} \text{ and } \tilde{\mathbf{W}}_i = \begin{bmatrix} 1 \\ \mathbf{W}_i \end{bmatrix},$$

for each $i \in [n]$. Then,

$$\begin{aligned} r_{1,n}(\alpha, \boldsymbol{\beta}, \sigma^2) &= \frac{1}{2} \log(2\pi) + \frac{1}{2} \log(\sigma^2) + \frac{1}{2n\sigma^2} \sum_{i=1}^n \left(Y_i - \tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{W}}_i \right)^2 \\ &= r_{1,n}(\tilde{\boldsymbol{\beta}}, \sigma^2). \end{aligned}$$

We can apply the Fermat rule and solve

$$\nabla r_{1,n}(\tilde{\boldsymbol{\beta}}, \sigma^2) = \begin{bmatrix} \frac{\partial}{\partial \tilde{\boldsymbol{\beta}}} r_{1,n}(\tilde{\boldsymbol{\beta}}, \sigma^2) \\ \frac{\partial}{\partial \sigma^2} r_{1,n}(\tilde{\boldsymbol{\beta}}, \sigma^2) \end{bmatrix} = \mathbf{0},$$

wheres

$$\begin{aligned} \frac{\partial}{\partial \tilde{\boldsymbol{\beta}}} r_{1,n}(\tilde{\boldsymbol{\beta}}, \sigma^2) &= \frac{1}{2n\sigma^2} \sum_{i=1}^n -2\tilde{\mathbf{W}}_i \left(Y_i - \tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{W}}_i \right) \\ &= -\frac{1}{n\sigma^2} \sum_{i=1}^n \tilde{\mathbf{W}}_i Y_i + \frac{1}{n\sigma^2} \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\} \tilde{\boldsymbol{\beta}}. \end{aligned}$$

The Fermat rule then implies that

$$\begin{aligned} \frac{1}{n\sigma^2} \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\} \tilde{\boldsymbol{\beta}} &= \frac{1}{n\sigma^2} \sum_{i=1}^n \tilde{\mathbf{W}}_i Y_i \\ \implies \tilde{\boldsymbol{\beta}} &= \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\}^{-1} \sum_{i=1}^n \tilde{\mathbf{W}}_i Y_i. \end{aligned}$$

We also have

$$\frac{\partial}{\partial \sigma^2} r_{1,n}(\tilde{\boldsymbol{\beta}}, \sigma^2) = \frac{1}{2\sigma^2} - \frac{1}{2n\sigma^4} \sum_{i=1}^n \left(Y_i - \tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{W}}_i \right)^2,$$

from which Fermat's rule implies

$$\begin{aligned} \frac{1}{2\sigma^2} &= \frac{1}{2n\sigma^4} \sum_{i=1}^n \left(Y_i - \tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{W}}_i \right)^2 \\ \implies \sigma^2 &= \frac{1}{n} \sum_{i=1}^n \left(Y_i - \tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{W}}_i \right)^2. \end{aligned}$$

Thus,

$$\left(\hat{\alpha}_n, \hat{\boldsymbol{\beta}}_n, \hat{\sigma}_n^2 \right) = \arg \min_{\alpha, \boldsymbol{\beta}, \sigma^2} r_{1,n}(\alpha, \boldsymbol{\beta}, \sigma^2),$$

where

$$\begin{bmatrix} \hat{\alpha}_n \\ \hat{\boldsymbol{\beta}}_n \end{bmatrix} = \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\}^{-1} \sum_{i=1}^n \tilde{\mathbf{W}}_i Y_i = \tilde{\boldsymbol{\beta}}_n, \quad (2.1.7)$$

and

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n \left(Y_i - \tilde{\boldsymbol{\beta}}_n^\top \tilde{\mathbf{W}}_i \right)^2. \quad (2.1.8)$$

Exercise 45. To apply Propositions 5 and 6, and Theorem 7, what assumptions do we need to place on $\mathbb{T} \subset \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}_{>0} \times \mathbb{R}^d \times \text{SPD}_d$? Does $\tilde{\beta}_n$ always exist, and what can we assume to guarantee its existence?

Remark 52. It is notable that when a joint density $p(w, y) = p_\theta(w, y)$ of $X = (W, Y) \in \mathbb{X} = \mathbb{W} \times \mathbb{Y}$ is characterized by a parameter $\theta = (\theta_1, \theta_2) \in \mathbb{T} = \mathbb{T}_1 \times \mathbb{T}_2$, where $p_W(w) = p_{\theta_1}(w)$ and $p(y|w) = p_{\theta_2}(y|w)$ are respectively characterized by θ_1 and θ_2 , we do not require direct estimation of θ_1 in order to draw inference about the properties of θ_2 and $p(y|w)$, and thus $P(Y|W = w)$ and $E(f(W, Y)|W = w)$, for any function $(w, y) \mapsto f(w, y)$. Thus, we can characterize $p(y|w)$ and $P(Y|W = w)$, individually to our characterization of $p_W(w)$ and P_W .

2.2 Regression

Let $X = (W, Y) \in \mathbb{X} = \mathbb{W} \times \mathbb{Y}$ be a random function from $(\Omega, \mathcal{F}, P)$ to

$$(\mathbb{W} \times \mathbb{Y}, \mathcal{B}(\mathbb{W}) \otimes \mathcal{B}(\mathbb{Y})),$$

where $\mathbb{W} \subset \mathbb{R}^d$ and $\mathbb{Y} \subset \mathbb{R}^{d_y}$. Given an IID sample $(X_i)_{i \in [n]} = ((W_i, Y_i))_{i \in [n]}$ with the same DGP P_X . In *regression analysis*, we are interested in estimating properties of the *conditional probability measure* $P(Y|W = w)$.

2.2.1 Maximum likelihood estimation

There are many properties of $P(Y|W = w)$ that we can consider, when $P_X \ll M_W \times M_Y$, the most obvious property to consider is the density of $P(Y|W = w)$:

the *conditional density function* $p(y|w)$, such as in Example 40. In Example 41, we showed that one can estimate the parameter relating to $p(y|w)$ and $p_W(w)$, separately, when conducting MLE, if the two factors of $p(w, y)$ are parameterized by a partition of the parameter space $\mathbb{T}$. A lingering question is then to consider what it means to use MLE to estimate the parameter for $p(y|w)$ alone, without considering $p_W(w)$. That is, suppose that

$$p(w, y) = p_\theta(y|w) p_W(w)$$

has the form, characterized by $\theta \in \mathbb{T}$, for some data generative $\theta_0 \in \mathbb{T}$. Then, how does our estimator

$$\hat{\theta}_n \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta)$$

relate to θ_0 , when

$$r_n(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_\theta(X_i),$$

with

$$\ell_\theta(x) = -\log \{p_\theta(y|w)\}.$$

To investigate this relationship, we consider the KL divergence between the data generative density function $p_{\theta_0}(x) = p_{\theta_0}(y|w) p_W(w)$ and the density $p_\theta(x) =$

$p_\theta(y|w)p_W(w)$, characterized by an arbitrary $\theta \in \mathbb{T}$:

$$\begin{aligned}
 \text{KL}(p_{\theta_0}, p_\theta) &= \int_{\mathbb{W}} \int_{\mathbb{Y}} p_{\theta_0}(w, y) \log \left\{ \frac{p_{\theta_0}(w, y)}{p_\theta(w, y)} \right\} dM_Y(y) dM_W(w) \\
 &= \text{term}_{\theta_0} - \int_{\mathbb{W}} \int_{\mathbb{Y}} p_{\theta_0}(w, y) \log p_\theta(w, y) dM_Y(y) dM_W(w) \\
 &= \text{term}_{\theta_0} - \int_{\mathbb{W}} \int_{\mathbb{Y}} p_{\theta_0}(w, y) \log p_W(w) dM_Y(y) dM_W(w) \\
 &\quad - \int_{\mathbb{W}} \int_{\mathbb{Y}} p_{\theta_0}(w, y) \log p_\theta(y|w) dM_Y(y) dM_W(w) \\
 &= \text{term}_{\theta_0} - \int_{\mathbb{W}} E[\log p_\theta(Y|W) | W = w] dP_W(w) \\
 &= \text{term}_{\theta_0} - E[\log p_\theta(Y|W)].
 \end{aligned}$$

This thus justifies our estimation of $-E[\log p_\theta(Y|W)]$ by $r_n(\theta)$, since, if there truly exists a data generative $\theta_0 \in \mathbb{T}$, then

$$\begin{aligned}
 \theta_0 &\in \arg \min_{\theta \in \mathbb{T}} \text{KL}(p_{\theta_0}, p_\theta) \\
 &= \arg \min_{\theta \in \mathbb{T}} -E[\log p_\theta(Y|W)],
 \end{aligned}$$

and thus, under the assumptions of Proposition 6, sequences $(\hat{\theta}_n)_{n \in \mathbb{N}}$, such that

$$\hat{\theta}_n \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta),$$

will converge to θ_0 , if θ_0 is the unique minimizer of $\text{KL}(p_{\theta_0}, p_\theta)$. Under *misspecification*, i.e., when there does not exist a $\theta_0 \in \mathbb{T}$, such that $p_X(x) = p_{\theta_0}(y|w)p_W(w)$ is the density of P_X , for any p_W , then the same observation as that of Section 2.1.2.4 applies, and the sequence $(\hat{\theta}_n)_{n \in \mathbb{N}}$ instead converge to the set

$$\arg \min_{\theta \in \mathbb{T}} \text{KL}(p_X, p_\theta),$$

in the sense that $\inf_{\theta_* \in \arg \min_{\theta \in \mathbb{T}} \text{KL}(\mathbf{p}_X, \mathbf{p}_\theta)} \left\| \hat{\theta}_n - \theta_* \right\| \rightarrow 0$, with probability 1.

Example 42. An often applied method that makes use of the observation regarding misspecification is the method of *maximum normal quasi-likelihood estimation* (or *MQLE*, for short). Here, we suppose that $\mathbf{p}_X(x) = \mathbf{p}(y|w) \mathbf{p}_W(w)$, for $\mathbb{W} \subset \mathbb{R}^d$ and $\mathbb{Y} \subset \mathbb{R}^d$, which we *model* (read *approximate*) with by $\mathbf{p}_\theta(x) = \mathbf{p}_\theta(y|w) \mathbf{p}_W(w)$, where

$$\mathbf{p}_\theta(y|w) = \phi(y; \alpha + \boldsymbol{\beta}^\top \mathbf{w}, \sigma^2), \quad \theta = (\alpha, \boldsymbol{\beta}, \sigma^2).$$

The **MQLE** then yields an estimate $\hat{\theta}_n$ characterized by (2.1.7) and (2.1.8), which converges almost surely to $\theta_* = (\alpha_*, \boldsymbol{\beta}_*, \sigma_*^2) = \arg \min_{\theta} \text{KL}(\mathbf{p}_X, \mathbf{p}_\theta)$, which can be interpreted as the parameters of the best fitting conditional probability model with mean conditional expectation

$$\mathbb{E}[Y | \mathbf{W} = \mathbf{w}] = \alpha_* + \boldsymbol{\beta}_*^\top \mathbf{w},$$

and variance σ_*^2 .

Example 43. Another method that can be viewed through the lens of MLE is *logistic regression*, which considers random variables $X = (W, Y)$, where $W \in \mathbb{W} \subset \mathbb{R}^d$ and $Y \in \mathbb{Y} = \{0, 1\}$. Here, we model the true joint density of $\mathbf{P}_X$ by $\mathbf{p}_\theta(x) = \mathbf{p}_\theta(y|w) \mathbf{p}_W(w)$, where

$$\mathbf{p}_\theta(y|w) = \pi(\theta; w)^y \{1 - \pi(\theta; w)\}^{1-y}.$$

That is, conditional on $W = w$, Y is a Bernoulli random variable with expected value (*success probability*) $\pi(\theta; w)$. Here, logistic regression specifies that the mean

takes the form

$$\pi(\theta; w) = \frac{\exp(\alpha + \beta^\top w)}{1 + \exp(\alpha + \beta^\top w)},$$

which is often referred to as the *logit* function in the traditional statistics literature.

Using IID data $(X_i)_{i \in [n]} = ((W_i, Y_i))_{i \in [n]}$ with the same measure as X , we define the MLE of the logistic regression model as

$$\hat{\theta}_n \in \arg \min_{\theta \in \mathbb{T}} \underbrace{-\frac{1}{n} \sum_{i=1}^n \log p_\theta(Y_i | W_i)}_{=r_n(\theta)}$$

where

$$r_n(\theta) = -\frac{1}{n} \sum_{i=1}^n Y_i (\alpha + \beta^\top W_i) + \frac{1}{n} \sum_{i=1}^n \{1 + \exp(\alpha + \beta^\top W_i)\}^{-1},$$

and $\theta = (\alpha, \beta) \in \mathbb{T} \subset \mathbb{R}^{1+d}$.

Exercise 46. A natural extension to the logistic regression model is the so-called *multinomial logistic regression model*, where $X = (W, Y)$, with $W \in \mathbb{W} \subset \mathbb{R}^d$ and $Y \in \mathbb{Y} = \{1, 2, \dots, K\}$, and the density of P_X is modeled by $p_\theta(x) = p_\theta(y|w) p_W(w)$, where

$$p_\theta(y|w) = \pi_1(\theta; w)^{[y=1]} \pi_2(\theta; w)^{[y=2]} \dots \pi_K(\theta; w)^{[y=K]}.$$

Propose an expression for the functions $\pi_k : \mathbb{T} \times \mathbb{W} \rightarrow \mathbb{R}$, such that conditional on $W = w$, Y has a *categorical distribution*, where we say that $U(\omega) : \Omega \rightarrow \{1, 2, \dots, K\}$ has a categorical distribution if P_U (absolutely continuous with respect to the counting measure on the integers), has density function

$$p_U(u) = \pi_1^{[y=1]} \pi_2^{[y=2]} \dots \pi_K^{[y=K]}.$$

Example 44. We continue with Example 43 by majorizing $r_n(\theta)$ by a function

$$\begin{aligned} \mathbf{m}(\theta, \tau) &= r_n(\tau) + \frac{1}{n} \sum_{i=1}^n \{\pi(\tau; W_i) - Y_i\} \mathbf{W}_i^\top (\boldsymbol{\theta} - \boldsymbol{\tau}) \\ &\quad + \frac{1}{8} (\boldsymbol{\theta} - \boldsymbol{\tau})^\top \left\{ \frac{1}{n} \sum_{i=1}^n \mathbf{W}_i \mathbf{W}_i^\top \right\} (\boldsymbol{\theta} - \boldsymbol{\tau}), \end{aligned}$$

using the quadratic upper bound majorizer from Example 27. Using $\mathbf{m}(\theta, \tau)$, we can generate an MM algorithm sequence $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ via the algorithm

$$\begin{aligned} \theta^{(\nu)} &= \mathfrak{A}(\theta^{(\nu-1)}) \\ &= \arg \max_{\theta \in \mathbb{T}} \mathbf{m}(\theta, \theta^{(\nu-1)}) \\ &= \theta^{(\nu-1)} - 4 \left\{ \sum_{i=1}^n \mathbf{W}_i \mathbf{W}_i^\top \right\}^{-1} \sum_{i=1}^n \mathbf{W}_i \left\{ \pi(\theta^{(\nu-1)}; W_i) - Y_i \right\}. \end{aligned}$$

Exercise 47. Verify that $\mathbf{m}(\theta, \tau)$ is in fact a majorizer of $r_n(\theta)$, in Example 44.

Remark 53. There is a case to be made regarding the fact that $\mathfrak{A}(\cdot)$ in Example 44, due to the potential singularity of the matrix

$$\sum_{i=1}^n \mathbf{W}_i \mathbf{W}_i^\top.$$

To remedy this potentiality, we can replace the majorizer $\mathbf{m}(\theta, \tau)$ by the *proximal-modified majorizer*

$$\mathbf{m}_\lambda(\theta, \tau) = \mathbf{m}(\theta, \tau) + \frac{\lambda}{2} \|\theta - \tau\|_2^2, \quad (2.2.1)$$

for some $\lambda > 0$. Using $\mathbf{m}_\lambda$ instead of $\mathbf{m}$, we can define the proximal-MM sequence $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ via the algorithm

$$\begin{aligned} \theta^{(\nu)} &= \mathfrak{A}_\lambda \left(\theta^{(\nu-1)} \right) \\ &= \arg \max_{\theta \in \mathbb{T}} \mathbf{m}_\lambda \left(\theta, \theta^{(\nu-1)} \right) \\ &= \theta^{(\nu-1)} - \left\{ \frac{1}{4} \sum_{i=1}^n \mathbf{W}_i \mathbf{W}_i^\top + \lambda \mathbf{I} \right\}^{-1} \sum_{i=1}^n \mathbf{W}_i \left\{ \pi \left(\theta^{(\nu-1)}; W_i \right) - Y_i \right\}, \end{aligned}$$

which is always computable, the sum of a positive semi-definite matrix and a positive definite matrix is always positive definite.

Exercise 48. Show that if $\mathbf{m}(\theta, \tau)$ is a majorizer of $r(\theta)$, then the proximal modification $\mathbf{m}_\lambda(\theta, \tau)$, defined by (2.2.1), is also a majorizer of $r(\theta)$, for any $\lambda > 0$.

2.2.2 Loss-based risk minimization

When considering the relationship between W and Y , we need not make any strong probabilistic hypotheses. We may simply make qualitative statements regarding the relationship between Y and W . For example, we can suppose that Y is *predicted* by W via a *regression model* $\mathfrak{R}(w; \theta)$, depending on parameter $\theta \in \mathbb{T}$, in the sense that

$$Y \approx \mathfrak{R}(W; \theta_0)$$

for some $\theta_0 \in \mathbb{T}$. We may then assess the goodness of our model by considering the a loss of the form

$$\ell_\theta(w, y) = \mathbf{D}(\mathfrak{R}(w; \theta), y),$$

for some *divergence* $\mathbf{D}$ on $\mathbb{R}$, in the sense that $(a, b) \mapsto \mathbf{D}(a, b)$, where

1. $D(a, b) = 0$ if and only if $a = b$, and
2. $D(a, b) \geq 0$, for all $a, b \in \mathbb{R}$.

Example 45. Some divergences related to typical losses for the case when $\mathbb{Y} = \mathbb{R}$ are as follows:

- l_q -norm loss: $D(a, b) = |a - b|^q$, for $q \geq 1$.
 - The most popular l_q -norm losses are the cases when $q = 1$ or $q = 2$, in which case the loss is called the *absolute difference loss* and the *square error (least square) loss*, respectively.
- Huber loss: for $\delta > 0$:

$$D(a, b) = \begin{cases} \frac{1}{2} (a - b)^2 & \text{if } |a - b| \leq \delta, \\ \delta (|a - b| - \frac{1}{2}\delta) & \text{if } |a - b| > \delta. \end{cases}$$

- Support vector regression loss: for $\varepsilon > 0$,

$$D(a, b) = \begin{cases} 0 & \text{if } |a - b| \leq \varepsilon, \\ |a - b| - \varepsilon & \text{if } |a - b| > \varepsilon. \end{cases} \quad (2.2.2)$$

Of course, the target when conducting *loss-based risk minimization* is to identify

$$\theta_* \in \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta),$$

where

$$\bar{r}(\theta) = \mathbb{E}[\ell_\theta(W, Y)] = \mathbb{E}[D(\mathfrak{R}(W; \theta), Y)],$$

which we estimate via an empirical risk minimizer

$$\hat{\theta}_n \in \arg \min_{\theta \in \mathbb{T}} r_n(\theta),$$

where

$$r_n(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_{\theta}(W_i, Y_i) = \frac{1}{n} \sum_{i=1}^n D(\mathfrak{R}(W_i; \theta), Y_i)$$

when we are provided with an IID sample $(X_i)_{i \in [n]}$, where each $X_i = (W_i, Y_i)$ has the same image measure P_X as $X = (W, Y)$.

Example 46. A popular loss minimization approach is *ordinary least squares*, where one suggests a linear regression function

$$\mathfrak{R}(w; \theta) = \alpha + \boldsymbol{\beta}^\top \mathbf{w},$$

where $\theta = (\alpha, \boldsymbol{\beta}) \in \mathbb{T} = \mathbb{R}^{1+d}$, and minimize the risk based on the l_2 -norm loss

$$\ell_{\theta}(w, y) = |\mathfrak{R}(w; \theta) - y|^2,$$

which yields the empirical risk:

$$r_n(\theta) = \frac{1}{n} \sum_{i=1}^n \left| Y_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i \right|^2,$$

where, for each $i \in [n]$

$$\tilde{\mathbf{W}}_i = \begin{bmatrix} 1 \\ \mathbf{w}_i \end{bmatrix}.$$

It is not difficult to verify that

$$\begin{aligned} \hat{\theta}_n &= \arg \min_{\theta \in \mathbb{T}} r_n(\theta) \\ &= \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\}^{-1} \sum_{i=1}^n \tilde{\mathbf{W}}_i Y_i, \end{aligned}$$

when $\sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top$ is non-singular, as per (2.1.7) from Example 40. It is also simple to characterize

$$\begin{aligned} \theta_* &= \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta) \\ &= \arg \min_{\theta \in \mathbb{T}} \mathbb{E} \left[\left| Y - \boldsymbol{\theta}^\top \tilde{\mathbf{W}} \right|^2 \right] \end{aligned}$$

by permitting ourselves to differentiate inside the expectation operator and solve the Fermat condition

$$\nabla \bar{r}(\theta) = \mathbb{E} \left[\nabla \left(\left| Y - \boldsymbol{\theta}^\top \tilde{\mathbf{W}} \right|^2 \right) \right] = \mathbf{0},$$

which implies

$$\mathbb{E} \left[-2\tilde{\mathbf{W}} \left(Y - \tilde{\mathbf{W}}^\top \boldsymbol{\theta} \right) \right] = \mathbf{0} \implies \boldsymbol{\theta}_* = \mathbb{E} \left[\tilde{\mathbf{W}} \tilde{\mathbf{W}}^\top \right]^{-1} \mathbb{E} \left[\tilde{\mathbf{W}} Y \right],$$

by the linearity of the expectation, and under the assumption that $\mathbb{E} \left[\tilde{\mathbf{W}} \tilde{\mathbf{W}}^\top \right]^{-1}$ is invertible. Thus, as long as $\mathbb{E} \left[\tilde{\mathbf{W}} \tilde{\mathbf{W}}^\top \right]^{-1}$ and $\mathbb{E} \left[\tilde{\mathbf{W}} Y \right]$ both exist, ordinary least squares is a sensible approach to estimating a linear regression model $\mathfrak{R}(w; \theta) = \alpha + \boldsymbol{\beta}^\top \mathbf{w}$ for relating Y to W , where the *risk minimizing regression model* can be explicitly expressed as $\mathfrak{R}(w, \theta_*)$.

Exercise 49. Prove that the integral of a convex function is convex, and use the fact to show that the risk in Example 46 is convex.

Remark 54. In Example 46, we permitted ourselves to differentiate under the expectation sign. Of course there are conditions that regulate our ability to swap these operations. Namely, for a function $(\omega, \theta) \mapsto X(\omega, \theta)$, which maps from $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ for each fixed $\theta \in \mathbb{T} \subset \mathbb{R}^p$, we make the assumptions:

1. The expected value $E[X(\cdot, \tau)]$ is defined and has finite value for each $\tau \in \mathbb{T}$.
2. There exists a positive random variable $L(\omega)$ such that $E[L] < \infty$, and for all $\theta_1, \theta_2 \in \mathbb{T}$ in a neighborhood of τ , and for almost every $\omega \in \Omega$, we have

$$|X(\omega, \theta_1) - X(\omega, \theta_2)| \leq L(\omega) \|\theta_1 - \theta_2\|.$$

3. For almost every $\omega \in \Omega$, $X(\omega, \cdot)$ is differentiable at τ .

If 1–3 are satisfied for some $\tau \in \mathbb{T}$, then

$$\nabla E[X(\cdot, \tau)] = E[\nabla X(\cdot, \tau)].$$

Of course, if 1–3 are true for all $\tau \in \mathbb{T}$ then we are permitted to swap the expectation and differentiation signs at any $\tau \in \mathbb{T}$.

Exercise 50. Enforce enough conditions on $\mathbb{T}$ and P_X to permit the application of Remark 54 to Example 46.

Example 47. We now consider a modified version of the ordinary least squares problem that can be thought of as a least-squares version of the support vector regression problem. Here, we shall again hypothesize that the relationship between W and Y is linear and use regression model

$$\mathfrak{R}(w; \theta) = \alpha + \beta^\top w,$$

for $\theta = (\alpha, \beta)$. Based on the support vector regression loss (2.2.2), we shall consider the loss function

$$\ell_{\theta}(w, y) = \begin{cases} 0 & \text{if } |\Re(w; \theta) - y| \leq \varepsilon, \\ |\Re(w; \theta) - y|^2 - \varepsilon^2 & \text{if } |\Re(w; \theta) - y| > \varepsilon, \end{cases}$$

for some $\varepsilon > 0$, which we can write more succinctly as

$$\ell_{\theta}(w, y) = [\max\{\varepsilon, |\Re(w; \theta) - y|\}]^2 - \varepsilon^2.$$

We are thus interested in estimating the risk minimizer

$$\theta_* \in \arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta),$$

where

$$\begin{aligned} \bar{r}(\theta) &= \mathbb{E}[\ell_{\theta}(W, Y)] \\ &= \mathbb{E}\left([\max\{\varepsilon, |\Re(W; \theta) - Y|\}]^2\right) - \varepsilon^2. \end{aligned}$$

To estimate θ_* , we are required to compute the minimizer

$$\hat{\theta}_n = \arg \min_{\theta \in \mathbb{T}} r_n(\theta),$$

where

$$r_n(\theta) = \frac{1}{n} \sum_{i=1}^n \left[\max\left\{\varepsilon, \left|Y_i - \theta^{\top} \tilde{\mathbf{W}}_i\right|\right\} \right]^2 - \varepsilon^2,$$

with $\tilde{\mathbf{W}}_i$ defined as in Example 46, for each $i \in [n]$.

Remark 55. To proceed further, we require a majorizer for functions of the form

$$r(\theta) = \max\{\varepsilon, |\theta|\}^2 - \varepsilon^2,$$

which exhibits two kinds of behavior depending on the value of $\theta \in \mathbb{R}$. Namely

$$r(\theta) = \begin{cases} 0 & \text{if } |\theta| \leq \varepsilon, \\ \theta^2 - \varepsilon^2 & \text{if } |\theta| > \varepsilon. \end{cases}$$

This implies that we need to capture both kinds of behavior when constructing a majorizer. One possible majorizer for $r(\theta)$ is

$$\begin{aligned} \mathbf{m}(\theta, \tau) &= \begin{cases} (\theta - \tau)^2 & \text{if } |\tau| \leq \varepsilon, \\ \theta^2 - \varepsilon^2 & \text{if } |\tau| > \varepsilon, \end{cases} \\ &= (\theta - \tau)^2 [\tau \leq \varepsilon] + \{\theta^2 - \varepsilon^2\} [\tau > \varepsilon] \end{aligned}$$

Exercise 51. Check that the function $\mathbf{m}(\theta, \tau)$ above is indeed a majorizer of $r(\theta)$.

Example 48. We continue from Example 47. We can apply Remark 55 to obtain the majorizer

$$\begin{aligned} \mathbf{m}(\theta, \tau) &= \frac{1}{n} \sum_{i=1}^n \left(Y_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i - Y_i + \boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i \right)^2 \left[\left| Y_i - \boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i \right| \leq \epsilon \right] \\ &\quad + \frac{1}{n} \sum_{i=1}^n \left\{ \left(Y_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i \right)^2 - \varepsilon^2 \right\} \left[\left| Y_i - \boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i \right| > \epsilon \right] \\ &= \frac{1}{n} \sum_{i=1}^n \left(\boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i \right)^2 \left[\left| Y_i - \boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i \right| \leq \epsilon \right] \\ &\quad + \frac{1}{n} \sum_{i=1}^n \left\{ \left(Y_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i \right)^2 - \varepsilon^2 \right\} \left[\left| Y_i - \boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i \right| > \epsilon \right] \\ &= \frac{1}{n} \sum_{i=1}^n \gamma(X_i; \tau) \left(\boldsymbol{\tau}^\top \tilde{\mathbf{W}}_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i \right)^2 \\ &\quad + \frac{1}{n} \sum_{i=1}^n \{1 - \gamma(X_i; \tau)\} \left\{ \left(Y_i - \boldsymbol{\theta}^\top \tilde{\mathbf{W}}_i \right)^2 - \varepsilon^2 \right\}, \end{aligned}$$

for $r_n(\theta)$, where we define

$$\gamma(X; \tau) = \left[\left| Y - \boldsymbol{\tau}^\top \tilde{\mathbf{W}} \right| \leq \epsilon \right].$$

An MM algorithm can be constructed by specifying an algorithm

$$\theta^{(\nu)} = \mathfrak{A} \left(\theta^{(\nu-1)} \right) = \arg \min_{\theta \in \mathbb{T}} \mathfrak{m} \left(\theta, \theta^{(\nu-1)} \right),$$

which we can obtain by solving the Fermat condition

$$\nabla \mathfrak{m} \left(\theta, \theta^{(\nu-1)} \right) = \mathbf{0},$$

where

$$\begin{aligned} & \nabla \mathfrak{m} \left(\theta, \theta^{(\nu-1)} \right) \\ &= -\frac{2}{n} \sum_{i=1}^n \gamma \left(\mathbf{X}_i; \theta^{(\nu-1)} \right) \tilde{\mathbf{W}}_i \left[\tilde{\mathbf{W}}_i^\top \boldsymbol{\theta}^{(\nu-1)} - \tilde{\mathbf{W}}_i^\top \boldsymbol{\theta} \right] \\ & \quad - \frac{2}{n} \sum_{i=1}^n \left\{ 1 - \gamma \left(\mathbf{X}_i; \theta^{(\nu-1)} \right) \right\} \tilde{\mathbf{W}}_i \left[Y_i - \tilde{\mathbf{W}}_i^\top \boldsymbol{\theta} \right]. \end{aligned}$$

The Fermat condition then implies that

$$\sum_{i=1}^n \left[\gamma \left(\mathbf{X}_i; \theta^{(\nu-1)} \right) \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \boldsymbol{\theta}^{(\nu-1)} + \left\{ 1 - \gamma \left(\mathbf{X}_i; \theta^{(\nu-1)} \right) \right\} \tilde{\mathbf{W}}_i Y_i \right] = \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\} \boldsymbol{\theta}$$

which yields the solution

$$\boldsymbol{\theta}^{(\nu)} = \left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\}^{-1} \sum_{i=1}^n \left[\gamma \left(\mathbf{X}_i; \theta^{(\nu-1)} \right) \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \boldsymbol{\theta}^{(\nu-1)} + \left\{ 1 - \gamma \left(\mathbf{X}_i; \theta^{(\nu-1)} \right) \right\} \tilde{\mathbf{W}}_i Y_i \right],$$

when $\left\{ \sum_{i=1}^n \tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i^\top \right\}^{-1}$ is invertible. Without a lot of effort, we can show that the

MM algorithm $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ has a convergent subsequence that is a global minimum of

r_n , since it is convex. Furthermore, with a bit more work, we can show that not only does $(\theta^{(\nu)})_{\nu \in \mathbb{N}}$ have a convergent subsequence, but that it must also be convergent to a global minimum of r_n , in its own right.

Remark 56. Of course, with some regularity on the probability measure P_X and the compactness restrictions on the parameter space $\mathbb{T}$, we can use the SAA theory from Section 1.3.6 to show that $\hat{\theta}_n$ converges a.s. to

$$\arg \min_{\theta \in \mathbb{T}} \bar{r}(\theta) = \arg \min_{\theta \in \mathbb{T}} E \left([\max \{ \varepsilon, |\mathfrak{R}(W; \theta) - Y| \}]^2 \right) - \varepsilon^2,$$

and that $\min_{\theta \in \mathbb{T}} r_n(\theta)$ converges a.s. to $\min_{\theta \in \mathbb{T}} \bar{r}(\theta)$. Another interesting question to ask is of course whether $\bar{r}(\hat{\theta}_n)$ converges to $\min_{\theta \in \mathbb{T}} \bar{r}(\theta)$, which can be answered in the affirmative, by considering the now famous inequality

$$\bar{r}(\hat{\theta}_n) - \inf_{\theta \in \mathbb{T}} \bar{r}(\theta) \leq 2 \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)|,$$

which leads to the probability statement that

$$P \left(\omega : \bar{r}(\hat{\theta}_n) - \inf_{\theta \in \mathbb{T}} \bar{r}(\theta) > \epsilon \right) \leq P \left(\omega : 2 \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)| > \epsilon \right).$$

Of course, our USLLN allows us to prove that

$$P \left(\omega : \lim_{n \rightarrow \infty} \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)| = 0 \right) = 1,$$

and thus

$$\lim_{n \rightarrow \infty} P \left(\omega : \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)| > \frac{\epsilon}{2} \right) = 0,$$

as a consequence, and hence $\bar{r}(\hat{\theta}_n)$ converges to $\inf_{\theta \in \mathbb{T}} \bar{r}(\theta)$, in probability, as $n \rightarrow \infty$.

But we can ask the further question as to whether we can make a more precise statement than this. Again, the answer is affirmative, as if $\mathfrak{R}$, $\mathbb{X} = \mathbb{W} \times \mathbb{Y}$ and $\mathbb{T}$ are such that

$$\ell_{\theta}(w, y) = [\max\{\varepsilon, |\mathfrak{R}(w; \theta) - y|\}]^2 - \varepsilon^2 \leq B$$

for every $\theta \in \mathbb{T}$ and $(w, y) \in \mathbb{X}$, and $\ell_{\theta}(w, y)$ has a *pseudo-dimension* of v , then for any P_X on $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$,

$$P \left(\omega : \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)| > \epsilon \right) \leq 4 \left(\frac{8enB}{\epsilon v} \right)^v \exp \left(-\frac{n\epsilon^2}{32} \right),$$

where the pseudo-dimension measures the “complexity” of the class of functions:

$$\mathcal{L} = \{\ell_{\theta}(w, y) : \theta \in \mathbb{T}, (w, y) \in \mathbb{X}\}.$$

For a rule of thumb, if we can compute $\ell_{\theta}(w, y)$ using a finite number of operations consisting of only the symbols $+$, $-$, $\times$, $/$, the indicator function $[\cdot \sim \cdot]$, where $\sim$ can be replaced by $<$, $\leq$, $>$, $\geq$, $=$, $\neq$, and the exponential function $\exp(\cdot)$, then $\mathcal{L}$ has a finite pseudo-dimension. Clearly, for our loss function, the pseudo-dimension is finite, via this criterion. Thus, combining everything together, we observe that

$$\begin{aligned} P \left(\omega : \bar{r}(\hat{\theta}_n) - \inf_{\theta \in \mathbb{T}} \bar{r}(\theta) > \epsilon \right) &\leq P \left(\omega : \sup_{\theta \in \mathbb{T}} |r_n(\theta) - \bar{r}(\theta)| > \frac{\epsilon}{2} \right) \\ &= 4 \left(\frac{16enB}{\epsilon v} \right)^v \exp \left(-\frac{n\epsilon^2}{128} \right). \end{aligned}$$

This implies that as n increases, the probability that $\bar{r}(\hat{\theta}_n) - \inf_{\theta \in \mathbb{T}} \bar{r}(\theta) > \epsilon$ becomes diminishingly smaller at a rate that is a ratio of a polynomial of finite degree v and an exponential in n . Therefore, for large n , the rate is effectively

exponential, but the largeness of n required to reach this asymptotically exponential rate depends on the complexity of the class $\mathcal{L}$ via the pseudo-dimension v .

Exercise 52. Use the inequality above to generate a $(1 - \alpha) \times 100\%$ confidence interval for $\bar{r}(\hat{\theta}_n) - \inf_{\theta \in \mathbb{T}} \bar{r}(\theta)$.

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